Between Weak and Contaminated: Identifying Monetary Policy Shocks with Imperfect Instruments[1]

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Abstract

No available instrument for monetary policy shocks is both strong and clean. Orthogonalising high-frequency surprises on pre-announcement information makes them cleaner but weakens their first-stage strength, and that weakness magnifies any residual contamination. Instruments that cover the whole policy decision, rather than the narrow window around the announcement, are strong but exposed to other contamination. I group instruments built from the same construction into families, whose members share their contamination but differ in relevance. Demeaning within each family removes the shared contamination, and pooling across families strengthens the first stage. In U.S. data, the pooled instrument is much stronger than the orthogonalised surprise and identifies a gradual decline in output and prices and a widening of credit spreads.

Keywords: monetary policy shocks; high-frequency identification; orthogonalisation; weak instruments; instrument families; instrument pooling; proxy-SVAR.

JEL classification codes: E43, E52, E58, C32, C36

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1 Introduction

What are the effects of monetary policy on the economy? The question is central to macroeconomics, yet estimates of how output and prices respond to a monetary tightening still vary widely across studies (Ramey, 2016; Brennan et al., 2025). The difficulty lies in identifying the causal effect of policy. Changes in the policy rate largely reflect the central bank's systematic response to economic conditions. The effect of policy can only be read from its unanticipated part, the structural monetary policy shock, which is not observed. A common approach estimates a structural vector autoregression (VAR) with external instruments, variables that supply identifying variation because they move with the policy shock and not with the other forces driving the economy (Mertens and Ravn, 2013; Stock and Watson, 2018). A useful instrument must be strong, moving the policy rate enough to identify its effects precisely. It must also be clean, free of the contamination that other news about the economy brings.

For monetary policy, the literature has built several candidate instruments, each with its own strengths and weaknesses. The most widely used are high-frequency surprises, the changes in interest-rate futures in a narrow window around Federal Open Market Committee (FOMC) announcements (Kuttner, 2001; Gürkaynak et al., 2005; Gertler and Karadi, 2015; Nakamura and Steinsson, 2018). Recently, these surprises have been shown to be predictable from information available before the announcement (Miranda-Agrippino and Ricco, 2021; Bauer and Swanson, 2023b). Two views compete. In one, the announcement reveals private information the Fed holds about the economy (Nakamura and Steinsson, 2018; Miranda-Agrippino and Ricco, 2021). In the other, the Fed responds to news that the market had not yet anticipated and fully priced (Schmeling et al., 2022; Bauer and Swanson, 2023a). In both, the predictability is a sign that the surprise responds to news, not only to policy. The standard remedy, orthogonalisation, projects the surprise on pre-announcement information and keeps the residual (Bauer and Swanson, 2023b). The remedy is necessary for the surprise to be orthogonal to the news that precedes the announcement, but it weakens the first-stage strength of the surprise (Bauer and Swanson, 2023b; Gürkaynak et al., 2026; Amodeo, 2026). The impulse responses identified with the orthogonalised surprise are then large and imprecise. The price level falls in the very month of the announcement, at odds with evidence that prices respond to monetary policy only gradually (Aruoba and Drechsel, 2024a; Buda et al., 2025). Other instruments have been proposed for the same shock, such as surprises cleaned of the information the Fed reveals (Jarociński and Karadi, 2020; Miranda-Agrippino and Ricco, 2021), narrative residuals of the Fed's decisions (Romer and Romer, 2004) and Cholesky-based innovations of a VAR (Christiano et al., 1999). Nothing guarantees that any of them is both strong, moving the policy rate enough, and clean of the news that drives the economy.

This paper asks how several imperfect instruments for the same shock can be combined, when none of them is assumed to be perfectly clean. The method it proposes is general, and the paper develops it for monetary policy in two steps. Every instrument is first orthogonalised on the same pre-announcement information, to remove its predictable component, following Bauer and Swanson (2023b). The instruments are then grouped into families and pooled. The pooling removes the contamination that the members of a family share, and it restores the first-stage strength that the orthogonalised surprise lacks.

Orthogonalisation is necessary but not sufficient. It removes the contamination that the chosen controls span. Any news the controls do not span survives in the instrument. Orthogonalisation also weakens the first-stage strength, because the policy rate reacts to the same pre-announcement

news. The strength lost depends on how closely the predictable component moves with the policy rate. Two equally predictable instruments can therefore lose very different shares of their first-stage strength. A weaker first stage ultimately magnifies the contamination that survives. When the orthogonalisation removes a larger share of the first-stage strength than of the contamination, the bias in the estimated impulse responses may not fall, and can even rise. The high-frequency surprise illustrates this mechanism. Orthogonalisation weakens its first-stage strength considerably.

The paper therefore turns to instruments that bring strength back to the identification. These instruments widen the window. Instead of the change in expectations in a narrow window around the announcement, they use the variation in the policy rate over the whole decision. A wider window also captures the part of the decision that the market prices before the announcement (Lucca and Moench, 2015; Bernile et al., 2016; Cieslak et al., 2019). The first, the Romer–Romer instrument, follows Romer and Romer (2004). It removes from the intended change in the federal funds rate the Fed's information at the time of the decision, measured by its Greenbook forecasts. The second, the Cholesky-based instrument, expresses the recursive restrictions of a VAR as an instrument (Sims, 1980; Bernanke and Blinder, 1992; Christiano et al., 1999). It removes the month's output and price innovations from the policy-rate innovation. Both instruments are partly predictable from information available before the announcement. They are therefore orthogonalised on the same controls as the surprise. Whereas the first-stage strength of the high-frequency surprise falls sharply after orthogonalisation, the wide-window instruments preserve more of it, because their predictable component moves less closely with the policy rate.

The wider window comes at a cost. The wide-window instruments gain strength, but they admit news that the narrow window excludes, such as data releases and financial developments later in the day or the month. Orthogonalised instruments may also retain pre-announcement news that the controls do not span. Instruments are therefore not necessarily free of contamination after orthogonalisation. The method proposed in this paper accepts this trade-off between strength and cleanliness. It removes the part of the residual contamination that similarly built instruments share.

Combining the instruments restores the strength that the orthogonalised surprise lacks. Simple averaging does not suffice. Averaging raises the first-stage strength and averages out the contamination specific to each instrument, as with factor-based instruments (Bai and Ng, 2010). The contamination that the instruments share remains no matter how many instruments are added.

This paper instead groups the instruments into families. A family is a set of instruments built such that they share their exposure to non-policy news. A family typically gathers instruments built from the same source, over the same window and with the same orthogonalisation. The members of a family share their contamination, but they may differ in their relevance for the policy shock. Demeaning the identifying moments, the covariances of each instrument with the VAR innovations, within each family removes the contamination the family shares. The relevance deviations that remain identify the impact direction of the policy shock, the direction in which it moves the endogenous variables on impact.

The estimator solves a minimum-distance problem (Newey and McFadden, 1994). It selects the impact direction, the relevance of each instrument and the common contamination of each family that best fit the estimated moments. The impact direction is then the leading eigenvector of the demeaned moments, the direction along which they vary most. In the population, the direction is identified when the members of each family share their contamination and relevance varies within at least one family. In a finite sample, recovering it also requires that relevance varies more within families than the contamination specific to individual instruments, the part that demeaning does

not remove. The leading direction is then the policy shock rather than a contamination direction. Geometrically, the moments of each family are shifted away from the impact direction of the policy shock by the family's contamination, and the leading principal component of the moments before demeaning therefore points in another direction. When the members of each family share their contamination, demeaning removes these shifts. The families then line up along the impact direction, and the leading principal component recovers it. An overidentification test evaluates the family assignment. Under its null hypothesis, every instrument points to the same direction once the contamination of its family is removed.

The same identification can be written as a constrained pooling of the instruments. The pooling weights maximise the strength of a single combined instrument, subject to the constraint that it cancels the contamination the families share. The weights are free within families, and the family totals are set to satisfy this constraint. The pooled instrument reproduces the impulse responses of the principal-component step exactly. The family averages add strength without reintroducing the shared contamination only when the families outnumber the non-policy shocks in the VAR. This condition is not needed to identify the direction.

The monetary policy instruments fall into different families because of how they are constructed. Two families are built from high-frequency surprises around FOMC announcements, one from near-dated futures and one from the factors of Swanson (2021). One family is built from Romer–Romer instruments and one from Cholesky-based instruments. Applied to these orthogonalised U.S. instruments, the pooled instrument is strong where the orthogonalised surprise is weak. It identifies a standard monetary contraction. Output falls gradually, with its peak decline after fifteen months. Consumer prices fall little on impact and decline gradually, and the credit spread widens on impact. Pooled without demeaning, the instruments are very strong, but they lean on the strongest family, inherit its contamination and point to a much milder contraction. The robustness exercises change the panel of instruments to limit the dependence on a single family. They add a fifth family of information-robust instruments, built from surprises cleaned of the information the Fed reveals. They also diversify the families with Treasury surprises at longer maturities and with narrative shocks built from Fed staff documents (Aruoba and Drechsel, 2024b), and repartition the instruments by window and horizon. These changes spread the identifying variation across more families, and in every exercise the impulse responses remain close to the baseline.

The baseline estimate is also robust to the specification of the VAR, with the shadow rate of Wu and Xia (2016) as the policy indicator and with other price measures. The VAR can be extended with additional variables, one at a time. Each larger system adds a family, as the added non-policy shock requires, and the pooled instrument is re-estimated for it. Housing starts, unemployment, stock prices and commodity prices then respond as expected after a monetary tightening (Bernanke and Gertler, 1995; Christiano et al., 1999; Bernanke and Kuttner, 2005; Anzuini et al., 2013). Finally, because the pooled instrument is a single series, it can be reused in other frameworks. In local projections (Jordà, 2005) and as an internal instrument in a VAR (Plagborg-Møller and Wolf, 2021), it gives impulse responses of the same sign as the baseline.

The paper contributes to three strands of the literature. The first contribution is to the literature on identification with many, weak or invalid instruments. Recent work relaxes the assumption of a single strong and valid instrument. It allows the instruments to be weak but keeps them valid (Angelini et al., 2024), bounds their exogeneity (Huang and Neri, 2026), or selects the valid ones as the largest group that agrees on the effect (Windmeijer et al., 2019, 2021; Lin et al., 2024). In this paper, the instruments are not assumed to be valid. The shock is identified from the structure of the

contamination, which is shared within families, rather than from the validity of any instrument. The family assumption parallels the condition under which invalid instruments still identify an effect, when their direct effects are unrelated to their relevance (Kolesár et al., 2015). The method also delivers more than a direction. Constrained pooling turns the panel into the strongest single instrument that is free of the contamination the families share, and the overidentification test makes the family assignment testable. Since the instruments enter a single pooled series, the many-instrument bias of overidentified estimators (Bekker, 1994; Newey and Windmeijer, 2009; Mikusheva and Sun, 2022) does not arise. The closest paper is Francis et al. (2026). They also combine several instruments into a composite, but as part of a general identification scheme, which maximises the correlation between each shock and its instrument and nests recursive identification. This paper instead builds the pooled instrument to cancel the contamination that families of instruments share, and uses it to identify a single shock.

The second contribution is to the literature on high-frequency identification and the predictability of the surprises. Beyond the information and Fed-response views, the predictability has been traced to the Fed's response to financial conditions (Chen, 2026), to shifts in the systematic component of policy (Hack et al., 2024), and to information effects whose importance varies over time (Hoesch et al., 2023). The remedies separate the policy shock from the information it reveals (Jarociński and Karadi, 2020; Miranda-Agrippino and Ricco, 2021) or project the surprise on pre-announcement controls, to remove the Fed's response to news that precedes the announcement (Bauer and Swanson, 2023a,b). The order of the projection and the frequency aggregation also matter (Amodeo, 2026). Bauer and Swanson (2023b) and Gürkaynak et al. (2026) note that the orthogonalisation weakens the instrument, and they treat this as a classical weak-instrument problem, to be met with more monetary events or with robust inference (Montiel Olea and Pflueger, 2013; Andrews et al., 2019; Montiel Olea et al., 2021; Jentsch and Lunsford, 2022). This paper extends the orthogonalisation to the wide-window instruments, built from the whole policy decision. It also shows that the weakness and the bias cannot be treated separately. Once the predictable variation is removed, the contamination that the controls do not span remains, and the weaker first stage magnifies it. The orthogonalised impulse responses can then be more biased than those of the raw surprise, a case of weakness meeting a failure of exogeneity (Bound et al., 1995).

The third contribution is empirical. It provides new estimates of the effects of U.S. monetary policy on output and prices. The literature identifies the policy shock in several ways. Some schemes impose sign restrictions (Uhlig, 2005) or narrative restrictions (Antolín-Díaz and Rubio-Ramírez, 2018). Others restrict the systematic component of policy (Arias et al., 2019) or exploit heteroskedasticity (Rigobon, 2003). This paper instead combines high-frequency, narrative and Cholesky-based instruments into a single estimate, which does not rest on any one of them. With the pooled instrument, prices decline only gradually, in line with evidence that they respond to monetary policy with a lag (Aruoba and Drechsel, 2024a; Buda et al., 2025).

The remainder of the paper is organised as follows. Section 2 shows that orthogonalisation removes the predictability of the high-frequency surprise, and documents the implications for the estimated impulse responses. Section 3 derives why a weaker first stage magnifies the contamination that survives orthogonalisation. Section 4 orthogonalises the wide-window instruments and shows that they keep more of their first-stage strength. Section 5 develops the method, and Section 6 applies it to U.S. monetary policy. Section 7 shows that the baseline estimate is robust to the construction of the families, to the VAR specification and to other frameworks. Section 8 concludes.

2 High-frequency identification and its orthogonalisation

The high-frequency surprise is the most widely used instrument for monetary policy shocks, and the starting point of this paper. It is predictable from information available before the announcement. The standard remedy orthogonalises it on pre-announcement controls. Once orthogonalised, the surprise delivers impulse responses with five features that are hard to reconcile with a valid instrument. The features are consistent with two costs of weaker first-stage strength, imprecision and bias. The bias comes from contamination that survives the orthogonalisation and is magnified by the weaker first stage. Let the central bank's policy rule be

$$i_t = f_t(\Omega_t) + \varepsilon_t^p, \tag{1}$$

with i_t the policy rate, Ω_t the information the central bank responds to and f_t its systematic response. The policy shock ε_t^p is the part of the policy rate that is not a systematic response to economic conditions. This paper identifies it with external instruments (Mertens and Ravn, 2013; Stock and Watson, 2018). An instrument is a series z_t that is correlated with the policy shock and, in the ideal case, with no other structural shock, collected in $\boldsymbol{\varepsilon}_t^{-p}$. The two covariance conditions are summarised as

$$\mathbb{E}[z_t\, \varepsilon_t^p] = \phi \neq 0 \quad \text{(relevance)}, \qquad \mathbb{E}[z_t\, \boldsymbol{\varepsilon}_t^{-p}] = \mathbf{0} \quad \text{(exogeneity)}. \tag{2}$$

Relevance makes the instrument informative about the shock. Exogeneity ensures that what it identifies is the policy shock and nothing else. The relative covariances between the instrument and the VAR residuals identify the impact column b_p, the response of each variable to the policy shock on impact, up to a normalisation.[2]

The high-frequency surprise is the change in short-term interest-rate futures in a narrow window around an FOMC announcement (Kuttner, 2001; Gürkaynak et al., 2005; Gertler and Karadi, 2015). For an announcement m, the surprise is the revision of the expected policy rate between the opening and the close of the window, $mps_m = \mathbb{E}_{m+\Delta}[i_m] - \mathbb{E}_{m-\Delta}[i_m]$, where $\mathbb{E}_{m-\Delta}$ and $\mathbb{E}_{m+\Delta}$ are the expectations priced just before the window opens and just after it closes.[3]

The narrow window isolates the announcement from other news, but it does not make the surprise unpredictable. The surprise still contains part of the Fed's systematic response to news the market had not yet priced. It is therefore predictable from information available before the announcement, and it must be orthogonalised on pre-announcement information.

[2] With u_t the vector of reduced-form VAR innovations and u_t^p the innovation in the policy rate, the identified impact vector, normalised such that the policy rate rises by one unit, is $\beta = b_p / b_{pp} = \mathbb{E}[u_t\, z_t] / \mathbb{E}[u_t^p\, z_t]$. It is the covariance of the instrument with each innovation relative to its covariance with the policy-rate innovation, which is the instrument's first stage. The response at horizon h is $\Phi_h \beta$, with Φ_h the moving-average coefficients of the VAR. Section 6.1 gives the empirical specification.

[3] The Bauer and Swanson (2023b) surprise is the first principal component of the changes in the first four quarterly Eurodollar futures (ED1 to ED4), which price the three-month rate over roughly the next year. i_m therefore stands for the expected path of the policy rate. The surprises are summed within the month, $z_t = \sum_{m \in t} mps_m$, to match the frequency of the VAR (Section 6.1).

2.1 Predictability and orthogonalisation

I borrow the orthogonalisation proposed in Bauer and Swanson (2023b) and apply the same procedure to every instrument in the paper. Every instrument is then orthogonalised on the same pre-announcement information, conditional on the choice of controls.

Two lines of work document the predictability of the surprise. Miranda-Agrippino and Ricco (2021) read it as an information effect, and project the surprise on the Fed's Greenbook forecasts and on its own lags. Bauer and Swanson (2023a,b) read it as the Fed's response to news the market had not yet fully priced, and project the surprise on publicly available data dated before the announcement. In their reading, the predictable part is the Fed's systematic response to that news, which the narrow window does not remove.

The orthogonalisation projects the surprise on a set $\mathbf{W}_m$ of six pre-announcement controls and keeps the residual.[4] Every control is measured before the announcement, and the market-based controls end the day before it. The projection can be applied announcement by announcement and then aggregated to the month (purge-then-aggregate, PTA; Amodeo, 2026), or to the monthly series (aggregate-then-purge, ATP; Bauer and Swanson, 2023b). The paper uses PTA for every instrument observed at the announcement level.[5] For a meeting m the orthogonalisation is

$$mps_m^{\perp} \;=\; mps_m - \text{Proj}\left(mps_m \mid \mathbf{W}_m\right),$$

where $\text{Proj}(y \mid \mathbf{X})$ denotes the linear projection of y on $\mathbf{X}$ and a constant. On the sample of this paper, the six controls jointly predict the surprise with $R^2 = 0.17$ over 1988–2019 (Appendix Table 3, first column).

The orthogonalised surprise is, by construction, orthogonal to the chosen set of controls. Orthogonalisation removes the predictability that these controls span. The contamination that remains comes from news on the announcement day or later in the window, and from predictable news the controls do not span. The orthogonalised surprise is therefore clean only if the contamination lies entirely in the span of the controls (Section 3.2). The controls end the day before the announcement by design. Controls dated on announcement days would also span the policy shock, and the projection would remove it from the instrument.

Orthogonalisation also weakens the first-stage strength. The predictable part of the surprise is the Fed's systematic response to news that the market had not yet priced. The policy rate moves with that response within the month. The predictable part is therefore correlated with the policy-rate innovation, and removing it removes part of the first stage. Richer controls remove more of the predictable variation, and they weaken the first-stage strength further. An instrument keeps its strength only to the extent that its relevance did not come from its predictable part (Section 3.2). The loss depends less on how predictable an instrument is than on how closely its predictable part moves with the policy rate (Section 4.3).

[4] The six controls of Bauer and Swanson (2023b) are the nonfarm-payrolls surprise relative to the survey median, employment growth over the year to the latest release, the S&P 500 over the sixty-five trading days to the day before, the second principal component of the one- to ten-year yield curve over three months, commodity prices over three months, and the skewness implied by ten-year Treasury options over the prior month.

[5] Amodeo (2026) shows that orthogonalising monthly surprises (ATP) can bias the orthogonalised surprise in months with more than one announcement. Under ATP, the monthly sum is projected on controls dated before the first announcement of the month. The news that arrives between the two announcements is then left in the second surprise. Under PTA, each announcement is projected on its own controls.

2.2 What the orthogonalised surprise delivers

Orthogonalisation aims at a plausibly exogenous instrument, but it has side effects. Figure 1 shows the impulse responses identified with the raw Bauer and Swanson (2023b) surprise and with its orthogonalised version (PTA) as external instruments, each with its 68% confidence band, in a monthly VAR in industrial production, consumer prices, the excess bond premium (Gilchrist and Zakrajšek, 2012) and the federal funds rate (Section 6.1). Five empirical features stand out.

(i) The orthogonalised instrument is weaker. Orthogonalisation weakens the first-stage strength of the surprise. Its first-stage F against the funds rate falls from 14.5 to 3.6. The part of the surprise that the controls predict is correlated with the policy-rate innovation. Most of the controls are financial, and the predictable part reflects the Fed's response to financial conditions that the market had not fully priced. Markets infer the Fed's reaction function mostly from macroeconomic news and underprice its response to financial conditions. Removing the predictable part removes first-stage strength along with it.

(ii) The confidence bands are wider. The weaker instrument identifies the impulse responses less precisely. At the peak decline in output, the 68% confidence band is four times as wide under the orthogonalised surprise as under the raw one.

(iii) The impulse responses are larger. The peak decline in output is 1.9% under the raw surprise and 6.6% under the orthogonalised one, and the credit spread rises on impact by 0.25 and 2.0 points. The true magnitudes are unknown, but these are large. On the same reduced form, they lie outside the range of five conventional identification schemes and of other high-frequency instruments (Online Appendix Figures OA.8 and OA.9).

(iv) Prices fall on impact. In the impulse responses identified with the orthogonalised surprise, the price level falls by 0.8% in the month of the announcement. A tightening usually reaches prices only gradually. It weakens aggregate demand, and firms' prices and workers' wages adjust to the resulting slack over time. An immediate fall in prices also sits uneasily with recent evidence on the timing of transmission, in which the aggregate price level takes several months at the earliest to respond (Aruoba and Drechsel, 2024a) and consumer prices decline only gradually even as spending reacts within weeks (Buda et al., 2025).

(v) The funds rate reverses. After the normalised one-point increase on impact, the funds-rate impulse response identified with the orthogonalised surprise turns negative from the sixth month. It reaches -1.5 points after seventeen months, one and a half times the initial tightening. Under the raw surprise, the funds-rate impulse response remains above zero throughout the response horizon.

The orthogonalisation was meant to make the instrument credible. It also made the instrument weaker, and the impulse responses it identifies became larger as its first stage weakened. A weaker first stage has two costs. The first is imprecision, the wider confidence bands of feature (ii). The second is bias. A weaker first stage magnifies any contamination that survives the orthogonalisation, and with it the bias of the identified impulse responses at every horizon (Section 3). Features (iii) to (v) are consistent with that bias. Section 6.2 revisits the five features under the pooled instrument.

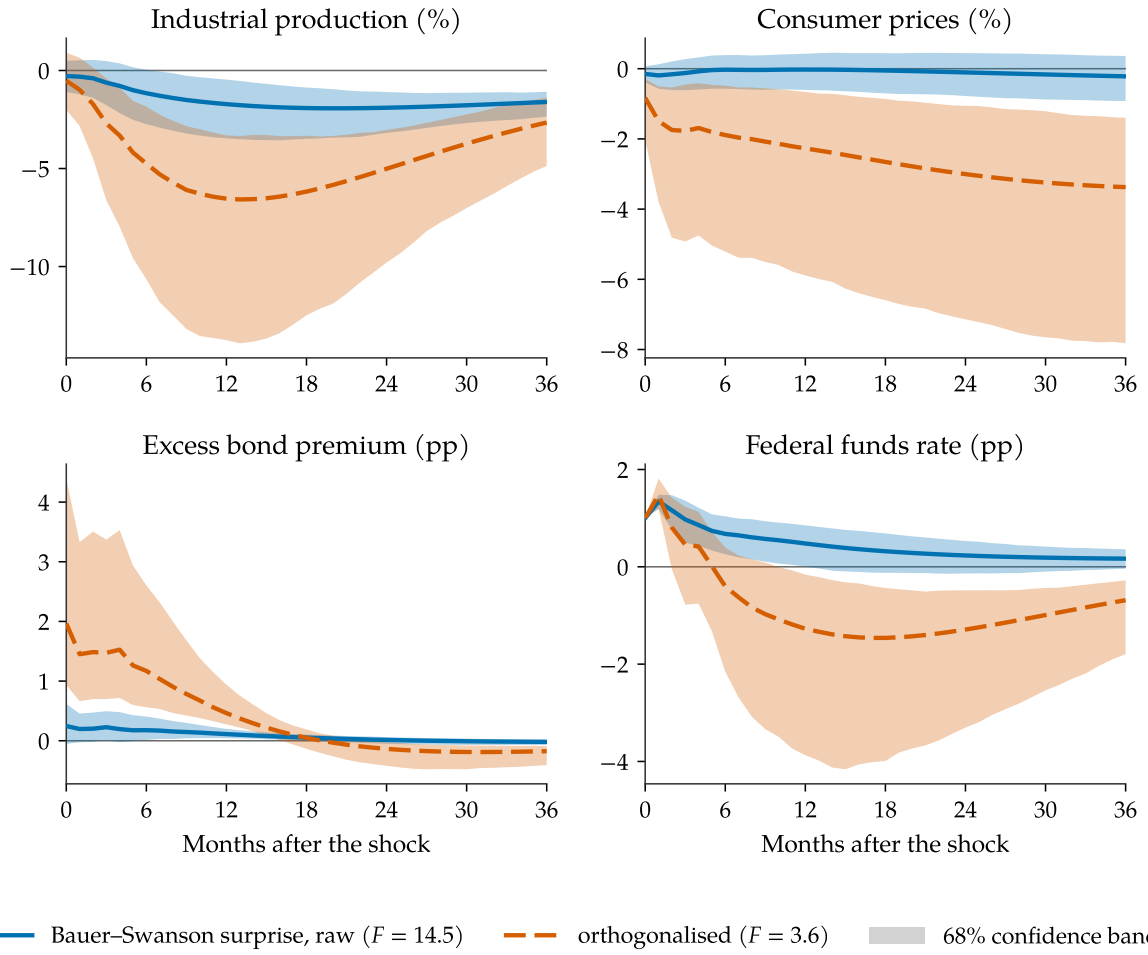


Figure 1: **Orthogonalising the surprise weakens it and enlarges its impulse responses.** Single-instrument impulse responses to a +1 percentage-point funds-rate shock in the monthly VAR of Section 6.1, identified with the Bauer and Swanson (2023b) surprise, raw (solid, $F = 14.5$) and orthogonalised announcement by announcement and then aggregated (PTA; dashed, $F = 3.6$), each with its 68% moving-block-bootstrap confidence band.

3 The contamination–strength trade-off

An instrument that carries residual contamination biases the impulse responses it identifies. A relevant instrument attenuates that bias, and a weak first stage magnifies it. The first stage is the instrument's covariance with the policy-rate innovation. Contamination and strength must therefore be judged together. An instrument with less residual contamination can still be more biased than a more contaminated one if it is much weaker. This is the contamination–strength trade-off.

The trade-off holds for any instrument, and it matters most for orthogonalisation. Orthogonalising an instrument on pre-announcement controls removes predictable variation and reduces its contamination. It can also weaken its first-stage strength considerably. The bias in the impulse

responses can then grow rather than shrink, the opposite of what the orthogonalisation intends. The section derives the bias, shows what orthogonalisation removes and how the bias that survives is magnified, and explains why weak-instrument-robust inference does not correct it.

3.1 Identification with a contaminated instrument

A single instrument may load on the non-policy shocks as well as on the policy shock. With it, the proxy-SVAR identifies the ratio of the instrument's covariance with each VAR innovation to its covariance with the policy-rate innovation. The bias follows from that ratio.

Let $y_t \in \mathbb{R}^n$ follow a stationary VAR with s lags, $y_t = c + \sum_{\ell=1}^{s} \Pi_\ell y_{t-\ell} + u_t$, with moving-average coefficients Φ_h ($\Phi_0 = I_n$). The response at horizon h to an impact vector b is $\Phi_h b$. The innovations are linked to orthogonal, unit-variance structural shocks by

$$u_t = B\boldsymbol{\varepsilon}_t, \qquad \boldsymbol{\varepsilon}_t = \begin{pmatrix} \varepsilon_t^p \\ \boldsymbol{\varepsilon}_t^{-p} \end{pmatrix}, \qquad \mathbb{E}\left[\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}_t'\right] = I_n, \qquad B = \left[\, b_p \;\; B_{-p} \,\right], \tag{3}$$

where ε_t^p is the policy shock, with impact vector b_p, and $\boldsymbol{\varepsilon}_t^{-p}$ the non-policy shocks, with impact matrix B_{-p}. With s_p the vector that selects the policy rate, the policy-rate innovation is

$$u_t^p := s_p' u_t = b_{pp}\, \varepsilon_t^p + b_{p,-p}' \boldsymbol{\varepsilon}_t^{-p}, \tag{4}$$

where $b_{pp} := s_p' b_p > 0$ is the policy rate's response to its own shock and $b_{p,-p}' := s_p' B_{-p}$, the policy-rate row of B_{-p}, is its within-period response to the non-policy shocks. The impact of a policy shock that raises the policy rate by one unit is $\beta := b_p / b_{pp}$. Let a potentially contaminated instrument z_t be

$$z_t = \phi\, \varepsilon_t^p + \boldsymbol{\psi}' \boldsymbol{\varepsilon}_t^{-p} + \omega_t, \qquad \mathbb{E}[\boldsymbol{\varepsilon}_t\, \omega_t] = \mathbf{0}, \tag{5}$$

with relevance $\phi > 0$, *contamination loading* $\boldsymbol{\psi} \in \mathbb{R}^{n-1}$ on the non-policy shocks, and a remainder ω_t uncorrelated with the period's shocks. The identifying moment is $\mathbb{E}[u_t z_t] = \phi\, b_p + B_{-p}\boldsymbol{\psi}$. The first stage and the proxy-SVAR estimand are

$$\pi := s_p'\, \mathbb{E}[u_t z_t] = \phi\, b_{pp} + b_{p,-p}' \boldsymbol{\psi}, \qquad \tilde{\beta} := \frac{\mathbb{E}[u_t z_t]}{s_p'\, \mathbb{E}[u_t z_t]}. \tag{6}$$

The first stage has two sources. One is the policy shock the instrument contains, ϕb_{pp}. The other is the non-policy news it carries, $b_{p,-p}'\boldsymbol{\psi}$, which also moves the policy rate within the period. The proxy-SVAR estimator replaces the covariances in $\tilde{\beta}$ with their sample counterparts, and converges in probability to $\tilde{\beta}$ as the sample grows. Its bias is the gap between $\tilde{\beta}$ and β, which is (Appendix A)

$$\tilde{\beta} = \beta + \frac{d}{\pi}, \qquad d := \underbrace{B_{-p}\boldsymbol{\psi}}_{\text{contamination}} - \underbrace{\beta\, b_{p,-p}'\boldsymbol{\psi}}_{\text{induced policy response}}, \qquad s_p' d = 0. \tag{7}$$

A contaminated instrument therefore identifies the true impact vector plus a bias in the direction of d. At horizon h, the bias of the impulse response is $\Phi_h d / \pi$. The contamination term in equation (7) is the impact of the non-policy shocks on which the instrument loads. These shocks also move the

policy rate within the period, and the proxy-SVAR attributes that movement to the policy shock. The induced policy response in equation (7) is the impact of a policy shock that moves the policy rate by the same amount. Hence, the contamination direction d is the contamination net of this induced policy response. Since the impact response of the policy rate is normalised, the bias falls entirely on the variables other than the policy rate ($s_p' d = 0$).

The first stage scales the bias, and a weaker first stage magnifies it. Neither contamination nor strength is therefore enough on its own to judge an instrument. Take two instruments for the same policy shock, $z_t^{(1)}$ and $z_t^{(2)}$, the first less contaminated, $\|d^{(1)}\| < \|d^{(2)}\|$. The first is still the more biased if its first stage is weaker by a larger factor, $\pi^{(1)}/\pi^{(2)} < \|d^{(1)}\|/\|d^{(2)}\|$. The contamination is not observed, and the comparison cannot be made directly in the data. It shows that a cleaner instrument is not necessarily a better one. A cleaner but weaker instrument is often taken to cost only precision. With any residual contamination, it also costs bias. A small reduction in $\boldsymbol{\psi}$ bought with a large reduction in π moves the estimate further from the truth.

Orthogonalisation aims to remove the predictable variation in an instrument, the empirical footprint of its contamination. It also affects its first-stage strength π. Section 3.3 shows that it reduces the bias only if it removes a larger share of the contamination than of the first-stage strength.

3.2 What orthogonalisation removes

Orthogonalisation removes the part of the instrument that the pre-announcement controls predict, the predictability that these controls span. The contamination that remains comes from news on the announcement day or later in the period, and from predictable news the controls do not span. Part of the predictable variation is contamination that moves the policy rate. Removing it reduces the contamination and weakens the first stage at the same time. Whether the bias falls depends on which of the two falls by the larger share.

The controls $\mathbf{W}_t$ are assumed orthogonal to the policy shock. The assumption rests on their construction. They are dated before the announcement, and hence do not respond to the policy shock it reveals. Their variation has two sources. Part of it is news from the period before the announcement, which reflects the period's non-policy shocks. The rest is news from earlier periods, because the controls are measured over windows that reach back before the period. Only the first part can change the moments of the instrument. News from earlier periods is uncorrelated with the period's VAR innovations, from which the lags have been removed. Controls that carried no news from the period would leave the moments unchanged.

The orthogonalisation is the population regression of z_t on $\mathbf{W}_t$. It splits the instrument into the fitted value and the residual, $z_t = \hat{z}_t + z_t^{\perp}$, with $\hat{z}_t = \text{Proj}(z_t \mid \mathbf{W}_t)$ the predictable part and $z_t^{\perp} = z_t - \text{Proj}(z_t \mid \mathbf{W}_t)$ the orthogonalised instrument. Because the controls are orthogonal to the policy shock, the fitted value contains none of it, and the orthogonalised instrument keeps the relevance ϕ of the raw instrument. Orthogonalisation changes only the contamination loading, which becomes $\boldsymbol{\psi}^{\perp}$. The first stage, $\pi^{\perp} := \text{Cov}(z_t^{\perp}, u_t^p)$, and the contamination direction keep the form of equation (7), with $\boldsymbol{\psi}^{\perp}$ in place of $\boldsymbol{\psi}$,

$$\pi^{\perp} = \phi\, b_{pp} + b_{p,-p}'\boldsymbol{\psi}^{\perp}, \qquad d^{\perp} = B_{-p}\boldsymbol{\psi}^{\perp} - \beta\, b_{p,-p}'\boldsymbol{\psi}^{\perp}, \qquad \tilde{\beta}^{\perp} = \beta + \frac{d^{\perp}}{\pi^{\perp}}.$$

The first stage keeps the same two parts as the raw first stage. The policy shock the instrument contains, ϕb_{pp}, is unchanged. Any change in first-stage strength therefore comes from the contamination that moves the policy rate, $b'_{p,-p}\boldsymbol{\psi}^{\perp}$.

Proposition 1 quantifies these changes and answers three questions. How much first-stage strength does orthogonalisation remove? How much of the contamination direction does it remove? And which of the two can be measured in the data?

Proposition 1 (What orthogonalisation removes)**.** *With the controls orthogonal to the policy shock:*

(i) *The share of the raw first-stage strength lost to orthogonalisation is*

$$\delta_{\pi} := 1 - \frac{\pi^{\perp}}{\pi} = \frac{b'_{p,-p}(\boldsymbol{\psi} - \boldsymbol{\psi}^{\perp})}{\pi} = \frac{\mathrm{Cov}(\hat{z}_t, u_t^p)}{\mathrm{Cov}(z_t, u_t^p)}. \tag{8}$$

It depends only on the contamination removed, and it is observable.

(ii) *The share of the contamination direction that orthogonalisation removes is*

$$\delta_d := 1 - \frac{\|d^{\perp}\|}{\|d\|} \leq 1. \tag{9}$$

Unlike δ_{π}, it is not observable. It is negative when the contamination direction is longer after orthogonalisation than before. This happens when orthogonalisation adds contamination (Lemma A.1), or when the contamination it removes was partly cancelling the effect of the contamination that remains.

Details are in Appendix A. Part (i) gives the cost of orthogonalisation. It weakens the first-stage strength only by removing contamination that moves the policy rate. The loss is positive when the policy rate and the instrument move in the same direction with the news the controls span. This is the expected case under the "Fed responds to news" reading (Bauer and Swanson, 2023a). The share lost, δ_{π}, is the share of the raw first stage carried by the part of the instrument the controls predict. It can be computed from the data. Since the predictable part carries no policy shock, δ_{π} is also the share of the raw first stage that came from the contamination orthogonalisation removes. Section 4.3 shows why equally predictable instruments lose very different shares of their first-stage strength.

Part (ii) gives the benefit. The bias depends on the contamination direction $d^{\perp}$, and computing δ_d requires the loadings $\boldsymbol{\psi}^{\perp}$, which are not observed. The cost of orthogonalisation is therefore measurable, and its benefit is not. Next, I show what this asymmetry does to the bias.

3.3 The magnification of the surviving bias

Proposition 1 gives two shares, the share of the contamination direction that orthogonalisation removes, δ_d, and the share of first-stage strength it removes, δ_{π}. Together they give the factor by which orthogonalisation scales the bias.

Proposition 2 (Magnification)**.** *Let the instrument be contaminated, $\boldsymbol{\psi} \neq \mathbf{0}$, with positive first stages before and after orthogonalisation, $\pi, \pi^{\perp} > 0$. The factor by which orthogonalisation scales the bias is*

$$\varrho := \frac{\|\tilde{\beta}^{\perp} - \beta\|}{\|\tilde{\beta} - \beta\|} = \frac{\|d^{\perp}\| / \|d\|}{\pi^{\perp} / \pi} = \frac{1 - \delta_d}{1 - \delta_{\pi}}, \tag{10}$$

and the denominator is observable by equation (8)*. Since* $\pi, \pi^{\perp} > 0$, $\delta_\pi < 1$, *and* δ_π *can have either sign. Moreover,*

$$\varrho > 1 \iff \delta_d < \delta_\pi, \qquad \varrho \leq \frac{1}{1-\delta_\pi} \text{ whenever } \delta_d \geq 0, \tag{11}$$

with equality in the bound if and only if $\delta_d = 0$*. If* $\delta_\pi \leq 0$*, then* $\varrho \leq 1 - \delta_d$*.*

Details are in Appendix A. Equation (10) has two factors. The first factor, $1/(1-\delta_\pi)$, is observable and multiplies the bias that survives orthogonalisation. It is the magnification when orthogonalisation removes none of the contamination direction, $\delta_d = 0$, and the largest magnification unless orthogonalisation lengthens the contamination direction, $\delta_d < 0$. The second factor, $1 - \delta_d$, is the share of the contamination direction that survives, and it depends on which non-policy news orthogonalisation removes. Orthogonalisation magnifies the bias if and only if it removes a smaller share of the contamination direction than of the first-stage strength. Removing contamination is therefore not enough. Magnification requires the controls to predict disproportionately the contamination that moves the policy rate. Section 4.3 finds that the predictable part of the high-frequency surprise moves with the policy rate more closely than the surprise itself. When orthogonalisation does not weaken the first-stage strength, $\delta_\pi \leq 0$, it never multiplies the bias that survives. It can then raise the bias only by adding contamination, $\delta_d < 0$.

An orthogonalised response larger than the raw one admits two readings. In the reading of Bauer and Swanson (2023b), orthogonalisation removes contamination that attenuated the raw estimate, $\delta_d > \delta_\pi$, and the orthogonalised estimate is closer to β. In the reading developed here, the contamination that survives is magnified by the weaker first stage. Two implications tell the readings apart. Under the second, the enlargement should be ordered by δ_π across instruments orthogonalised on the same controls, and an instrument that addresses the same contamination without weakening its first-stage strength should not enlarge the impulse responses. Section 4.4 examines both, using an identity that holds under both readings. A larger orthogonalised response is therefore not necessarily a better estimate of β. It may be residual contamination magnified by a weaker first stage.

Figure 2 illustrates the magnification, with parameter values chosen for illustration, not estimated. Panel A plots ϱ against δ_π, one curve for each value of δ_d. The dotted line $\varrho = 1$ separates the two outcomes. Above it, in the shaded region, orthogonalisation magnifies the bias. Below it, orthogonalisation reduces the bias. Read along one curve, the share of the contamination direction removed is fixed and the loss of strength rises. The curve crosses the dotted line where the two shares are equal, $\delta_\pi = \delta_d$, marked by a circle. To the left of that point, orthogonalisation removes a larger share of the contamination than of the strength, and the bias falls. To the right, it removes a larger share of the strength, and the bias is magnified. The curve $\delta_d = 0$ crosses at $\delta_\pi = 0$. When orthogonalisation removes none of the contamination direction, any loss of strength magnifies the bias, by the bound $1/(1-\delta_\pi)$. This bound rises slowly at first and then steeply. A larger δ_d scales the curve down. The same loss of strength then magnifies the bias less. For the orthogonalised surprise, $\delta_\pi \approx 0.54$ (Section 4.3), and any $\delta_d < 0.54$ magnifies the bias.

Panel B shows what this does to the impulse response of a non-policy variable, in a stylised two-variable VAR. The raw instrument overstates the decline of the variable. Orthogonalisation removes a quarter of the contamination but half of the first-stage strength, $\delta_d = 0.25$ and $\delta_\pi = 0.5$, and hence $\varrho = 1.5$. The orthogonalised impulse response lies further from the truth than the raw one and shows a larger decline.

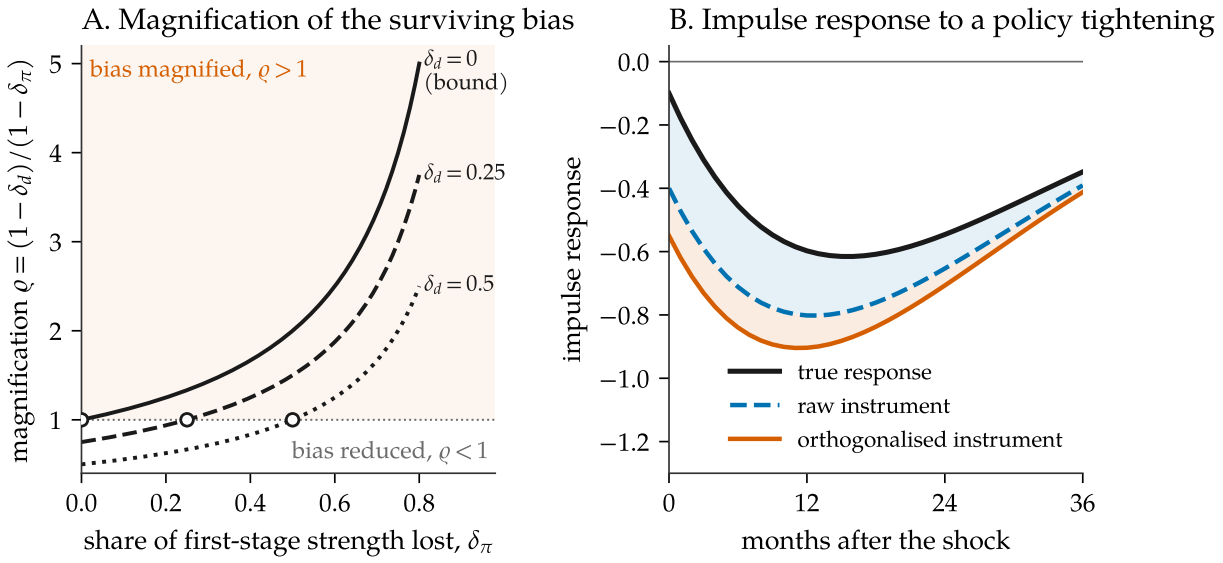


Figure 2: **The magnification of the surviving bias.** Panel A: the factor $\varrho = (1-\delta_d)/(1-\delta_\pi)$ of equation (10) by which orthogonalisation scales the bias, against the share of first-stage strength it removes, δ_π. Each curve holds the share of the contamination direction removed, δ_d, fixed, and crosses the dotted line $\varrho = 1$ at $\delta_\pi = \delta_d$ (circles). The curve $\delta_d = 0$ is the bound in equation (11). In the shaded region, $\varrho > 1$, orthogonalisation magnifies the bias. At a given δ_π, points between the curve $\delta_d = 0$ and the dotted line have $0 < \delta_d < \delta_\pi$, and points above the curve $\delta_d = 0$ have $\delta_d < 0$. Panel B: an illustrative example. Population impulse responses of a non-policy variable to a policy shock that raises the policy rate by one unit, in a bivariate VAR(1) in that variable and the policy rate. True response $\Phi_h\beta$; raw instrument $\Phi_h(\beta + d/\pi)$; orthogonalised instrument $\Phi_h(\beta + d^\perp/\pi^\perp)$, with $\delta_\pi = 0.5$, $\delta_d = 0.25$ and hence $\varrho = 1.5$. The contamination direction satisfies $s_p'd = 0$ and loads on the non-policy variable.

Orthogonalisation leaves the surprise weak, and the standard response to a weak first stage is weak-instrument-robust inference.[6] An Anderson–Rubin-type confidence set has correct coverage for any instrument strength (Montiel Olea et al., 2021; Jentsch and Lunsford, 2022). It addresses weakness, not contamination (Andrews et al., 2019). Built on $z_t^\perp$, it covers the estimand of the contaminated instrument, $\beta + d^\perp/\pi^\perp$. It is therefore centred on the wrong value if some residual contamination survives, and it inherits the magnification of Proposition 2. Wider confidence bands repair neither the bias nor its magnification. Only an instrument that keeps its first-stage strength under the same orthogonalisation can reduce them. Section 4 turns to such instruments.

4 Widening the window

Neither version of the high-frequency surprise is satisfactory. Raw, it is predictable and contaminated. Orthogonalised, it becomes weak, and any contamination that survives is magnified by the weaker first stage. An ideal instrument would be both unpredictable from pre-announcement information and strongly correlated with the innovation in the policy rate.

[6] Weakness is a bias problem in its own right. Under weak identification, the IV estimate is biased toward its least-squares counterpart, which in a proxy-SVAR is the recursive response obtained by ordering the policy rate first in a Cholesky decomposition of the residual covariance matrix (Stock and Watson, 2018).

The narrow window also limits how much of the policy shock the surprise contains. The surprise measures only the part of the decision revealed inside the window. By the rule in equation (1), the expectation at the opening of the window is $\mathbb{E}_{m-\Delta}[i_m] = \mathbb{E}_{m-\Delta}[f_m(\Omega_m)] + \mathbb{E}_{m-\Delta}[\varepsilon_m^p]$, and its second term may not be zero. On the eve of a meeting, and on the morning of the announcement, press reports and informal signals from policymakers can reveal part of the decision itself (Lucca and Moench, 2015; Cieslak et al., 2019). When the decision deviates from the rule, what is revealed in this way is part of the policy shock. It is not a response to the economy, and it moves the policy rate in the month of the decision. Because the market prices it before the window opens, it is already in the expectation at the opening of the window, and the surprise misses it. An instrument built on the decision itself keeps it, whenever the market learned it. If a share a of the shock's variance is revealed before the window opens, the surprise's relevance is only $1 - a$ times that of the decision. Widening the window therefore adds policy shock, not only contamination.

The window can also close too early. Markets take at least forty minutes to price FOMC statements, and up to an hour at long maturities. The conventional thirty-minute window can therefore miss part of the reaction, and the length of the window changes what the surprises capture (Tran, 2026).

Another remedy is to add events rather than widen the window. Bauer and Swanson (2023b); Swanson and Jayawickrema (2023) add Fed Chair speeches, press conferences and congressional testimony to the FOMC announcements, which raises the first-stage strength of the surprise. The added strength may come from contamination rather than from the policy shock. Bauer and Swanson (2023b) note that speeches and testimonies can carry larger information effects. A speech mostly conveys the Fed's view of the outlook and how it intends to respond to it. In terms of the rule in equation (1), this is news about the information the Fed responds to or about its systematic response, not a policy shock. The longer windows of these events can also let other news in.

Widening the window from the announcement to the decision addresses both shortfalls of the narrow window. The two instruments proposed below are reaction-function residuals, each the change in the policy rate that the policymaker's information set does not explain. Neither construction removes all of the pre-announcement information that predicts the high-frequency surprise. Both remain predictable from the controls of Bauer and Swanson (2023b) (Appendix Table 3). For consistency, they are orthogonalised on the same pre-announcement controls as the surprise.

Orthogonalisation deals with only one source of contamination, and widening the window adds a second. Figure 3 separates the two. The first is predictability. The controls are dated before each announcement and capture news the market had not yet priced. Orthogonalisation removes the part of that news the controls span. The second is news that arrives on the day of the decision or later in the window, such as data releases or financial developments. Orthogonalisation cannot remove it. Controls dated on the day of the announcement would also span the policy shock, and would remove it along with the contamination. Exposure to this news increases with the length of the window. It is limited for the high-frequency surprise and larger for the two wider-window instruments.

No instrument is therefore clean after orthogonalisation. The news that survives is not specific to each instrument, though. Instruments built from the same source over the same window share it. Section 5 groups such instruments into families and removes what they share, the part of the contamination that orthogonalisation cannot reach. The first instrument widens the window to the whole decision, with the narrative approach of Romer and Romer (2004). The second widens it to

the whole month, with the VAR's own policy-rate innovation.

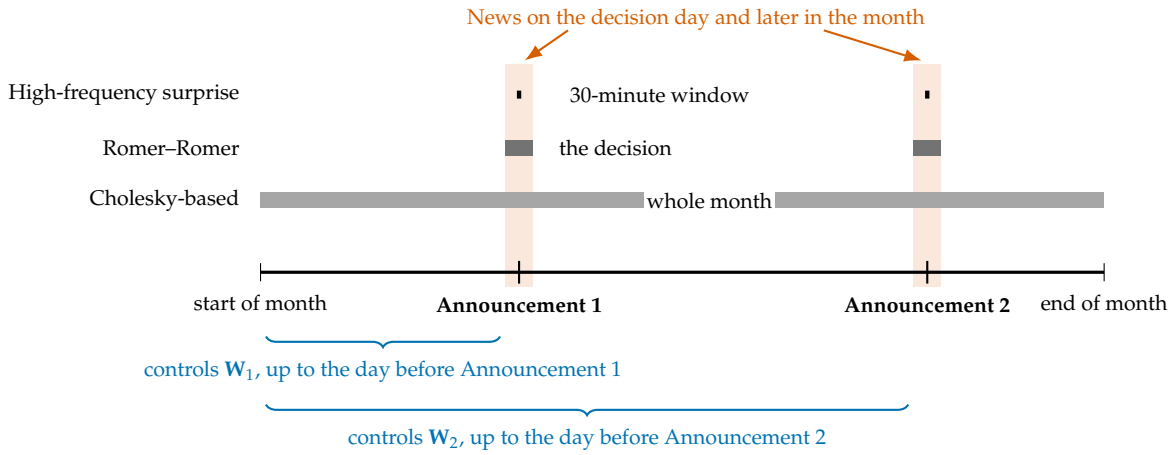


Figure 3: **Two sources of contamination within a month.** A month with two monetary policy announcements. Grey bars: the window over which each instrument measures the decision. The Romer–Romer instrument is the decision itself, net of the Fed's response to its Greenbook forecasts, whenever the market learned it. Blue braces: the pre-announcement controls used to orthogonalise each announcement (PTA), which end the day before it. Shaded bands: the day of each decision. Orange: news on the decision day and later in the month, inside the window of the wider instruments and outside every control set. Orthogonalisation removes the predictable part of the instrument, the news in the controls. It cannot remove news that arrives on the decision day or later in the window, which the wider instruments admit; the within transform of Section 5 removes the part that a family of instruments shares.

4.1 From the announcement to the decision: the Romer–Romer instrument

To widen the window to the whole monetary policy decision, I revisit the narrative instrument of Romer and Romer (2004). Measuring the whole decision rather than the announcement window aims to capture the part of the policy shock that the high-frequency surprise misses. The Romer–Romer construction suits that aim for two reasons. Because its left-hand side is the decision itself, it keeps the part of the shock revealed before the announcement. Because it removes the Fed's systematic response with the Fed's own forecasts, less of its relevance should rest on what the pre-announcement controls predict.

The Romer–Romer instrument, Z_m^{RR}, is the portion of the FOMC's intended change in the funds-rate target at meeting m, Δi_m, that the Fed's own information set does not explain. The information set contains the pre-meeting target level i_{m^-} and the real-time Greenbook forecasts $\mathbf{G}_m$. Romer and Romer (2004) give the details of the construction, and the series is extended here to 2019. The orthogonalised instrument controls for the same information set and, jointly, for the pre-announcement controls $\mathbf{W}_m$ of Bauer and Swanson (2023b),

$$Z_m^{\mathrm{RR},\perp} = \Delta i_m - \mathrm{Proj}\left(\Delta i_m \mid i_{m^-}, \mathbf{G}_m, \mathbf{W}_m\right). \tag{12}$$

The projection can be computed jointly or in two steps.[7] The Romer–Romer instrument is pre-

[7] By the Frisch–Waugh–Lovell theorem, the joint projection is a two-step residual, namely the Romer–Romer instrument

dictable from the pre-announcement controls ($R^2 = 0.11$, Appendix Table 3), mostly through the yield-curve slope and Treasury skewness. The predictability has two potential sources.

The first is *timing*. The Greenbook closes about six days before the meeting. Information that arrives in between, such as an employment report or a move in financial markets, can influence the Committee's decision but does *not* enter the forecasts $\mathbf{G}_m$. If the Committee reacts to it at the meeting, the residual Z_m^{RR} still carries a systematic response, visible as predictability from the controls $\mathbf{W}_m$.

The second is that the two sets cover different information. The Greenbook forecasts output, inflation and unemployment, while four of the six controls are market prices, namely the S&P 500, the slope of the yield curve, commodity prices and Treasury skewness. The controls therefore load more on financial conditions, and they span different structural shocks from the Greenbook forecasts. The two information sets overlap, but neither contains the other.

The joint orthogonalisation in equation (12) removes both the systematic response that the Greenbook spans and the response to news that arrives after the Greenbook closes. After orthogonalisation, the instrument keeps most of its first-stage strength. Its first-stage F falls from 37.3 to 29.6 (Section 4.3). It is not clean. News that arrives after the controls are measured survives, as does any response that neither information set captures. This contamination differs from that of the high-frequency surprise, because the instrument is built from different information over a different window.

4.2 To the whole month: the Cholesky-based instrument

The Romer–Romer instrument widens the window from the announcement to the decision. The Cholesky-based instrument widens it further, to the whole month. It removes the central bank's systematic response within the VAR, rather than with the Fed's forecasts. Its purpose is to add first-stage strength to the identification. The cost is a window that no longer surrounds the announcement. The instrument admits contamination from news later in the month, which differs from the contamination of the high-frequency surprise and of the Romer–Romer instrument. Like the other instruments, it is orthogonalised on the same controls, such that it does not carry the response to pre-announcement information.

The construction has two steps. The first removes what past data predict. The VAR is estimated and its policy-rate residual is kept, the innovation u_t^p, the part of the month's policy rate not predicted by the lags of the system. It contains the whole decision, and every other shock that moves the policy rate within the month.

The second step removes the Fed's systematic reaction within the month. The policy rate may respond to output and prices in the same month. The instrument keeps only the variation in the policy rate beyond that reaction.

Order the VAR as output (o), prices (c, for the CPI), the policy rate (p) and the excess bond premium (f), with innovations u_t^i and impact responses b_{ij} of variable i to shock j, and let ε_t^f be the financial shock.

The timing restrictions of the recursive identification (Sims, 1980; Bernanke and Blinder, 1992; Christiano et al., 1999) can be written as an instrument. It is the policy-rate innovation residualised

orthogonalised on the part of the controls that its own information set does not already explain, $Z_m^{\mathrm{RR},\perp} = Z_m^{\mathrm{RR}} - \mathrm{Proj}(Z_m^{\mathrm{RR}} \mid \tilde{\mathbf{W}}_m)$ with $\tilde{\mathbf{W}}_m = \mathbf{W}_m - \mathrm{Proj}(\mathbf{W}_m \mid i_{m^-}, \mathbf{G}_m)$. The same holds for the Cholesky-based instrument of Section 4.2, whose own information set is the month's output and price innovations, (u_t^o, u_t^c).

on the contemporaneous output and price innovations,

$$Z_t^{\text{CHOL}} = u_t^p - \text{Proj}\left(u_t^p \mid u_t^o, u_t^c\right) = b_{pp}\varepsilon_t^p + b_{pf}\varepsilon_t^f, \tag{13}$$

with the effective federal funds rate as the policy variable.

Two restrictions of the recursive ordering (o, c, p, f) give the second equality in equation (13). First, the policy shock does not move output and prices within the month, $b_{op} = b_{cp} = 0$. The innovations u_t^o and u_t^c are then orthogonal to ε_t^p, and residualising on them removes none of the policy shock. Second, the financial shock does not move output and prices within the month either, $b_{of} = b_{cf} = 0$. The projection then removes exactly the output and price shocks. The recursive ordering would also restrict the policy rate not to respond to the financial shock within the month, $b_{pf} = 0$. This restriction is relaxed. The Fed can respond to financial news within the month, as the predictability of the surprises from financial information suggests (Bauer and Swanson, 2023a,b). The instrument therefore carries the Fed's systematic response to the financial shock, $b_{pf}\varepsilon_t^f$. By construction, it is uncorrelated with the month's output and price innovations. Used alone, it identifies no response of output or prices on impact. This is the recursive restriction, and it is valid if the policy shock does not move them within the month.

The Cholesky-based instrument remains predictable from the controls ($R^2 = 0.06$, $p = 0.017$, Appendix Table 3), through the payroll surprise, the slope of the yield curve and Treasury skewness. One source is the financial term in equation (13). If the Fed responds to financial conditions within the month, $b_{pf} \neq 0$, the timing restrictions leave that systematic response in the instrument. Most of the controls are financial market variables, and they predict this response. The instrument is therefore orthogonalised on the same controls,

$$Z_t^{\text{CHOL},\perp} = u_t^p - \text{Proj}\left(u_t^p \mid u_t^o, u_t^c, \mathbf{W}_t\right),$$

computed jointly or in two steps (footnote 7). The instrument is built from monthly VAR innovations. Its orthogonalisation therefore uses monthly controls, dated before the month's announcement, and aggregates before orthogonalising (ATP). It removes the part of $b_{pf}\varepsilon_t^f$ that the controls span, the financial exposure the ordering was meant to exclude. The part the controls do not span remains in the instrument.

The VAR innovations cover the whole month, including the days after the decision, and both projections inherit that window. The recursive restriction then requires that the policy shock not move output and prices before the end of the month. Such a monthly transmission lag is consistent with the gradual response of prices (Aruoba and Drechsel, 2024a). The controls, by contrast, end the day before the meeting. Variation in u_t^p dated after the announcement lies outside their span, and orthogonalisation cannot remove it. This variation contains the rest of the policy shock, which the instrument should keep. It also contains any other news that moves the policy rate later in the month. This news is the cost of the wider window, a contamination that the narrower instruments exclude.

Two further features of the orthogonalised instrument matter. First, $Z_t^{\text{CHOL},\perp}$ is defined only on 1988–2019, where the controls exist. On this subsample, the recursive zero restrictions no longer hold exactly, and the projection on the controls moves the instrument further away from them. The instrument therefore no longer forces the impact responses of output and prices to zero. In the estimates, they are small. Second, the monthly controls exist only in months with an FOMC

announcement in the Bauer–Swanson sample. $Z_t^{\mathrm{CHOL},\perp}$ is therefore zero in the other months, which carry the largest raw innovations of the window (Figure 10).

Both wider-window instruments keep the whole decision, at the cost of more contamination from news within their window. In exchange, they keep much of their first-stage strength once orthogonalised.

4.3 First-stage strength after orthogonalisation

Two questions follow. Do the wider-window instruments keep their first-stage strength after the orthogonalisation that removes predictability, where the high-frequency surprise does not? And why do the three instruments lose such different shares of it? All versions of an instrument are on the same months, and the raw wider-window instruments are re-estimated on the window of their orthogonalised versions, such that each comparison isolates the controls.

The pattern is the one Section 3 anticipates. Orthogonalisation weakens every instrument, but by very different amounts. The high-frequency surprise, reasonably strong when raw ($F = 14.5$), falls to $F = 3.6$. The Romer–Romer instrument falls from $F = 37.3$ to 29.6, and the Cholesky-based instrument from $F = 966$ to 534. Both wider-window instruments survive the same orthogonalisation that collapses the high-frequency surprise. Table 1 reports the results. The first-stage F varies with the policy variable and the way of orthogonalising, but the ordering of the instruments and the collapse of the orthogonalised surprise do not (Appendix Table 4).

All three instruments are predictable from the same controls. Predictability alone therefore does not explain the difference. By Proposition 1(i), the share of first-stage strength that the orthogonalisation removes, δ_π, is observable. Appendix B shows that it is the product of two observable numbers,

$$\delta_\pi = \sqrt{R^2}\,\frac{\hat{\rho}}{\rho}, \qquad \hat{\rho} := \mathrm{Corr}(\hat{z}_t, u_t^p), \quad \rho := \mathrm{Corr}(z_t, u_t^p), \tag{14}$$

where R^2 is that of the orthogonalisation regression and $\hat{z}_t$ its fitted value. The first factor is how predictable the instrument is. The second, $\hat{\rho}/\rho$, is the correlation of the predictable part with the policy-rate innovation, relative to that of the instrument as a whole. Only the part of the predictable component driven by the month's shocks covaries with the policy-rate innovation. A predictable component driven by earlier news, or by news to which the policy rate does not respond within the month, lowers $\hat{\rho}/\rho$ (Appendix equation (31)).

Table 1 reports the two factors for each instrument. The high-frequency surprise is the most predictable of the three, but predictability is not what sets it apart. If the predictable part of each instrument moved with the policy rate as closely as the instrument as a whole, the three would lose 0.41, 0.32 and 0.24 of their first-stage strength. They lose 0.54, 0.15 and 0.13. The difference is in the second factor.

The predictable part of the surprise moves with the policy rate more closely than the surprise itself, $\hat{\rho}/\rho = 1.33$. The controls predict news the market had not yet priced, to which the Fed responds systematically. The Fed's systematic response moves the policy rate. Removing it takes more than half of the surprise's covariance with the policy-rate innovation, and three quarters of its first-stage F. More than half of the raw surprise's first stage therefore came from the contamination that orthogonalisation removes. The contamination that survives can then be magnified up to 2.2

Table 1: The loss of first-stage strength and the magnification bound. R^2: the orthogonalisation regression on the six pre-announcement controls (Appendix Table 3). $\hat{\rho}/\rho$ and δ_π: equation (14), from the raw and orthogonalised first-stage F (Appendix B). First-stage F: HC0, each instrument alone for the funds-rate innovation in the baseline VAR, on the 383 months within 1988m1–2019m12 on which all its versions are observed. Orthogonalisation: PTA for the surprise, PTA-FWL for the Romer–Romer instrument and ATP-FWL for the Cholesky-based instrument. Magnification bound: $1/(1-\delta_\pi)$ (Proposition 2).

			Strength lost	First-stage F		Magnification
	R^2	$\hat{\rho}/\rho$	δ_π	raw	orthogonalised	bound
High-frequency surprise	0.165	1.33	0.54	14.5	3.6	2.2
Romer–Romer instrument	0.105	0.46	0.15	37.3	29.6	1.18
Cholesky-based instrument	0.060	0.52	0.13	966	534	1.14

times.

For the two wider-window instruments, $\hat{\rho}/\rho$ is about one half. Their predictable parts are correlated with the policy-rate innovation only half as strongly as the instruments themselves. Their construction already removes most of the systematic response, with the Greenbook forecasts for the Romer–Romer instrument and with the month's output and price innovations for the Cholesky-based instrument. The controls may still predict older information, or news to which the policy rate does not respond within the month. Either would lower $\hat{\rho}/\rho$. Removing it costs a small share of their covariance with the policy rate. The contamination it removes accounted for 15% and 13% of their raw first stage, and the magnification is bounded by 1.18 and 1.14. The first-stage F of the Cholesky-based instrument falls by more, close to half, mostly because its orthogonalised series is zero in months without an FOMC announcement (Appendix B). The loss of first-stage strength therefore depends less on how predictable an instrument is than on how its predictable component moves with the policy rate, the second factor in equation (14).

Orthogonalisation can also add contamination (Lemma A.1). This requires the controls and the instrument to share news from earlier periods. For a high-frequency surprise, that would not happen if the market priced all older information before the window opens. The surprise would then be unpredictable from any pre-announcement data. It is not. It is predictable from controls measured over windows that reach back well before the announcement, such as stock returns over the previous quarter and employment growth over the previous year. Bauer and Swanson (2023a) attribute this predictability to the market learning gradually how the Fed responds to the economy. The surprise then contains a response to news that is weeks or months old, which is the older news it shares with the controls.

Keeping strength is not evidence that the orthogonalisation cleans an instrument. A predictable part dated before the month carries little strength, and the orthogonalisation can then add contamination (Lemma A.1). The controls also end the day before the announcement, and they span only part of the news that precedes it. Orthogonalisation cannot remove contamination that arrives on the announcement day or later in the window, nor contamination the controls do not span. Strength instead limits how far the contamination that survives distorts the estimated impulse responses.

4.4 Strength limits the surviving bias

Neither wider-window instrument is clean. The Romer–Romer instrument carries any news that neither the Greenbook nor the controls span. The Cholesky-based instrument carries the financial exposure the controls do not span, and any news later in the month. Each is contaminated in a direction of its own, possibly more than the orthogonalised surprise. Both, however, keep their first-stage strength. By the bias identity in equation (7), that strength reduces how much the contamination that survives distorts the impulse responses. Their first-stage F is at least eight times that of the orthogonalised surprise, and the contamination that survives is magnified far less.

Figures 11 and 12 of Appendix G report the impulse responses to the two instruments, raw and orthogonalised. Each raw instrument is re-estimated on the months of its orthogonalised version, such that the two differ only by the orthogonalisation. Figure 1 is the corresponding figure for the surprise. Under either wider-window instrument, raw or orthogonalised, output falls gradually, with a peak decline of 1.4 to 1.7% after two to three years, about a quarter of the peak decline under the orthogonalised surprise. Prices rise by 0.1 to 0.4% at two years. This price puzzle is a well-documented symptom of contamination (Ramey, 2016). It points to contamination that survives in each wider-window instrument, which orthogonalisation on pre-announcement controls cannot remove by construction.

The first-stage strength that remains after orthogonalisation determines how far the impulse responses move. The high-frequency surprise loses three quarters of its first-stage F, and its output decline after two years more than doubles, from 1.9% to 5.0%. The Romer–Romer instrument loses a fifth of its first-stage F, and its output decline after two years grows only from 1.3% to 1.7%. The Cholesky-based instrument loses close to half of its first-stage F but remains strong, and its impulse responses are essentially unchanged.

An identity shows why the loss of strength governs how far the impulse responses move. The raw instrument is the sum of its predictable part, $\hat{z}_t$, and the orthogonalised instrument. Its first stage splits in the same way, with a share δ_π from the predictable part. The raw estimand is therefore a weighted average of the estimand with the predictable part as the instrument, $\tilde{\beta}_{\hat{z}} := \mathbb{E}[u_t \hat{z}_t] / \mathbb{E}[u_t^p \hat{z}_t]$, and the orthogonalised estimand,

$$\tilde{\beta} = \delta_\pi \, \tilde{\beta}_{\hat{z}} + (1 - \delta_\pi) \, \tilde{\beta}^{\perp}, \qquad \tilde{\beta}^{\perp} - \tilde{\beta} = \frac{\delta_\pi}{1 - \delta_\pi} (\tilde{\beta} - \tilde{\beta}_{\hat{z}}).$$

Orthogonalisation removes one term of this average. The raw estimand does not change. The orthogonalised estimand therefore lies on the opposite side of it from $\tilde{\beta}_{\hat{z}}$, at $\delta_\pi / (1 - \delta_\pi)$ times the distance between the two, at every horizon. This ratio is the observable shift factor. The news removed sets the direction in which the impulse responses move. The share of first-stage strength that this news carried sets how far they move.

The predictable parts of the high-frequency surprise and of the Romer–Romer instrument identify nearly the same impulse responses, those to news of a stronger economy. Output and prices rise, and credit spreads narrow. Orthogonalisation therefore moves the impulse responses of both instruments in the same direction. The distances differ. The predictable part carries about half of the first-stage strength of the surprise, and less than a sixth of that of the Romer–Romer instrument (Table 1). The impulse responses of the surprise move slightly further than the distance between its raw impulse responses and those to the removed news. The impulse responses of the Romer–Romer instrument move less than a fifth of that distance.

The size of the move therefore reflects the first-stage strength an instrument loses, not the contamination it sheds. If orthogonalisation left both instruments clean, their orthogonalised impulse responses would agree. The output decline is three to five times larger under the orthogonalised surprise. At least one of the two remains contaminated. By Proposition 2, any remaining contamination is magnified more for the surprise, which keeps less than half of its first-stage strength.

The two wider-window instruments are also contaminated along separate directions, since they rest on distinct information sets and windows. Section 6 exploits this structure. It groups instruments that share a contamination direction into families and removes what the members of each family share.

5 Identification from many instruments

Pooling the instruments raises the first-stage strength. A simple pool, however, does not make the instruments clean. No single instrument can be assumed valid, and the pool inherits the contamination of its members. Contamination that several instruments share does not average out as instruments are added. The estimator below therefore demeans before it pools. It groups instruments that share their exposure to non-policy shocks into families, removes the contamination each family shares, and only then combines them. Its target is the direction of the impact of the policy shock on the variables of the VAR, from which the causal impulse responses to that shock follow. The differences in relevance within families then identify the direction of the policy shock. The estimator is the leading eigenvector of the demeaned moments, and a single pooled instrument then delivers its first stage.

5.1 The instrument panel

Each instrument considered so far may load on non-policy shocks. The instruments also differ in relevance for the policy shock, because they are built from different sources, windows and maturities. The estimator uses this dispersion in relevance rather than assuming it away, because relevance carries the information about b_p. Recall that the reduced form is the VAR,

$$y_t = c + \sum_{\ell=1}^{s} \Pi_\ell \, y_{t-\ell} + u_t, \qquad u_t = B\varepsilon_t, \qquad \mathbb{E}\left[\varepsilon_t \varepsilon_t'\right] = I_n, \qquad B = \left[\, b_p \;\; B_{-p} \,\right],$$

with unit-variance structural shocks and residual covariance matrix $\Sigma_u := \mathbb{E}[u_t u_t'] = BB'$. The estimator recovers the direction of b_p, and the impact vector $\beta = b_p / b_{pp}$ of equation (4) is obtained at the end by normalising the policy coordinate to one. With a single strong and valid external instrument, its covariance with the VAR innovations is proportional to the impact column b_p. This covariance alone identifies the direction of b_p.

Suppose instead there are m candidate instruments, none of which is assumed to be valid. The direction of b_p must then be recovered from contaminated moments. Let each candidate instrument z_{jt} be

$$z_{jt} = \phi_j \, \varepsilon_t^p + \boldsymbol{\psi}_j' \boldsymbol{\varepsilon}_t^{-p} + \omega_{jt}, \qquad g_j := \mathrm{Cov}(u_t, z_{jt}) = \phi_j \, b_p + B_{-p} \boldsymbol{\psi}_j \in \mathbb{R}^n, \qquad j = 1, \ldots, m, \tag{15}$$

with relevance ϕ_j, contamination loading $\boldsymbol{\psi}_j \in \mathbb{R}^{n-1}$ and measurement noise ω_{jt} orthogonal to all

shocks. The identifying moment g_j is the covariance of the VAR innovations with instrument j. It has two components. One is the policy-shock impact vector scaled by relevance, $\phi_j b_p$. The other is the impact of the non-policy shocks the instrument loads on, $B_{-p}\boldsymbol{\psi}_j$. Stacking the moments into the matrix $G := [g_1, \ldots, g_m] \in \mathbb{R}^{n\times m}$, $\phi := (\phi_1, \ldots, \phi_m)'$ and $\Psi := [\boldsymbol{\psi}_1, \ldots, \boldsymbol{\psi}_m]' \in \mathbb{R}^{m\times(n-1)}$,

$$G = \underbrace{b_p \phi'}_{\text{relevance term}} + \underbrace{B_{-p}\Psi'}_{\text{contamination term}} . \tag{16}$$

Without a restriction on the contamination term, the moments cannot separate it from the relevance term, because b_p and the columns of the contamination term both lie in the column space of G. The restriction used here is that instruments in the same family share their contamination.

5.2 Families and the within transform

A family is a group of instruments built similarly, from the same source and over the same window. Examples are high-frequency surprises at several maturities around the same announcements, or several orthogonalisations of the same narrative series. Instruments in a family are exposed to similar non-policy news. Therefore, they share the mean of their contamination, while their relevance to the shock of interest can differ within a family. The difference in relevance is what the estimator uses for identification.

The main assumption is that the instruments can be grouped into C such families. Instrument j belongs to family c, written $j \in c$, and $|c|$ is the size of the family.

Assumption 1 (Common within-family contamination)**.** *Instruments can be grouped into C families. Within family c,*

$$\boldsymbol{\psi}_j = \bar{\boldsymbol{\psi}}_c + \boldsymbol{\nu}_j, \qquad j \in c, \tag{17}$$

where the idiosyncratic contamination $\boldsymbol{\nu}_j$ satisfies (a) zero mean within the family, $\mathbb{E}[\boldsymbol{\nu}_j \mid c] = \mathbf{0}$, and (b) no relation to relevance, $\mathbb{E}[\boldsymbol{\nu}_j \mid \boldsymbol{\phi}_c, c] = \mathbb{E}[\boldsymbol{\nu}_j \mid c]$, with $\boldsymbol{\phi}_c := \{\phi_k\}_{k\in c}$ the relevances of the family's members.

Assumption 1 splits each instrument's contamination loading into two parts, a family-common part $\bar{\boldsymbol{\psi}}_c$, shared by all members of the family, and an idiosyncratic part $\boldsymbol{\nu}_j$. The common part contributes $d_c := B_{-p}\bar{\boldsymbol{\psi}}_c$ to the moment of every member, the family-common contamination direction. Under condition (a), the members of a family share the mean of their contamination. For the high-frequency surprises, for example, the contamination after orthogonalisation is any news in the announcement window that the controls cannot span. Surprises measured over the same window share some of this news (Figure 3). Orthogonalisation cannot remove this contamination, which the controls do not span. The within transform is designed to remove it.

Condition (b) rules out a systematic link between relevance and idiosyncratic contamination within a family. On average, a more relevant member is not contaminated differently from a less relevant one. The condition holds given the relevances of all members of the family. After demeaning, the policy shock enters each moment only through the deviation of the instrument's relevance from its family's average, $\phi_j - \bar{\phi}_c$. The direction of b_p is identified from how the demeaned moments vary with these deviations. If the idiosyncratic contamination also varied with them, it would enter the demeaned moments in the same way as the policy shock. The estimator would then recover a mix of the target and the contamination.

Substituting the assumption, each moment is $g_j = \phi_j b_p + d_c + B_{-p}\nu_j$ for $j \in c$. Demeaning within families, with $\tilde{g}_j := g_j - \bar{g}_c$, where $\bar{g}_c = |c|^{-1}\sum_{k\in c} g_k$,[8] and likewise $\tilde{\phi}_j$ and $\tilde{\nu}_j$,

$$\tilde{g}_j = \tilde{\phi}_j b_p + B_{-p}\tilde{\nu}_j. \tag{18}$$

The common contamination d_c cancels exactly within families. The demeaned moment is the policy-shock term $\tilde{\phi}_j b_p$ plus the idiosyncratic contamination. The policy-shock term survives the within transform only if relevance differs within families. If all members of a family were equally relevant, demeaning would remove the policy-shock term along with the common contamination. The direction of b_p is the direction along which the demeaned moments move with the relevance deviations $\tilde{\phi}_j$. The total within-family dispersion of relevance adds up, over all instruments, the squared deviation of each instrument's relevance from its family's average,

$$\sigma^2_\phi := \sum_{c=1}^{C}\sum_{j\in c}\tilde{\phi}_j^2 = \sum_c\sum_{j\in c}(\phi_j - \bar{\phi}_c)^2. \tag{19}$$

It grows with the number of families that contain instruments of unequal relevance. It measures the strength of the signal the target leaves in the demeaned moments. It plays the role of a first stage for the direction. The direction is identified when this dispersion is positive and large relative to the idiosyncratic contamination.

Figure 4 is an illustrative example. It shows why the demeaning step is needed, with two families of six instruments and two variables. Panel (a) plots each instrument's moment, its covariance with the two VAR innovations. Each family lies along a line parallel to the true direction b_p, shifted away from the origin by its common contamination d_c. Pooling the moments without demeaning mixes the two shifts with the target. The first principal component of the raw moments therefore points in neither direction, and a simple pool recovers the wrong direction. Panel (b) plots the same moments, on the same axes, after each family's mean has been subtracted. Demeaning moves each family by minus its own mean. Therefore, both families now lie around the origin, along b_p. The first principal component of the demeaned moments is the true direction. It is recovered because relevance varies within each family. The moments of a family spread along b_p, and that spread survives demeaning. Appendix D shows, in a simulated three-variable VAR, the impulse responses that the same kinds of direction imply.

5.3 The estimator

The estimator, called the within estimator, works in two stages. It first whitens the moments. It then demeans them within families and finds the direction along which they vary most.

Whitening expresses each moment in terms of the structural shocks rather than the VAR innovations. The innovations are measured in the units of the variables, and they are correlated because each shock moves several variables at once. A covariance with the innovations therefore mixes the shocks. A non-policy shock can move the variables in proportions close to those of the policy shock, and its contamination then resembles relevance. Rescaling the innovations by the inverse square root of their covariance matrix, $\Sigma_u^{-1/2}$, removes both the units and the correlation.

[8] In the application, the family means weight each instrument by the precision of its moment (Section 6).

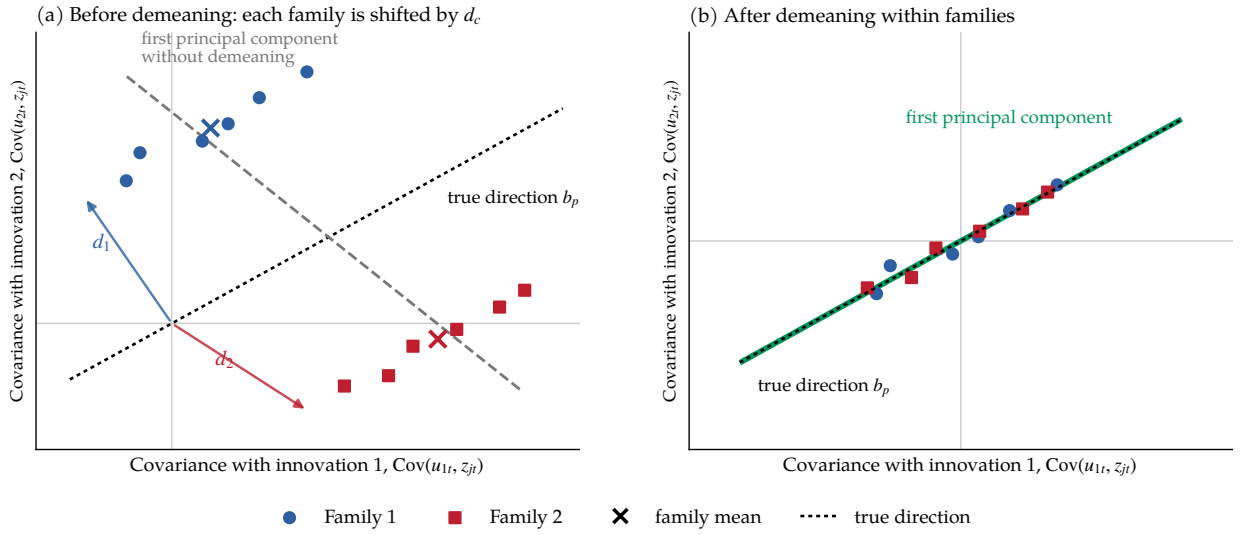


Figure 4: **Demeaning within families recovers the direction of the policy shock.** Simulated moments of two families of six instruments in a two-variable system, $g_j = \phi_j b_p + d_c + e_j$. Panel (a): the moments, with each family's mean (cross), the contamination d_c that the family shares (arrows), the true direction b_p (dotted) and the first principal component of all the moments without demeaning (dashed). Panel (b): the same moments on the same axes, after each family's mean is subtracted, with their first principal component (solid) and the true direction (dotted). Relevance ϕ_j varies within each family. The idiosyncratic contamination e_j is drawn uncorrelated with relevance within each family, as Assumption 1(b) requires. The estimator also whitens the moments, a step the figure omits.

This is the metric of generalised least squares. Let $\mathcal{Q} := \Sigma_u^{-1/2} B = [\, q_p \;\; \mathcal{Q}_{-p} \,]$. Since $\Sigma_u = BB'$,

$$\mathcal{Q}'\mathcal{Q} = B'\Sigma_u^{-1}B = B'(BB')^{-1}B = I_n.$$

The whitened innovations, $\Sigma_u^{-1/2} u_t = \mathcal{Q}\varepsilon_t$, therefore have unit variance, are uncorrelated, and are a rotation of the structural shocks. The whitened moment is the covariance of the instrument with them, $g_j^w := \mathrm{Cov}(\Sigma_u^{-1/2} u_t, z_{jt}) = \Sigma_u^{-1/2} g_j$. It collects the covariances of the instrument with the structural shocks, in rotated coordinates,

$$g_j^w = \phi_j\, q_p + \mathcal{Q}_{-p}\boldsymbol{\psi}_j, \qquad q_p'\mathcal{Q}_{-p} = \mathbf{0}', \quad \|q_p\| = 1. \tag{20}$$

The first term is the target direction q_p, the whitened impact of the policy shock, scaled by relevance. The second term is the contamination. The two terms do not overlap, because the shocks behind them are uncorrelated, and the size of each no longer depends on the units of the variables. After demeaning, the idiosyncratic contamination therefore adds noise to the estimated direction, but under condition (b) of Assumption 1 it does not shift the direction systematically towards the contamination.

The estimator is derived directly in these whitened coordinates. Let $\hat{g}_j^w := \hat{\Sigma}_u^{-1/2} \hat{g}_j$ denote the whitened estimated moments. By equation (20), they satisfy $\hat{g}_j^w = \phi_j q_p + d_c^w + e_j^w$ for $j \in c$, with $d_c^w := \mathcal{Q}_{-p}\bar{\boldsymbol{\psi}}_c$ the family's common contamination and e_j^w collecting the idiosyncratic contamination

$\mathcal{Q}_{-p}\boldsymbol{\nu}_j$ and the sampling error. The estimator chooses the family effects, the relevances and the direction that bring this structure closest to the moments, by least squares:

$$Q(q, \{\phi_j\}, \{d_c^w\}) = \sum_{c=1}^{C} \sum_{j \in c} \left\| \hat{g}_j^w - \phi_j q - d_c^w \right\|^2. \tag{21}$$

Written in terms of the original moments, this is weighted least squares with weight $\hat{\Sigma}_u^{-1}$, the metric of generalised least squares.

For a given direction, the criterion is quadratic in the family effects $\{d_c^w\}$ and in the relevances $\{\phi_j\}$. Therefore, it can be minimised in three steps. The first two remove the family effects and the relevances in closed form. The last finds the direction as the leading eigenvector of a matrix.

Step 1: Remove the family contamination, for given $(q, \{\phi_j\})$. Only the terms of family c involve d_c^w. The first-order condition for d_c^w, with $\bar{g}_c^w$ and $\bar{\phi}_c$ the family means of $\hat{g}_j^w$ and ϕ_j, gives

$$\hat{d}_c^w(q, \phi) = \bar{g}_c^w - \bar{\phi}_c\, q, \qquad \text{hence} \qquad \hat{g}_j^w - \phi_j q - \hat{d}_c^w = \tilde{g}_j^w - \tilde{\phi}_j\, q, \tag{22}$$

with the within transform $\tilde{g}_j^w := \hat{g}_j^w - \bar{g}_c^w$. In the rest of this section, $\bar{g}_c^w$, $\tilde{g}_j^w$ and the matrix $\widetilde{G}^w$ below are computed from the estimated moments. Estimating the family effects removes each family's common contamination. However, it also removes the family's mean relevance $\bar{\phi}_c$, as the two enter the criterion only through the sum $\bar{\phi}_c q + d_c^w$. Only the relevance deviations $\tilde{\phi}_j = \phi_j - \bar{\phi}_c$ survive, and they identify the direction of the policy shock.

Step 2: Estimate the relevance deviations, for given q. Substituting equation (22) into equation (21) gives the within criterion,

$$Q^w(q, \{\tilde{\phi}_j\}) := \min_{\{d_c^w\}} Q = \sum_j \left\| \tilde{g}_j^w - \tilde{\phi}_j q \right\|^2,$$

which separates across j. For fixed q each term is a scalar least-squares problem. Its solution, $\hat{\tilde{\phi}}_j(q) = q' \tilde{g}_j^w / q'q$, is the projection coefficient of the demeaned moment on the candidate direction q. It is the relevance deviation that q implies for instrument j, its relevance relative to the average of its family. The minimised value of term j measures the part of $\tilde{g}_j^w$ that q does not explain. Summing these over instruments gives the concentrated criterion, a function of the candidate direction q alone,

$$Q^c(q) := \min_{\{\tilde{\phi}_j\}} Q^w = \sum_j \|\tilde{g}_j^w\|^2 - \frac{q' \widetilde{G}^w \widetilde{G}^{w\prime} q}{q'q}, \qquad \widetilde{G}^w := [\tilde{g}_1^w, \ldots, \tilde{g}_m^w]. \tag{23}$$

The first term does not depend on q. The second is a Rayleigh quotient, invariant to the scale and the sign of q. The estimator therefore identifies a direction only, and the normalisation of equation (4) sets its scale.

Step 3: Find the direction. Minimising Q^c is the same as maximising the Rayleigh quotient of $\hat{S} := \widetilde{G}^w \widetilde{G}^{w\prime}$, over $q \neq \mathbf{0}$. By the Rayleigh–Ritz theorem, the maximum is the largest eigenvalue of $\hat{S}$, attained at the corresponding eigenvector. The eigenvector is unique up to sign when $\hat{\lambda}_1 > \hat{\lambda}_2$,

the two largest eigenvalues of $\hat{S}$. Hence

$$\hat{q} = \arg\min_{q} Q^c(q) \quad \Longleftrightarrow \quad \hat{S}\,\hat{q} = \hat{\lambda}_1\,\hat{q}, \qquad Q^c(\hat{q}) = \sum_j \|\bar{g}_j^w\|^2 - \hat{\lambda}_1. \tag{24}$$

The estimate $\hat{q}$ is the first principal component of the whitened demeaned moments. Undoing the whitening gives the impact vector in the original units, $\hat{b}_p = \hat{\Sigma}_u^{1/2}\hat{q}$, which is then normalised such that the policy rate rises by one unit, $\hat{\beta} = \hat{b}_p/(s_p'\hat{b}_p)$.

Substituting $\hat{q}$ back into the projection coefficient of Step 2 and into equation (22) returns the fitted relevance deviations $\hat{\tilde{\phi}}_j$ and family effects $\hat{d}_c^w$. Therefore, the three steps deliver every parameter of equation (21). Online Appendix OA.1 gives the derivations in full. The ratio of the two largest eigenvalues, the eigengap $\hat{\lambda}_1/\hat{\lambda}_2$, measures how much more the demeaned moments vary along $\hat{q}$ than along any other axis. It measures how sharply the data identify the direction of the policy shock, as the first-stage F measures the strength of a single instrument. It is reported with every estimate.

5.4 Identification

The estimator returns a direction, the proportions in which the policy shock moves the variables of the VAR on impact. The target is the direction of b_p, the true impact of the policy shock. Its scale is set at the end by the normalisation of equation (4). Whitening rescales b_p one-to-one, $b_p = \Sigma_u^{1/2}q_p$. Therefore, identifying q_p in whitened coordinates is the same as identifying b_p in the original units, and this subsection states the results for b_p. Two questions decide whether the estimated direction is the target. The first is whether the moments determine b_p at all, in population. The second is whether the leading eigenvector recovers it in a finite panel of instruments. The first is identification, the second recovery. The two paragraphs below answer them in turn.

Which restriction identifies the target. Consider two descriptions of the data. In the first, the policy shock has impact b_p and instrument j has contamination loading $\boldsymbol{\psi}_j$. In the second, the policy shock has impact $b_p + B_{-p}\alpha$, a mix of its true impact and the impact of the non-policy shocks, and each loading is reduced by $\phi_j\alpha$, in proportion to the instrument's relevance. Both descriptions give every instrument the same moment (Lemma C.1).[9] The data therefore cannot tell them apart. The reason is economic. If the instruments that move the policy rate more also carry more non-policy news, in fixed proportion, that news looks like part of the effect of the policy shock. Since the shift moves each loading in proportion to relevance, it is called a relevance-proportional shift.

Demeaning removes part of the shift. The part proportional to the family's average relevance is common to the family, and the within transform removes it together with the common contamination. The rest moves the idiosyncratic contamination of each instrument by $-\tilde{\phi}_j\alpha$, in proportion to its relevance deviation. A restriction that only requires the idiosyncratic contamination to average to zero within each family does not rule this out, because the relevance deviations also average to

[9] Under the shift, $b_p \mapsto b_p + B_{-p}\alpha$ and $\boldsymbol{\psi}_j \mapsto \boldsymbol{\psi}_j - \phi_j\alpha$. Therefore, $g_j = \phi_j(b_p + B_{-p}\alpha) + B_{-p}(\boldsymbol{\psi}_j - \phi_j\alpha) = \phi_j b_p + B_{-p}\boldsymbol{\psi}_j$ is unchanged. With $\phi_j = \bar{\phi}_c + \tilde{\phi}_j$, the change in the loading splits into $-\bar{\phi}_c\alpha$, common to the family and absorbed by d_c, and $-\tilde{\phi}_j\alpha$, which moves the deviation ν_j. Since $\sum_{j\in c}\tilde{\phi}_j = 0$, the shifted deviations still average to zero within each family. Their mean conditional on the relevances is $-\tilde{\phi}_j\alpha$, which is zero for every j only if $\alpha = \mathbf{0}$ when relevance varies within a family.

zero. Condition (b) of Assumption 1 rules it out, because it excludes idiosyncratic contamination that moves with relevance.

Proposition 3 (Identification). (i) *Under Assumption 1 and* $\sigma_\phi^2 > 0$, b_p *is identified up to scale.* (ii) *Under the restriction that the deviations average to zero within each family,* $\mathbb{E}[\bar{\boldsymbol{\nu}}_c \mid c] = \mathbf{0}$, *alone,* b_p *is not identified, at any* m *and in population.* (iii) *If instead* $\boldsymbol{\nu}_j = \alpha_0 \tilde{\phi}_j + \boldsymbol{\eta}_j$, *with* $\alpha_0 \in \mathbb{R}^{n-1}$ *fixed and* $\boldsymbol{\eta}_j$ *satisfying equation* (17), *every estimator identifies* $b_p + B_{-p}\alpha_0$, *not* b_p.

Details are in Appendix C.4. Identification is therefore lost through contamination that moves with relevance within families. Part (iii) shows that condition (b) cannot be weakened. If every family shares the same relevance-correlated component α_0, the moments are those of the shifted target $b_p + B_{-p}\alpha_0$, and no estimator can tell the two apart. If the component differs across families, each family points to its own direction, and the overidentification test of Section 5.5 can detect it.

Recovery in a finite panel. Proposition 3 concerns what the moments reveal in population. The estimator works with a finite panel, in which each instrument also carries its own idiosyncratic contamination.

Within a family, the estimator compares instruments that carry the same common contamination but move the policy rate by different amounts. The difference between their moments reflects the policy shock, scaled by the difference in their relevance. With idiosyncratic contamination, each comparison also contains the difference in the news that the two instruments carry on their own. This noise can affect the estimate in two ways.

First, in a given panel the idiosyncratic news can happen to be larger for the more relevant instruments. The estimated direction then tilts towards the contamination. Condition (b) makes this tilt zero on average, and it shrinks as instruments are added. It is a source of sampling uncertainty rather than bias, and it enters the uncertainty of the direction as a second variance component (Online Appendix OA.2).

Second, the idiosyncratic news can be large relative to the differences in relevance. The estimator picks the direction along which the demeaned moments vary most. When relevance differs enough within families, that direction is the target, because the moments vary with the relevance deviations along b_p. When the idiosyncratic contamination varies more along some other direction, that direction becomes the leading one, and the estimator picks a contamination direction. The target therefore remains the leading direction if the within-family dispersion of relevance exceeds the spread of the idiosyncratic contamination in any single direction. This is the ordering condition, $\sigma_\phi^2 > \lambda_{\max}(N)$. Here N adds up, over instruments, the outer products of the demeaned idiosyncratic contamination, and its largest eigenvalue, $\lambda_{\max}(N)$, is the spread of that contamination along the direction in which it is largest (Appendix C.5). The condition compares the spread along the target, σ_ϕ^2, with the largest spread along any other direction. It plays for the panel the role that a strong first stage plays for a single instrument.

Adding instruments does not by itself secure the ordering condition. Both sides grow with the number of instruments. The condition holds when, within families, the instruments differ more in how strongly they move the policy rate than in the news they carry on their own.

When the ordering condition fails, the estimator returns a contamination direction. A longer time sample does not help. More observations estimate each moment more precisely but the idiosyncratic contamination is part of the moments themselves. The fit of the estimator cannot reveal it, since

the leading direction always fits best (Lemma C.2). The signs of the estimated relevances can reveal it (Appendix C.6). Appendix C.5 states these results formally (Proposition C.2). None of the conditions requires any instrument to be exogenous. Identification of the target in population rests on two: common within-family contamination (Assumption 1) and relevance that varies within families ($\sigma_\phi^2 > 0$). Recovery of the target by the leading eigenvector in a finite panel adds a third, the ordering condition.

5.5 The family partition

Identification also places requirements on the partition of the instruments into families. It needs enough instruments within families and enough families. These are necessary conditions on the design of the panel. The first concerns the direction. Each family spends one instrument on its common contamination. Therefore, only the remaining $m - C$ instruments carry information about the direction. The second concerns strength. The within transform removes the common contamination of each family for any number of families, but it also discards each family's average relevance. Enough families allow that average relevance to be used as well, without bringing the common contamination back.

A combination of the family levels removes the common contamination only if the family contaminations cancel each other. It still carries idiosyncratic contamination, but not the common part. This requires more families than the number of distinct directions the family contaminations occupy. With $n - 1$ non-policy shocks, there are at most $n - 1$ such directions.

Proposition 4 (The two order conditions)**.** Order condition 1 (O1). Enough instruments within families. *Identification of the direction requires*

$$m - C \;\geq\; 1,$$

and the rank condition $\sigma_\phi^2 > 0$, that at least one family contains two instruments of unequal relevance. When order condition 1 fails, the direction is not identified.
Order condition 2 (O2). Enough families to use the levels. *Let*

$$r := \operatorname{rank}[\bar{\boldsymbol{\psi}}_1, \ldots, \bar{\boldsymbol{\psi}}_C] \;\leq\; \min(C,\, n-1) \tag{25}$$

be the number of distinct directions of the family contaminations. Order condition 2 is $C > r$. A clean combination with nonzero family totals exists if and only if it holds. Since $r \leq n - 1$, a sufficient condition, for any realised rank, is

$$C \;>\; n - 1.$$

When order condition 2 fails, the direction is still identified, and the estimate remains free of the common contamination. Only the strength that can be used cleanly falls, to σ_ϕ^2, the part carried by the relevance deviations.

Details are in Appendix C.4. Order condition 1 counts the instruments within families and governs the direction. Its count, $m - C \geq 1$, is implied by $\sigma_\phi^2 > 0$, and it sets the degrees of freedom of the fit. Order condition 2 counts the families and governs strength. With too few families, the level of relevance, usually the larger part, is lost.

Order condition 2 also becomes harder to meet as the VAR grows. Each added variable adds a non-policy shock, and hence one more direction the family contaminations may occupy, while

the number of families is fixed by the instruments available. It is assessed on the family-average contamination of the C families, estimated from the family means of the moments.

Testing the family structure. The family structure has testable implications. Under exact block structure, the idiosyncratic contamination is zero, and once each family's common contamination is removed every instrument points to the same direction, up to sampling error. The number of restrictions this places on the moments follows from counting. The moments supply nm moment conditions, n for each instrument. The unknowns are the direction of b_p, with $n-1$ free entries once its scale is fixed, the m relevances, and the C family contaminations d_c. Each d_c is a combination of the $n-1$ columns of B_{-p}, and hence also has $n-1$ free entries. The moments therefore exceed the unknowns by

$$nm - (n-1) - m - (n-1)C = (n-1)(m-C-1).$$

Each family spends one instrument on its common contamination, and one more instrument is spent on the direction. Each remaining instrument supplies $n-1$ overidentifying restrictions, one for each direction orthogonal to the target. Order condition 1 is the requirement that this count be non-negative.

The overidentification test evaluates these restrictions. Its statistic adds up the squared within-family residuals, standardised by their sampling covariance,

$$J_T = \sum_{i=1}^{n-1} \hat{e}_i' \big(\hat{A}\hat{V}\hat{A}'\big)^{+} \hat{e}_i \;\sim\; \chi^2_{(n-1)(m-C-1)},$$

where $\hat{e}_i$ collects the residuals of the m instruments in the ith of the $n-1$ directions orthogonal to $\hat{q}$. The residuals are what remains of the whitened moments after within-family demeaning and removal of the fitted relevance deviations. $\hat{A}$ is the linear map from the moments to these residuals, of rank $m-C-1$. $\hat{V}$ is the sampling covariance of the moments, estimated from the instruments residualised on the VAR regressors. Its off-diagonal elements matter, because instruments built from the same announcements or the same VAR are highly correlated. The pseudo-inverse, $(\cdot)^{+}$, handles the rank of $\hat{A}\hat{V}\hat{A}'$. Details are in Online Appendix OA.2. The null hypothesis is exact block structure. With idiosyncratic contamination, the residuals also carry that contamination. J_T and its within-family part then reject even when Assumption 1 holds. The between-family part is made robust to this by adding the estimated variance of the idiosyncratic contamination to $\hat{V}$, and it remains valid without exact block structure. This variance, $\hat{\Sigma}_\nu$, is estimated as the part of the variance of the residuals that sampling error alone cannot explain (Online Appendix OA.2). It is also used to assess the ordering condition. The statistic splits into a within-family part, which asks whether the members of each family agree, and a between-family part, which asks whether the families agree with each other (Appendix C.6). The between-family part detects relevance-correlated contamination that differs across families, but not a component common to all families (Proposition 3(iii)). Online Appendix OA.2 bounds the impulse responses against such a component. The test is reported with every estimate in Section 6.

Choosing and testing the partition. The partition is chosen on economic grounds, before the moments are formed. Families follow the construction of the instruments (Section 5.2). Instruments measured at a different point in time are exposed to different news and form families of their own.

A high correlation between two instruments is not a reason to group them, because correlation mostly reflects their common response to the policy shock, which is relevance. Each family needs at least two instruments of unequal relevance, and three where possible.

The partition can also be estimated from the data. Once the target direction is removed, instruments of the same family cluster around their family's common contamination. An algorithm searches over these clusters, chooses the number of families and assigns each instrument to one (Online Appendix OA.3.1). In the simulations of Online Appendix Section OA.4, the algorithm recovers the number of families once the sample is long enough. It recovers the assignment of every instrument only without idiosyncratic contamination (Online Appendix Proposition OA.3.1). The partition based on the construction of the instruments is therefore used for estimation, and the estimated partition is used to verify it.

Errors in the partition are not symmetric. Splitting a family costs only precision. Merging two families that differ both in mean contamination and in mean relevance makes contamination move with relevance inside the merged family, and biases the direction (Online Appendix Lemma OA.3.1). When in doubt, split families. Splitting costs only precision, while merging can bias the direction.

5.6 A clean pooled instrument

The within estimator identifies the direction from the within-family moments. Pooling reaches the same direction another way. It combines the instruments into a single instrument whose contamination cancels. This pooled instrument gives the same impulse responses as the within estimator. It also delivers an instrument series. Therefore, its first-stage strength can be reported.

How the instruments are combined matters. When instruments share contamination within families, a simple average does not identify the direction. It adds strength, but it keeps the common contamination of each family in full, however many instruments it averages (Appendix C.7). Panel (a) of Figure 4 illustrates the consequence. Without demeaning, the leading direction of the moments differs from the target. Constrained pooling chooses the weights to remove the common contamination.

Constrained pooling. Constrained pooling seeks the strongest combination of the instruments whose contamination is zero. A combination $w'z_t = \sum_j w_j z_{jt}$ inherits the three parts of its members, equation (15), each combined with the same weights,

$$w'z_t = \underbrace{(w'\phi)}_{\text{relevance}}\,\varepsilon_t^p + \underbrace{(\Psi' w)'}_{\text{contamination}}\,\boldsymbol{\varepsilon}_t^{-p} + \underbrace{w'\omega_t}_{\text{noise}}, \qquad \operatorname{Var}(w'\omega_t) = w'\Sigma_\omega w.$$

Its strength is the concentration parameter $\mu^2(w) = T(w'\phi)^2/(w'\Sigma_\omega w)$, the population counterpart of the first-stage F. It is relevance squared over noise. The program maximises it subject to zero contamination,

$$w^\star = \arg\max_w \frac{(w'\phi)^2}{w'\Sigma_\omega w} \quad \text{s.t.} \quad \Psi' w = \mathbf{0}. \tag{26}$$

The constraint restricts the contamination of the combination to zero. Column k of Ψ gives each instrument's loading on non-policy shock k. Therefore, $\Psi' w$ is the loading of the combination on each shock. The weights that satisfy the constraint form the *kernel* of Ψ', $\ker(\Psi') := \{w \in \mathbb{R}^m :$

$\Psi' w = \mathbf{0}\}$, the set of clean combinations. Two kinds of combination are clean. The first gives weights that sum to zero within each family. It removes the common contamination of each family, as demeaning does. The second gives the same weight to all members of a family, with family totals $a_c := \sum_{j \in c} w_j$ chosen such that the contaminations of the families cancel each other, $\sum_c a_c \bar{\boldsymbol{\psi}}_c = \mathbf{0}$.

With homoskedastic independent noise, $\Sigma_\omega = \sigma_\omega^2 I$, and the normalisation $w'\phi = 1$, the solution is

$$w^\star = \frac{P_\perp \phi}{\|P_\perp \phi\|^2}, \qquad \mu^2(w^\star) = T \, \|P_\perp \phi\|^2 / \sigma_\omega^2, \tag{27}$$

where $P_\perp := I - \Psi(\Psi'\Psi)^+\Psi'$. The optimal weights are the part of the relevances that the contamination loadings, the columns of Ψ, do not explain. Clean strength is the strength of this best combination with zero contamination, $\mu^2(w^\star)$. It is proportional to $\|P_\perp \phi\|^2$, the squared length of the part of relevance left after the contamination loadings are removed.[10] An instrument helps the pool only to the extent that its relevance is not explained by its contamination.

The program gives the strength of the best clean instrument, but not where that strength comes from. Within a family, the weights are free, and the program sets them in proportion to relevance. Across families, the weighted sum of the family contaminations must cancel, $\sum_c a_c \bar{\boldsymbol{\psi}}_c = \mathbf{0}$, which requires order condition 2. Without the constraint, the strengths of the instruments add (Appendix C.8). The constraint determines how much of that sum survives, and from which source.

The strength of the best clean instrument splits into two parts (Appendix C.3). The first comes from the relevance deviations. It equals σ_ϕ^2, identifies the direction, and is clean for any partition. The second comes from the family levels. It adds strength only, and only when there are more families than contamination directions, which is order condition 2. A family whose members are near-copies of one series adds almost nothing to σ_ϕ^2. Its level can still add strength when order condition 2 holds. A family can therefore add strength without adding identifying variation.

In practice, Ψ is unobserved, and the fitted direction supplies a stand-in. The part of each whitened moment orthogonal to $\hat{q}$ estimates its contamination. The feasible weights are the estimated relevances with their part along these estimated contaminations removed, and the pooled instrument is $z_t^\star := \sum_j \hat{w}_j z_{jt}$. It gives the same impulse responses as the within estimator, and it adds an instrument series and its first stage. Because it is built from $\hat{q}$, it also inherits any error in the fitted direction. Details are in Appendix C.1.

6 Empirical application to high-frequency monetary surprises

This section applies the within estimator to U.S. monetary policy. Each instrument is orthogonalised on pre-announcement controls, grouped into families, demeaned within families and pooled.

The panel holds twelve U.S. monetary policy instruments in four families of three. The pooled instrument is strong. Its first stage is above the conventional threshold for a single instrument (Stock and Yogo, 2005) and seven times that of the orthogonalised surprise. Under the pooled instrument, output falls, the headline CPI falls at every horizon, and the credit spread widens on impact. Relative to the impulse responses estimated with the orthogonalised surprise, the confidence bands are narrower. The fall in prices on impact shrinks and is no longer significant, and the impulse responses of output and the credit spread are smaller.

[10] Appendix C.8 derives equation (27) and gives the solution for general noise, $w^\star = \Sigma_\omega^{-1}\phi^\perp / (\phi^{\perp\prime}\Sigma_\omega^{-1}\phi^\perp)$, with $\phi^\perp$ the residual of ϕ on the columns of Ψ in the Σ_ω^{-1} metric. Relative to the unconstrained pool, whose weights are ϕ itself, the constraint costs $\|\phi\|^2 - \|P_\perp \phi\|^2$, the part of relevance collinear with the contamination.

6.1 Data, the instrument panel, and the partition

The baseline model is a monthly VAR with six lags and a constant,

$$y_t = c + \sum_{\ell=1}^{6} \Pi_\ell \, y_{t-\ell} + u_t, \qquad \mathbb{E}[u_t u_t'] = \Sigma_u, \qquad u_t = B\varepsilon_t,$$

with $y_t = (\mathrm{IP}_t, \mathrm{CPI}_t, \mathrm{EBP}_t, i_t)'$: the logs of industrial production and of the consumer price index, the excess bond premium of Gilchrist and Zakrajšek (2012), and the federal funds rate as policy variable. The VAR is estimated from January 1973 to December 2019. The identifying moments $\hat{g}_j$ are formed on each instrument's own support between January 1988 and December 2019, the period over which the Bauer and Swanson (2023b) controls are available.

Impulse responses are normalised to raise the federal funds rate by one percentage point on impact, and are reported over 36 months. First-stage strength is the heteroskedasticity-robust F of the regression of the policy-rate innovation on the instrument. The estimates use Algorithm 1 with precision weights.[11] Confidence bands are 68% median-recentred percentile intervals from a residual moving-block bootstrap (Jentsch and Lunsford, 2022) with 1,000 draws (Online Appendix equation (OA.6)). Each draw resamples the VAR residuals and the instruments jointly and rebuilds the data from the estimated VAR. The VAR coefficients, Σ_u, the moments, the precision weights and the direction are then re-estimated. Only the partition is held fixed.

Instrument families. The baseline panel holds twelve instruments (Appendix Table 5) in four families of three. Every instrument is orthogonalised on the six Bauer and Swanson (2023b) controls, at the announcement level and then summed within the month (PTA), jointly with the instrument's own first stage (PTA-FWL), or after monthly aggregation (ATP).

- *High-frequency surprises.* The changes in the three-month-ahead federal funds future and in the four-quarter-ahead Eurodollar future around FOMC announcements (FF4 and ED4, from the Acosta et al. (2025) database, starting in February 1994.), and the Bauer and Swanson (2023b) surprise, each orthogonalised at the announcement level (PTA).
- *Swanson factors.* Swanson (2021) extracts three factors from the high-frequency responses of futures and Treasury yields to FOMC announcements and rotates them to separate a change in the federal funds rate, forward guidance and asset purchases. The family uses the factors of Swanson (2024), each orthogonalised at the announcement level (PTA).
- *Romer–Romer.* The Romer and Romer (2004) instrument (Section 4.1), in the same three orthogonalisations.
- *Cholesky-based instruments.* The Cholesky innovations of Section 4.2 for the federal funds rate and the one- and two-year Treasury yields, orthogonalised after monthly aggregation (ATP-FWL).

The assignment is economic. Each family shares a source, a window and the controls its members are orthogonalised on. The controls end the day before each announcement and span only part of the news that precedes it. Orthogonalisation therefore leaves the contamination that arrives on

[11] Each instrument is weighted by $1/\hat{v}_j$, the sampling variance of its moment, and contributes on its own support. The weights make the estimate invariant to the units of each family, and they give more influence to the moments measured precisely.

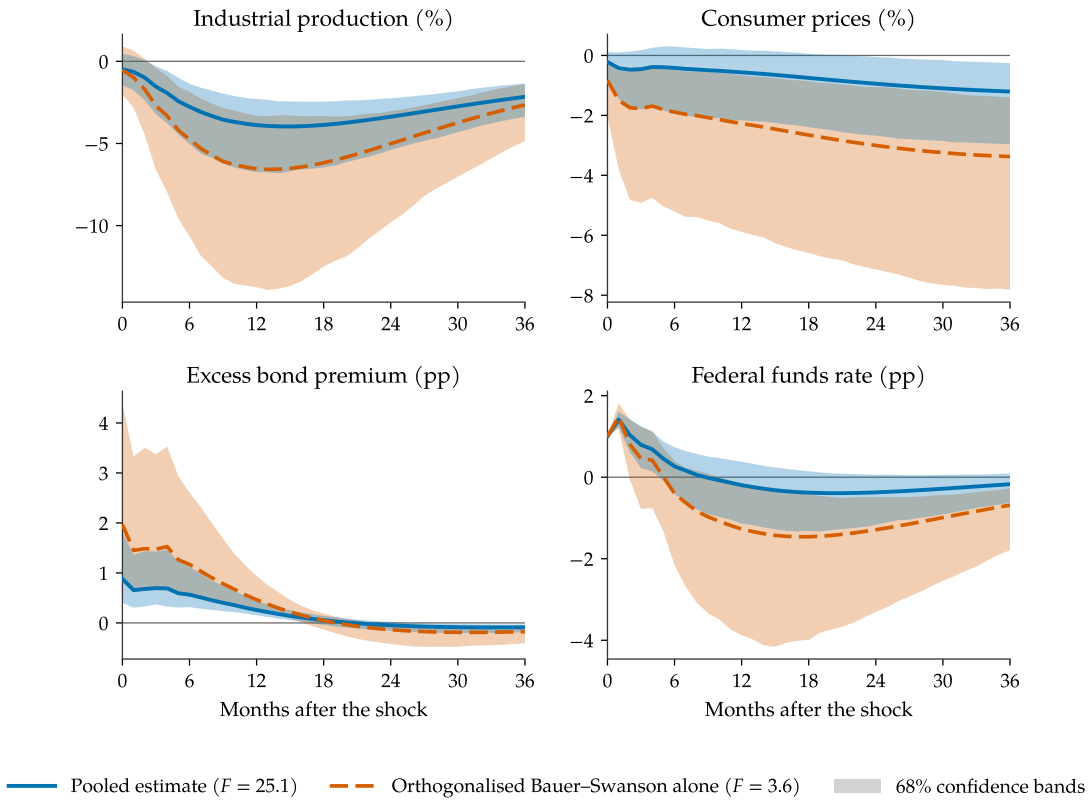


Figure 5: **The pooled monetary policy shock.** Impulse responses to a +1 percentage-point shock to the federal funds rate. Solid: the baseline pooled estimate from twelve orthogonalised instruments in four families ($F = 25.1$). Dashed: the orthogonalised Bauer and Swanson (2023b) surprise alone ($F = 3.6$). Shaded: 68% median-recentred percentile confidence bands from the moving-block bootstrap, 1,000 draws, with the VAR, the moments and the weights re-estimated in each draw.

the announcement day or later in the window, and the contamination the controls do not span. After orthogonalisation, a family therefore shares the news inside its window. The window is the announcement window for the high-frequency and Swanson families, the whole decision on the day of the meeting for the Romer–Romer family, and the month for the Cholesky-based instruments. Its members differ in relevance through their maturity, their factor, or the way they are orthogonalised. In addition, since each member is orthogonalised on the same controls, the pooled instrument is also unpredictable from the spanned variation (Appendix Table 3).

6.2 The baseline impulse responses

Figure 5 shows the baseline estimate, the impulse responses identified by the pooled instrument, against those of the orthogonalised surprise alone. Table 2 compares the two, and the raw surprise, on the five features of Section 2.2. Pooling moderates each of them.

The first stage rises to $F = 25.1$, seven times that of the orthogonalised surprise, and the confidence bands narrow by more than half. Output falls by 0.5% on impact and by 4.0% at its peak

Table 2: The five features of Section 2.2. Raw and Orthogonalised (PTA): the Bauer and Swanson (2023b) surprise, raw and orthogonalised announcement by announcement; Pooled baseline estimate. Impulse responses to a +1 pp funds-rate shock; confidence bands are 68% moving-block bootstrap intervals.

		Raw	Orthogonalised (PTA)	Pooled
(i) First stage	F, funds rate	14.5	3.6	25.1
(ii) Precision	width of the 68% confidence band, output at its peak decline	2.5	10.6	4.3
	width of the 68% confidence band, prices at 24 months	1.2	6.1	2.6
(iii) Magnitude	peak output decline, % (month)	−1.9 (20)	−6.6 (13)	−4.0 (15)
	credit spread on impact, pp	+0.25	+1.97	+0.88
(iv) Prices on impact	price level in the month of the shock, %	−0.15	−0.83	−0.22
(v) Funds-rate path	first month below zero	–	6	9
	lowest value, pp (month)	+0.17 (36)	−1.46 (17)	−0.39 (20)

decline, after fifteen months, between the raw surprise (1.9%) and the orthogonalised one (6.6%). Prices fall gradually, by 0.9% after two years, and their fall on impact is no longer significant. The excess bond premium rises by 0.9 points on impact, less than half its rise under the orthogonalised surprise, and the funds rate turns negative later and less deeply.

These estimates reflect what each family brings to the pool. Strength is spread across the families. In the unit-invariant form of the pooled instrument, the Cholesky-based and Romer–Romer families each supply a third of the first stage, and the Swanson family only an eighth. The direction comes mostly from the Swanson family, which carries most of the identifying variation (Section 6.2). Its three factors separate the surprise in the funds rate from forward guidance and asset purchases. These components move the funds rate by very different amounts, and this dispersion in relevance pins down the direction. On its own, the Swanson family identifies a sharp contraction, and the Cholesky-based family a mild one with higher prices. The baseline lies between the two.

Pooling moderates the five features of the orthogonalised surprise for the reason of Section 3. The orthogonalised surprise has a weak first stage, and its impulse responses magnify any contamination that survives the orthogonalisation. The pooled instrument is seven times stronger, and the contamination that survives in it is magnified far less. Demeaning also removes the news that each family shares, which the controls do not span and orthogonalisation therefore leaves in place. The impact responses show the difference most, as the fall in prices and the jump in the credit spread both shrink. The credit spread still rises on impact. A monetary tightening lowers the value of borrowers' collateral and of lenders' capital, and the premium that lenders demand rises (Gertler and Karadi, 2015). The smaller rise in the premium in turn explains the milder reversal of the funds rate. The Fed eases in response to tighter credit conditions (Caldara and Herbst, 2019). The funds rate crosses zero at nine months, after the premium has peaked on impact and before output falls most, at fifteen months.

Diagnostics. Two tests evaluate the identification. The first asks whether the family structure fits the data. Its null hypothesis is exact block structure: once each family's common contamination is removed, every instrument points to the same direction, up to sampling error. The overidentification test of Section 5.5 does not reject it ($p = 0.85$), neither within families ($p = 0.59$) nor between them ($p = 0.90$).[12] Under the null hypothesis, the instruments of a family differ only in

[12]The statistic uses the full sampling covariance of the moments, since instruments built from the same announcements or the same VAR are correlated up to 0.98. It is $J_T = 14.5$ on $(n-1)(m-C-1) = 21$ degrees of freedom, and splits

how strongly they move the policy rate, not in the news they carry, and families built from different sources and windows point to the same policy shock. With three instruments per family, the test has limited power. Its large p-value therefore means only that the family structure is not rejected.

The second test, a test of no common direction, asks whether the instruments share a direction at all. Its null hypothesis is that the demeaned moments contain only sampling noise, with no relevance deviations within families. The leading eigenvalue of the within scatter measures how much the demeaned moments vary along the estimated direction. Under the null, a value as large as in the data occurs in 0.8% of draws, and the null is rejected (Online Appendix OA.2). The instruments share a direction, and every family points to it.

What identifies the shock. The direction and the strength come from different families. Most instruments are weak on their own. The Romer–Romer instruments are moderately strong, and the Cholesky-based instruments are by far the strongest (left panel of Appendix Figure 13), as they are functions of the VAR's own innovations.

The right panel of Appendix Figure 13 shows where the identifying variation comes from. The Swanson factors supply 71% of the identifying variation and the Cholesky family 28% (Appendix Table 7). The two families play different roles. The Cholesky family brings strength and tempers the magnitude of the impulse responses. The Swanson family supplies most of the dispersion in relevance within families, which sets the direction.

The premise that the three factors share their contamination is testable inside the family. The within-family test J_c, the overidentification statistic of family c alone, has the null hypothesis that the family's members share their contamination, such that once it is removed they identify a single direction (Appendix C.6). For the Swanson family, $J_c = 4.4$ on three degrees of freedom ($p = 0.22$), and $p \geq 0.35$ for every other family. The null hypothesis is not rejected for any family.

The other two families contribute less to the direction. The Romer–Romer family holds three orthogonalisations of one series, which are highly correlated. Therefore, relevance varies little within it. The three high-frequency surprises also move the funds rate by similar amounts. Both families still matter for strength. Their family levels enter the pooled instrument and add to its first stage. In the unit-invariant form of the pooled instrument, the two families together supply more than half of the first stage.

Dropping one family at a time confirms these roles. With only three families left, order condition 2 would fail. The exercise therefore adds the information-robust family of Section 7.1 as a fifth family and drops each family in turn from the five (Appendix Table 9 and Appendix Figure 14). Dropping the high-frequency or the Romer–Romer family leaves the estimate essentially unchanged. Dropping the Cholesky-based family makes the impulse responses larger and the pooled instrument weak ($F = 4.5$). The Cholesky-based instruments therefore do not create the result. They strengthen the pool and temper its magnitude. Dropping the Swanson family gives a milder and later contraction, close to the Cholesky-based benchmark, which orders the funds rate after output and prices.

into a within-family part (10.3 on 12) and a between-family part (4.2 on 9), which asks whether the families agree on the direction (Appendix C.6).

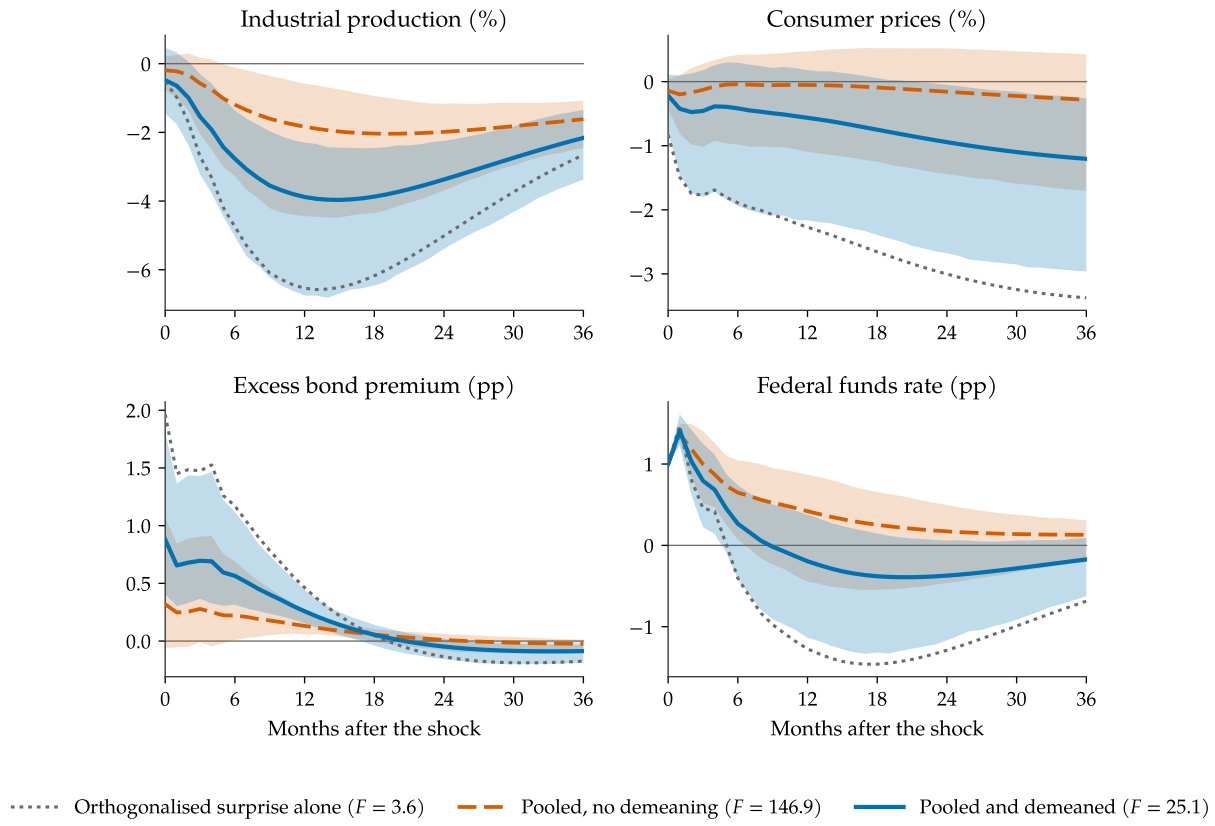


Figure 6: **What each step contributes.** The orthogonalised Bauer and Swanson (2023b) surprise alone (dotted, $F = 3.6$), the twelve instruments pooled without demeaning (dashed, $F = 146.9$), and the baseline (solid, $F = 25.1$). Same VAR, moment window, weights and bootstrap as Figure 5; 68% confidence bands.

6.3 Why pooling needs demeaning

Pooling alone is not enough. A pool of contaminated instruments is strong, but it keeps the contamination its members share. The baseline therefore combines two steps, and each accomplishes a different task. Pooling supplies strength, and strength narrows the confidence bands. Demeaning removes contamination, and contamination shifts the impulse responses away from the truth. A pool that is strong but not demeaned can therefore give narrow confidence bands around the wrong impulse responses.

Pooling the twelve instruments without demeaning illustrates this risk (Figure 6). The pooled instrument is very strong ($F = 146.9$), but it leans on its strongest family, the Cholesky-based instruments, and inherits their contamination. Output falls by only 2.0% at its peak decline, prices rise, and the funds rate never turns negative. These impulse responses are close to those the Cholesky-based family identifies on its own (Appendix Figure 12). Demeaning costs most of the strength, down to $F = 25.1$, and the peak decline in output doubles, to 4.0%. The contamination removed is the news inside each family's window, which orthogonalisation on pre-announcement controls cannot reach.

The null hypothesis that the pools with and without demeaning estimate the same direction is

not rejected ($p = 0.87$). However, the demeaning step halves the peak decline in output. Strength alone is therefore not a guide to an instrument's validity.

6.4 What demeaning removes

Demeaning removes the contamination that each family shares. Looking at each family separately shows which families carry that contamination, and in which direction it moves each variable. Each family's average instrument, used on its own, is compared with the pooled estimate. The difference measures the family's common contamination in units of impulse responses.

Each family's mean moment has two parts. One is the family's average relevance times the impact of the policy shock, and the other is the family's common contamination, d_c. The within transform discards both. Losing the first costs strength, which constrained pooling partly recovers (Section 5.6). Losing the second removes contamination. Used on its own, the family's average instrument identifies, up to sampling error,

$$\tilde{\beta}_c = \beta + \frac{d_c - d_{c,p}\,\beta}{\bar{\phi}_c^p + d_{c,p}}, \tag{28}$$

with $\bar{\phi}_c^p$ the family's mean relevance in units of the policy rate and $d_{c,p}$ the funds-rate coordinate of d_c. Relevance moves every variable in the proportions of β, and it cancels once the funds-rate coordinate is set to one. Only the contamination that does not move with the target shifts the estimate, and the family's own first stage scales it. A weak family therefore magnifies its contamination, as detailed in Section 3.3. Figure 7 plots the shift $\tilde{\beta}_c - \beta$, mapped into impulse responses on impact and after twelve months. Each bar is measured relative to the pooled estimate, not to the true impulse responses. Reading it as a bias therefore relies on block structure.

The families fall into two groups (Figure 7). The two wide-window families, Romer–Romer and Cholesky-based, point toward a *milder* contraction. Their means would have raised the impulse responses of output and prices and lowered that of the credit spread. Their instruments are measured over longer windows, which also contain news other than the policy decision, and the instruments attribute part of that news to the policy shock. The high-frequency family points the other way, toward a *stronger* contraction. Its common contamination also raises the credit-spread response on impact (Figure 7). This is consistent with news in the announcement window that moves interest rates and credit spreads together. The family's weak first stage also magnifies that contamination. The Swanson family, also built from surprises in the announcement window, carries the least contamination.

The two wide-window families also carry the largest common contamination relative to their relevance. For each family, the fraction of its mean moment that lies off the pooled direction measures how contaminated it is (Appendix Table 7). The fraction is computed family by family, and the fractions therefore do not add up to one. It is largest for the Cholesky-based and Romer–Romer families. This is the cost that Section 4 anticipated. Widening the window restores first-stage strength, but it admits news that a narrow window excludes (Figure 3).

The two groups point in opposite directions, which is why the pooled estimate sits between the single instruments (Appendix Table 8). The contributions do not sum to the gap between the pools with and without demeaning in Figure 6, for two reasons. First, the pool without demeaning does not average the four families. It weights them by their strength and leans on the Cholesky-based

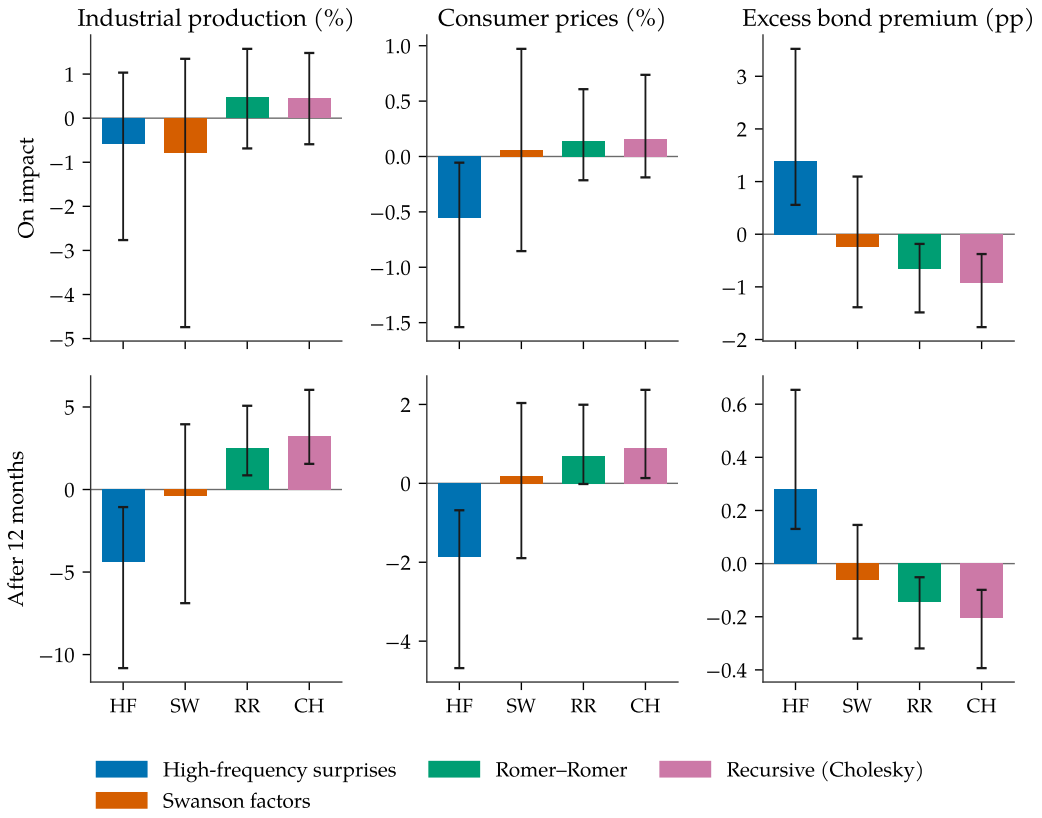


Figure 7: **The family-common component removed by the within transform.** For each family, the impulse response to a +1 percentage-point funds-rate shock identified by the family's precision-weighted mean moment $\bar{g}_c$ (its average instrument used alone), minus the pooled response, equation (28). Top row: on impact; bottom row: after twelve months. Whiskers: 68% median-recentred percentile intervals, 1,000 draws, with the VAR, the moments, the weights, and the pooled direction re-estimated in each draw. HF: high-frequency surprises; SW: Swanson factors; RR: Romer–Romer; CH: Cholesky-based.

family, whose bar points toward a milder contraction. Second, each contribution bar divides the family's contamination by its own family's first stage. The pool without demeaning instead divides the combined contamination by the combined first stage. Since the scaling differs, the effects of the families on the impulse responses differ.

7 Robustness of the baseline estimate

The baseline rests on choices made in the empirical application, and the data cannot fully validate them. They concern which instruments enter, how they are grouped into families, how the VAR is specified, and which estimator is used. The exercises below change one choice at a time and keep the baseline families, weights, and windows unless stated otherwise. The panel of instruments comes first, because the identification rests on it.

7.1 Alternative panels of instruments

The identification rests on the panel of instruments and on its partition into families. If the baseline reflected one particular choice of instruments, a different panel would give a different estimate. The exercises below add a family, diversify the families and repartition the instruments.

A fifth family. An information-robust family (IR) can be added as a fifth family. Its three members start from a rate surprise in the announcement window and remove the information that the announcement reveals about the economy. The Miranda-Agrippino and Ricco (2021) surprise does so with the Fed's own forecasts. It is the residual of the three-month federal funds future surprise (FF4) on the Greenbook forecasts and their revisions. The other two members use the poor man's sign restriction of Jarociński and Karadi (2020). A monetary tightening should lower stock prices. When the rate surprise and the S&P 500 surprise in the same window move in the same direction, the announcement is read as news about the economy, and the surprise is set to zero. Every member is first orthogonalised at the announcement level on the Bauer and Swanson (2023b) controls (PTA). The sign restriction is then applied to the orthogonalised FF4 surprise and to the orthogonalised Bauer and Swanson (2023b) surprise. In both, 35% of the orthogonalised announcements still move with stock prices and are set to zero. Details are in Appendix E.3. Its members are built from announcement-window surprises, like the high-frequency family. Removing the information effect, however, changes the news they carry, and with it their contamination.

The between-family test asks whether the five families agree on the direction. Its null hypothesis is that they do. It does not reject ($p = 0.79$). The IR family therefore points to the same direction as the baseline families, and its contamination shows no difference from theirs that moves with relevance. Adding the IR family moves the estimate little. The first stage is $F = 21.2$, and output falls by 4.3% at its peak decline, against 4.0% in the baseline. Appendix Table 9 reports the panels of this subsection.

A fifth family also isolates the role of each family. Each family is dropped in turn, and the direction and the pooled instrument are re-estimated from the remaining four. The impulse responses then show the contribution of each family to the estimation. With four families left, order condition 2 still holds, unlike when one of the four baseline families is dropped (Appendix Figure 14). The Swanson family sets the depth of the contraction. Without it, output falls by about half as much, and prices barely move. The Cholesky-based family supplies strength and tempers the magnitude. Without it, the pooled instrument is weak ($F = 4.5$), and the impulse responses are larger. The high-frequency and Romer–Romer families confirm the direction. Without either, the impulse responses are those of the five-family estimate. Their members agree with the other families, and they supply overidentifying restrictions to the tests of block structure. In the baseline, each of them is also needed for order condition 2.

Diversified families. In the baseline, the Swanson family carries most of the identifying variation. Adding instruments within families can reduce this dependence. The goal is to introduce dispersion in relevance within each family. Each family must still group instruments exposed to the same non-policy news, and the demeaning step still removes their common exposure. Two families are diversified (Appendix E.3).

- *Maturity ladder.* The two-, five- and ten-year Treasury surprises in the announcement window, added to the high-frequency family. They share the family's window, and their relevance falls

with maturity.

- *Romer–Romer family* (RR′). In the baseline, the family holds three orthogonalisations of the same instrument (PTA, PTA-FWL and ATP). Their relevance along the pooled direction is nearly the same. The family carries 31% of the first-stage strength but adds little dispersion in relevance. RR′ keeps the Romer–Romer instrument orthogonalised before aggregation (PTA) and replaces the two other versions by two specifications of the Aruoba and Drechsel (2024b) shock. Like the Romer–Romer instrument, each is the residual of a regression of the change in the intended funds rate on the Greenbook forecasts. The regression also includes sentiment indicators extracted from Fed staff documents. The two specifications differ in whether the regressors enter with lags. The Aruoba and Drechsel (2024b) shocks cover only scheduled meetings from 1982 to 2008, a shorter support than the baseline window.

All new instruments are orthogonalised on the same controls as the baseline instruments. Each new family adds dispersion in relevance. The Treasury surprises share the announcement window of the near-dated futures and are exposed to the same news, but longer maturities respond less to the policy rate. Demeaning within the family removes the common news and keeps these differences in relevance, which point to the policy direction without the Swanson factors. The Aruoba and Drechsel (2024b) shocks remove more of the predictable part of the change in the funds rate. Their relevance along the pooled direction, 0.004 and 0.009, is lower than that of the Romer–Romer instrument, 0.013. A family's share is its contribution to the variation that pins down the direction. With the three changes together, the share of the Swanson family falls from 71% to about half. The impulse responses remain close to the baseline (Appendix Figure 15). Without the Swanson family, they now stay much closer to the baseline than in the baseline panel, because the other families carry identifying variation of their own. The dependence on the Swanson family can thus be reduced without altering the baseline estimates.

Repartitioning the instruments. The two previous exercises add instruments to the panel. The instruments can also be repartitioned into different families, which tests whether the results depend on the baseline partition. The new partition keeps the instruments of the previous exercises, except the minutes factors, and forms five families, each holding instruments built from the same data over the same window. The five families are the near-term futures (the baseline high-frequency family), the Treasury curve (the two-, five- and ten-year Treasury surprises), the IR family, the narrative instruments (the Romer–Romer instrument and the two Aruoba and Drechsel (2024b) shocks) and the Cholesky-based instruments. The first three share the announcement window, the narrative instruments the decision day and the Cholesky-based instruments the month. The Swanson factors no longer form a family of their own. The target, path and asset-purchase factors measure the surprise in the current rate, the expected path of policy over the next year or so and the long end of the curve, respectively. The first two therefore join the near-term futures, and the third joins the Treasury curve.

The impulse responses nearly coincide with the baseline (Appendix Figure 16). The peak decline in output and the fall in prices after two years differ from the baseline by less than a tenth and a fifth of a percentage point, respectively, and both lie well inside the baseline confidence bands. The new partition passes the same tests as the baseline. Exact block structure is not rejected ($p = 0.64$), and the test of no common direction (Section 6.2) rejects as strongly as in the baseline ($p = 0.008$). The target factor remains the main source of identifying variation. Within the near-term family,

it moves the funds rate far more than the futures surprises do, and this dispersion in relevance pins down the direction. The asset-purchase factor contributes less than in the baseline, because its relevance deviation is now measured against the Treasury surprises. The role of the Swanson factors in the identification thus falls, as a by-product of the new partition. Without any factor, the near-dated futures and the Treasury surprises differ too little in relevance to identify the direction, and the estimate is close to the Cholesky-based benchmark. Moving the path factor to the Treasury curve leaves the estimate unchanged.

The estimate survives the addition of a family, the diversification of the families, and a variation in family formation. In all three exercises, the impulse responses remain broadly consistent with the baseline.

7.2 The VAR specification

The exercises below return to the baseline panel of four families and change the composition of the VAR. They replace the federal funds rate by the shadow rate as the policy indicator, replace the price level by inflation[13] and core prices, and include, one at a time, the unemployment rate, commodity prices, the S&P 500 and housing starts, to trace the transmission of the tightening to the rest of the economy.

Policy indicator. From December 2008 to December 2015 the federal funds rate was at its effective lower bound, and these months contribute little identifying variation. Policy still acted through forward guidance and asset purchases, which moved the yield curve. The shadow rate of Wu and Xia (2016) is constructed from the yield curve and can turn negative at the lower bound. Used as the policy indicator, it keeps moving in those months, which then add identifying variation. The results remain consistent with the baseline (Appendix Figure 17). Output and prices fall, and the credit spread rises, with impulse responses slightly larger than under the funds rate. The pooled instrument is weaker but remains strong ($F = 12.7$).

Inflation and core prices. The orthogonalised surprise implies an immediate fall in the price level, a pattern at odds with the gradual transmission of monetary policy (feature (iv) of Section 2.2). The price measure shows whether this pattern depends on how prices enter the VAR. In the baseline, prices enter as the log level of the consumer price index. Appendix Figure 18 replaces it in turn with year-over-year and month-over-month inflation, and with core CPI, which excludes food and energy. Each replacement changes one variable but keeps four variables. Hence, the baseline families, order conditions and partition still apply.

The pooled instrument is strong for every price measure, and the orthogonalised surprise reports the larger price effect in every case. The two also differ in the timing of the price response. Under the orthogonalised surprise, prices and inflation fall in the month of the announcement. The pooled instrument attenuates this impact effect, and prices fall gradually as the tightening weakens demand. The impact fall that remains comes mostly from food and energy, whose prices move quickly in response to a monetary tightening. Core prices barely move on impact under the pooled instrument, and the core level falls below zero only after a long delay. This timing is consistent with the gradual transmission of monetary policy to prices. Under every measure, prices eventually fall.

[13] Year-over-year inflation is $100(\log P_t - \log P_{t-12})$ and month-over-month inflation $100\Delta \log P_t$, with P_t the price index.

The timing and the size of the fall differ between the pooled instrument and the orthogonalised surprise.

Transmission to the rest of the economy. A larger VAR traces the transmission of the policy shock to the rest of the economy. Within the pooling method, each additional variable brings a non-policy shock of its own, which may contaminate the instruments. A larger VAR therefore requires more families to meet order condition 2. Each larger system includes the IR family of Section 7.1 as a fifth family ($C = 5 > n - 1 = 4$). Hence, the pooled instrument can remove the contamination from every non-policy shock in the system. Each panel of Figure 8 is a separate five-variable VAR, the baseline four variables and the one shown, estimated with the baseline weights and windows on the baseline sample, January 1973 to December 2019. The four variables shown are the unemployment rate, a commodity-price index, the S&P 500 and housing starts.

The impulse responses follow the conventional transmission of a monetary tightening. Financial markets react first, and the S&P 500 falls on impact (Bernanke and Kuttner, 2005). Housing, the most interest-sensitive component of demand, contracts next, ahead of output (Bernanke and Gertler, 1995). Unemployment rises and commodity prices fall more gradually, as demand weakens (Christiano et al., 1999; Anzuini et al., 2013). Monetary policy affects these variables only with a lag. In all four systems, the impulse responses of output and prices are close to the baseline, and exact block structure is not rejected.

The first-stage strength varies across systems, from 9.7 to 18.7, for two reasons. First, the policy-rate innovation changes, because it is the part of the funds rate that the lags of all the VAR's variables cannot predict. Second, the weights of the pooled instrument change, because they are set to cancel the contamination from the new non-policy shock. When the instruments move with that shock on impact, cancelling it also removes part of their relevance. Stock prices react within minutes of the announcement, and the S&P 500 gives the weakest system. Slow-moving real variables give the strongest (Appendix Table 10).

Appendix Figure 19 extends the exercise to eight further variables, in the same five-variable systems. Financial conditions tighten on impact, while demand, the labour market and credit contract more gradually, consistent with the lagged effects of monetary policy on real activity. In every system, output and prices fall. Appendix Table 10 reports these systems and the other candidate variables screened.

7.3 Reusing the pooled instrument in other frameworks

The constrained pooling delivers a series, the pooled instrument $z_t^\star$, which can be reused in any framework that takes an instrument. Two such frameworks are an internal-instrument VAR (Plagborg-Møller and Wolf, 2021) and local projections (Jordà, 2005). In the internal-instrument VAR, the pooled instrument is ordered first, and the shock is identified from a Cholesky decomposition of the extended covariance matrix. In local projections, each outcome variable is regressed, horizon by horizon, on the policy rate instrumented by the pooled instrument. Appendix E.4 gives the specifications of both.

The pooled instrument exists only on the moment window, from January 1988 to December 2019. Both frameworks use it as a regressor, and their sample is restricted to that window. For a consistent comparison, the external-instrument VAR is re-estimated on the same window, with the baseline variables, lags and normalisation. Only the framework then changes across the three.

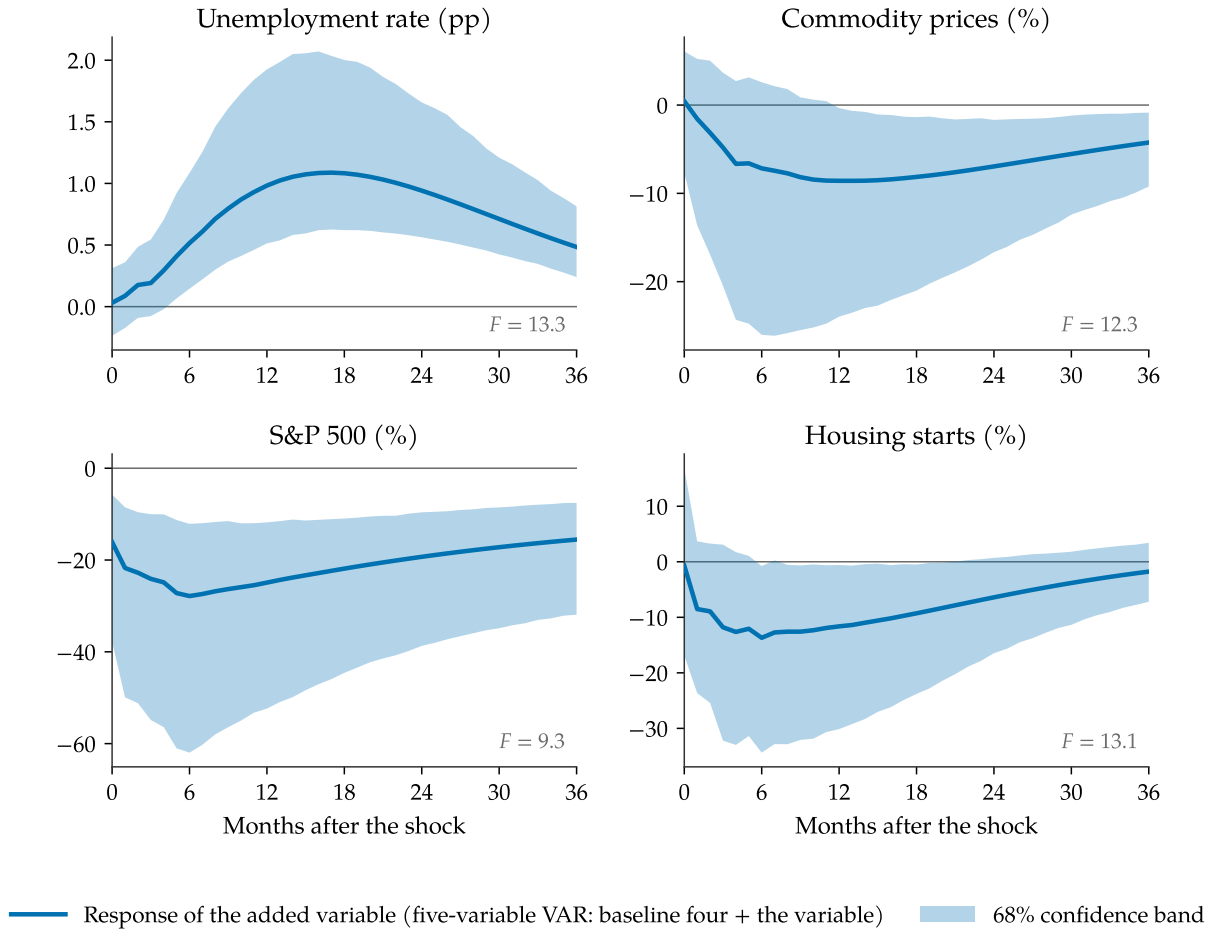


Figure 8: **Larger systems, one added variable at a time.** Each panel is estimated separately, as a five-variable VAR in the baseline four variables plus the variable shown, with the baseline families and the IR family as a fifth, and the baseline precision weights, windows and bootstrap; the panel plots only the impulse response of the added variable to a +1 percentage-point funds-rate shock, with its 68% confidence band.

Appendix Figures 20 and 21 show the three estimates, with the baseline, estimated from 1973, for reference.

The pooled instrument remains strong in all three specifications, and the estimates agree on impact. Output falls, and the excess bond premium rises by similar amounts. The transmission that follows differs in timing and depth. The external-instrument VAR extrapolates the dynamics from its six lags, which smooths the impulse responses at longer horizons. Local projections estimate each horizon separately, without this extrapolation. Their impulse responses are therefore more irregular and less precise, and their output decline is earlier and deeper. The internal-instrument VAR lies between the two, with a decline of similar depth to the external estimate but reached sooner. The signs agree across the three frameworks. Output falls and the credit spread rises in all of them. On the baseline sample, prices fall at every horizon.

8 Conclusion

No available instrument for monetary policy is both strong and clean, and the identification proposed in this paper does not require one. It groups the instruments into families whose members share their exposure to non-policy news but differ in how strongly they move the policy rate. Demeaning the identifying moments within families removes the contamination each family shares. The relevance deviations that remain identify the direction of the policy shock. Pooling across families restores the first-stage strength that each instrument may lack.

The method answers a trade-off that orthogonalisation creates. Orthogonalising an instrument on pre-announcement information is necessary to remove its predictable variation. However, it weakens the first-stage strength, and it cannot remove news that the controls do not span. The contamination that survives orthogonalisation is then magnified by the weaker first stage. The high-frequency surprise is the extreme case. It loses most of its first-stage strength, and its impulse responses are large in magnitude. The paper extends the orthogonalisation to instruments built from wider windows around the policy decision, a Romer and Romer (2004) instrument and a Cholesky-based instrument. They are partly predictable, but they retain most of their first-stage strength under the same orthogonalisation. They may, however, admit news that the narrow window excludes. Orthogonalisation and pooling are therefore complements. The first removes what was predictable before the announcement. The second removes the news that a family shares within its window, and it restores the strength that the first takes away.

On twelve orthogonalised U.S. instruments in four families, the pooled instrument is strong where the orthogonalised surprise is weak, and it identifies a standard monetary contraction. Output falls gradually, consumer prices decline slowly and the credit spread widens. The estimate survives the addition of a family, the diversification of the families and a different formation of the families. The pooled instrument can also be reused in other frameworks, such as an internal-instrument VAR and local projections.

The pooling method is not specific to monetary policy. It suits any shock that several instruments identify. Oil supply shocks are a natural candidate. They are instrumented by high-frequency surprises around OPEC announcements (Känzig, 2021), by measures of exogenous supply disruptions (Kilian, 2008), and by oil price increases above their recent peak (Hamilton, 2003; Ramey and Vine, 2011). Fiscal policy offers other applications. Government spending shocks are instrumented by narrative military news (Ramey, 2011; Ramey and Zubairy, 2018) and by the excess returns of defence contractors (Fisher and Peters, 2010), and tax shocks by narrative tax changes (Romer and Romer, 2010; Mertens and Ravn, 2013).

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Appendix

A Proofs for Section 3

Derivation of the bias identity in equation (7). With the shocks orthogonal and of unit variance, and ω_t uncorrelated with them, the identifying moment is $\mathbb{E}[u_t z_t] = \phi\, b_p + B_{-p}\boldsymbol{\psi}$, and its policy-rate entry is $\pi = \phi\, b_{pp} + b'_{p,-p}\boldsymbol{\psi}$. Since $b_p = b_{pp}\beta$, the policy term is $\phi\, b_p = \phi b_{pp}\,\beta = (\pi - b'_{p,-p}\boldsymbol{\psi})\beta$. Substituting it in the moment gives $\mathbb{E}[u_t z_t] = \pi\beta + B_{-p}\boldsymbol{\psi} - \beta\, b'_{p,-p}\boldsymbol{\psi} = \pi\beta + d$. Dividing by π gives $\tilde{\beta} = \beta + d/\pi$. Since $s'_p\beta = 1$, $s'_p d = b'_{p,-p}\boldsymbol{\psi} - b'_{p,-p}\boldsymbol{\psi} = 0$. □

The controls and the orthogonalisation regression. The controls are dated before the announcement and are assumed orthogonal to the policy shock. They decompose into a part that loads on the period's non-policy shocks and a remainder,

$$\mathbf{W}_t = A'\boldsymbol{\varepsilon}_t^{-p} + \boldsymbol{\zeta}_t, \qquad \mathbb{E}[\boldsymbol{\varepsilon}_t\boldsymbol{\zeta}_t'] = \mathbf{0}, \tag{29}$$

where A collects the covariances between the period's non-policy shocks and the controls, and the remainder $\boldsymbol{\zeta}_t$ is news from earlier periods. The shocks have unit variance and are uncorrelated with $\boldsymbol{\zeta}_t$, and hence the variance of the controls is $\Sigma_W = A'A + \Sigma_\zeta$, with $\Sigma_\zeta := \mathrm{Var}(\boldsymbol{\zeta}_t)$. Σ_W is assumed nonsingular. Let $\sigma_{\zeta\omega} := \mathbb{E}[\boldsymbol{\zeta}_t\omega_t]$ be the covariance between the news from earlier periods in the controls and in the instrument.

Lemma A.1 (The contamination that survives orthogonalisation)**.** *With the controls orthogonal to the policy shock, the contamination loading of the orthogonalised instrument is*

$$\boldsymbol{\psi}^\perp = \underbrace{\left(I - A\Sigma_W^{-1}A'\right)\boldsymbol{\psi}}_{\text{contamination left behind}} - \underbrace{A\Sigma_W^{-1}\sigma_{\zeta\omega}}_{\text{added contamination}}. \tag{30}$$

The added contamination is non-zero only if the controls and the instrument share news from earlier periods, $\sigma_{\zeta\omega} \neq \mathbf{0}$. Without that shared news, $\|\boldsymbol{\psi}^\perp\|^2 \leq \|\boldsymbol{\psi}\|^2 - \mathrm{Var}(\hat{z}_t)$, and validity improves relative to the raw instrument, strictly if the controls predict any contamination.

Proof. The loading follows in four steps.

1. *Covariance of the controls with the instrument.* Multiplying equation (29) by equation (5),

$$\mathbb{E}[\mathbf{W}_t z_t] = \mathbb{E}\left[(A'\boldsymbol{\varepsilon}_t^{-p} + \boldsymbol{\zeta}_t)(\phi\,\varepsilon_t^p + \boldsymbol{\psi}'\boldsymbol{\varepsilon}_t^{-p} + \omega_t)\right] = A'\boldsymbol{\psi} + \sigma_{\zeta\omega}.$$

   Only two products survive. $\mathbb{E}[\boldsymbol{\varepsilon}_t^{-p}\boldsymbol{\varepsilon}_t^{-p\prime}] = I$ gives $A'\boldsymbol{\psi}$, and $\mathbb{E}[\boldsymbol{\zeta}_t\omega_t]$ gives $\sigma_{\zeta\omega}$. The others vanish because the shocks are orthogonal to each other, and ω_t and $\boldsymbol{\zeta}_t$ are uncorrelated with the shocks.

2. *Regression coefficient.* $\boldsymbol{c} := \Sigma_W^{-1}\,\mathbb{E}[\mathbf{W}_t z_t] = \Sigma_W^{-1}(A'\boldsymbol{\psi} + \sigma_{\zeta\omega})$.

3. *Fitted value and residual.* The fitted value splits into a part in the period's shocks and a part in earlier news,

$$\hat{z}_t = \boldsymbol{c}'\mathbf{W}_t = \hat{z}_t^c + \hat{z}_t^\ell, \qquad \hat{z}_t^c := (A\boldsymbol{c})'\boldsymbol{\varepsilon}_t^{-p}, \qquad \hat{z}_t^\ell := \boldsymbol{c}'\boldsymbol{\zeta}_t.$$

Subtracting it from equation (5),

$$z_t^{\perp} = \phi\, \varepsilon_t^{p} + (\boldsymbol{\psi} - A\boldsymbol{c})'\boldsymbol{\varepsilon}_t^{-p} + (\omega_t - \boldsymbol{c}'\boldsymbol{\zeta}_t).$$

The last term is uncorrelated with the shocks. The orthogonalised instrument therefore keeps the relevance ϕ, and its contamination loading is $\boldsymbol{\psi}^{\perp} = \boldsymbol{\psi} - A\boldsymbol{c}$.

4. *Substituting the coefficient* from step 2 gives equation (30).

Bound. Suppose $\boldsymbol{\sigma}_{\zeta\omega} = \mathbf{0}$, such that $\boldsymbol{c} = \Sigma_W^{-1} A'\boldsymbol{\psi}$, or $A'\boldsymbol{\psi} = \Sigma_W \boldsymbol{c}$. Three facts follow.

(a) $\boldsymbol{\psi}' A\boldsymbol{c} = \boldsymbol{c}'\Sigma_W \boldsymbol{c} = \mathrm{Var}(\hat{z}_t)$, by $A'\boldsymbol{\psi} = \Sigma_W \boldsymbol{c}$.

(b) $\|A\boldsymbol{c}\|^2 = \mathrm{Var}(\hat{z}_t^{c})$, because the non-policy shocks have identity variance.

(c) $\mathrm{Var}(\hat{z}_t^{c}) \leq \mathrm{Var}(\hat{z}_t)$. The two parts of $\hat{z}_t = \hat{z}_t^{c} + \hat{z}_t^{\ell}$ are uncorrelated, because $\boldsymbol{\zeta}_t$ is uncorrelated with the shocks. Hence $\mathrm{Var}(\hat{z}_t) = \mathrm{Var}(\hat{z}_t^{c}) + \mathrm{Var}(\hat{z}_t^{\ell}) \geq \mathrm{Var}(\hat{z}_t^{c})$.

Expanding the square and using (a) to (c),

$$\|\boldsymbol{\psi}^{\perp}\|^2 = \|\boldsymbol{\psi}\|^2 - 2\,\boldsymbol{\psi}' A\boldsymbol{c} + \|A\boldsymbol{c}\|^2 \leq \|\boldsymbol{\psi}\|^2 - 2\,\mathrm{Var}(\hat{z}_t) + \mathrm{Var}(\hat{z}_t) = \|\boldsymbol{\psi}\|^2 - \mathrm{Var}(\hat{z}_t).$$

The bound is strictly below $\|\boldsymbol{\psi}\|^2$ whenever $\mathrm{Var}(\hat{z}_t) > 0$, that is whenever the controls predict any contamination, $A'\boldsymbol{\psi} \neq \mathbf{0}$. Conversely, a clean instrument, $\boldsymbol{\psi} = \mathbf{0}$, becomes contaminated, $\boldsymbol{\psi}^{\perp} = -A\Sigma_W^{-1}\boldsymbol{\sigma}_{\zeta\omega}$, whenever this term is non-zero. □

The first term is the contamination the controls leave behind. Without the remainder, $\Sigma_{\zeta} = \mathbf{0}$ and, with A of full column rank, $A\Sigma_W^{-1}A' = A(A'A)^{-1}A'$ is the projection onto the span of A. The first term would then be the part of $\boldsymbol{\psi}$ outside that span. The remainder inflates Σ_W and shrinks the regression coefficients, and part of the contamination inside the span also survives. The noisier the controls, the larger that part.

The second term is the contamination orthogonalisation adds. News from earlier periods shared by the controls and the instrument, $\boldsymbol{\sigma}_{\zeta\omega}$, raises the regression coefficients. Subtracting the controls then also subtracts their loading on the period's non-policy shocks, A, in proportion. This is news from the period that the instrument did not carry. The term vanishes if $\boldsymbol{\sigma}_{\zeta\omega} = \mathbf{0}$.

From the two terms to the bias. Write equation (30) as $\boldsymbol{\psi}^{\perp} = L\boldsymbol{\psi} - \boldsymbol{a}$, with $L := I - A\Sigma_W^{-1}A'$ and $\boldsymbol{a} := A\Sigma_W^{-1}\boldsymbol{\sigma}_{\zeta\omega}$. By equation (7), the contamination direction is linear in the loading, $d = M\boldsymbol{\psi}$ with $M := B_{-p} - \beta\, b_{p,-p}'$, which does not depend on the instrument. It therefore splits into the same two terms, the contamination left behind, d_L, and the added contamination, d_A,

$$d^{\perp} = d_L + d_A, \qquad d_L := ML\boldsymbol{\psi}, \qquad d_A := -M\boldsymbol{a},$$

and so does the first stage, $\pi^{\perp} = \phi\, b_{pp} + b_{p,-p}'L\boldsymbol{\psi} - b_{p,-p}'\boldsymbol{a}$. The bias of the orthogonalised instrument is

$$\tilde{\beta}^{\perp} - \beta = \frac{d_L + d_A}{\pi^{\perp}} = \frac{d_L + d_A}{\pi\,(1 - \delta_{\pi})}.$$

The added contamination enters the bias exactly as the contamination left behind does. It is the impact of the added news net of the induced policy response, magnified by the same weaker first

stage. The two terms can reinforce or offset each other, because the sign of d_A depends on $\sigma_{\zeta\omega}$ and not on ψ. The added news also moves the first stage, through $-b'_{p,-p}a$.

A clean instrument, $\psi = 0$, is unbiased before orthogonalisation, with $\pi = \phi\, b_{pp}$ and $d = 0$. After orthogonalisation, $\psi^\perp = -a$, and

$$\tilde{\beta}^\perp - \beta = \frac{d_A}{\pi^\perp} = -\frac{MA\Sigma_W^{-1}\sigma_{\zeta\omega}}{\phi\, b_{pp} - b'_{p,-p}A\Sigma_W^{-1}\sigma_{\zeta\omega}}.$$

Orthogonalisation then biases an unbiased instrument. The bias is non-zero only if the controls carry news from the period, $A \neq 0$, and share news from earlier periods with the instrument, $\sigma_{\zeta\omega} \neq 0$. Proposition 2 excludes this case, $\psi \neq 0$, because the magnification factor is not defined when the raw bias is zero.

Proof of Proposition 1 (what orthogonalisation removes). First stage. With $u_t^p = b_{pp}\varepsilon_t^p + b'_{p,-p}\varepsilon_t^{-p}$ and $z_t^\perp$ from step 3, $\pi^\perp = \mathbb{E}[z_t^\perp u_t^p] = \phi\, b_{pp} + b'_{p,-p}\psi^\perp$. The last term of $z_t^\perp$ drops out because it is uncorrelated with the shocks.

(i) Share lost. Relevance is unchanged, and hence, by the first stage, $\pi - \pi^\perp = b'_{p,-p}(\psi - \psi^\perp)$. Dividing by π gives the first expression for δ_π. Covariance is additive over $z_t = \hat{z}_t + z_t^\perp$, and hence $\pi - \pi^\perp = \mathrm{Cov}(\hat{z}_t, u_t^p)$. Dividing by π gives the second.

(ii) Contamination direction. As above, $d = M\psi$ with $M = B_{-p} - \beta\, b'_{p,-p}$, which does not depend on the instrument. Since the orthogonalised instrument keeps the relevance ϕ, $d^\perp = M\psi^\perp$. Since norms are non-negative, $\delta_d \leq 1$. δ_d can be negative even when $\|\psi^\perp\| < \|\psi\|$. M weighs the loadings unequally, and removing a component of ψ that offset the rest can lengthen d. □

Proof of Proposition 2 (magnification). Since $\beta = b_p/b_{pp}$, the contamination direction can be written $d = B\big(-b'_{p,-p}\psi/b_{pp},\ \psi'\big)'$. The impact matrix B is invertible, and hence $d \neq 0$ whenever $\psi \neq 0$, and $\|d\| > 0$. By equation (7), $\tilde{\beta} - \beta = d/\pi$ and $\tilde{\beta}^\perp - \beta = d^\perp/\pi^\perp$. Because $\pi, \pi^\perp > 0$, the norms factor as

$$\varrho = \frac{\|d^\perp\|/\pi^\perp}{\|d\|/\pi} = \frac{\|d^\perp\|/\|d\|}{\pi^\perp/\pi} = \frac{1-\delta_d}{1-\delta_\pi},$$

using the definitions of δ_d and δ_π. Positivity of $\pi^\perp/\pi$ gives $\delta_\pi < 1$, and $\|d^\perp\| \geq 0$ gives $\delta_d \leq 1$.

Condition in equation (11). The denominator $1 - \delta_\pi$ is positive. Hence, $\varrho > 1$ iff $1 - \delta_d > 1 - \delta_\pi$, that is $\delta_d < \delta_\pi$. If $\delta_d \geq 0$, the numerator is at most one. Hence, $\varrho \leq 1/(1-\delta_\pi)$, with equality iff $\delta_d = 0$. If $\delta_\pi \leq 0$, then $1 - \delta_\pi \geq 1$, and since $1 - \delta_d \geq 0$, $\varrho \leq 1 - \delta_d$. □

B The loss of first-stage strength: derivations

This appendix derives equation (14) and the related identities, and shows how Table 1 computes the loss of first-stage strength δ_π from reported statistics. Let $R^2 := \mathrm{Var}(\hat{z}_t)/\mathrm{Var}(z_t)$ be the R^2 of the orthogonalisation regression, with ρ and $\hat{\rho}$ as in equation (14). The fitted value and the residual of the regression are orthogonal, and hence $\mathrm{Var}(\hat{z}_t) = R^2\ \mathrm{Var}(z_t)$ and $\mathrm{Var}(z_t^\perp) = (1 - R^2)\,\mathrm{Var}(z_t)$.

The loss of first-stage strength. Equation (14) of the main text is

$$\delta_\pi = \sqrt{R^2}\,\frac{\hat{\rho}}{\rho}. \tag{14}$$

Derivation. By Proposition 1(i), $\delta_\pi = \operatorname{Cov}(\hat{z}_t, u_t^p)/\operatorname{Cov}(z_t, u_t^p)$. Writing each covariance as a correlation times the two standard deviations,

$$\delta_\pi = \frac{\hat{\rho}\,\sqrt{\operatorname{Var}(\hat{z}_t)}\,\sqrt{\operatorname{Var}(u_t^p)}}{\rho\,\sqrt{\operatorname{Var}(z_t)}\,\sqrt{\operatorname{Var}(u_t^p)}} = \frac{\hat{\rho}}{\rho}\sqrt{\frac{\operatorname{Var}(\hat{z}_t)}{\operatorname{Var}(z_t)}} = \sqrt{R^2}\,\frac{\hat{\rho}}{\rho}.$$

The loss is the size of the predictable part, $\sqrt{R^2}$, times how closely it moves with the policy rate relative to the instrument as a whole, $\hat{\rho}/\rho$.

Decomposing the loss of first-stage strength. Only the part of the predictable component in the period's shocks, $\hat{z}_t^c$ of Appendix A, covaries with u_t^p. With $s_c := \operatorname{Var}(\hat{z}_t^c)/\operatorname{Var}(\hat{z}_t)$ and $\hat{\rho}_c := \operatorname{Corr}(\hat{z}_t^c, u_t^p)$,

$$\delta_\pi = \sqrt{s_c\,R^2}\,\frac{\hat{\rho}_c}{\rho}. \tag{31}$$

Derivation. The rest of the predictable component, $\hat{z}_t - \hat{z}_t^c = \boldsymbol{c}'\boldsymbol{\zeta}_t$, is uncorrelated with u_t^p. Using the variance of $\hat{z}_t^c$, which is s_c times that of $\hat{z}_t$,

$$\delta_\pi = \frac{\operatorname{Cov}(\hat{z}_t^c, u_t^p)}{\operatorname{Cov}(z_t, u_t^p)} = \frac{\hat{\rho}_c\sqrt{s_c R^2\,\operatorname{Var}(z_t)\operatorname{Var}(u_t^p)}}{\rho\sqrt{\operatorname{Var}(z_t)\operatorname{Var}(u_t^p)}} = \sqrt{s_c\,R^2}\,\frac{\hat{\rho}_c}{\rho}.$$

Predictability weakens the first-stage strength only through the part in the period's shocks, and only to the extent that this part moves with the policy rate.

The correlation after orthogonalisation. Let $\rho^\perp := \operatorname{Corr}(z_t^\perp, u_t^p)$ be the correlation of the orthogonalised instrument with the policy-rate innovation. It is related to the raw correlation by

$$\rho^\perp = \frac{1-\delta_\pi}{\sqrt{1-R^2}}\,\rho. \tag{32}$$

Derivation. By Proposition 1(i), the covariance of the orthogonalised instrument with u_t^p is $1-\delta_\pi$ times that of the raw instrument. Dividing by the standard deviations, and using the variance of $z_t^\perp$ above,

$$\rho^\perp = \frac{(1-\delta_\pi)\operatorname{Cov}(z_t, u_t^p)}{\sqrt{1-R^2}\,\sqrt{\operatorname{Var}(z_t)\operatorname{Var}(u_t^p)}} = \frac{1-\delta_\pi}{\sqrt{1-R^2}}\,\rho.$$

When strength falls. The correlation falls if and only if

$$\rho^{\perp} < \rho \iff \delta_{\pi} > 1 - \sqrt{1 - R^2} \approx \tfrac{1}{2} R^2. \tag{33}$$

Derivation. Since $\rho > 0$, equation (32) gives $\rho^{\perp} < \rho$ if and only if $1 - \delta_{\pi} < \sqrt{1 - R^2}$. The approximation uses $\sqrt{1 - R^2} \approx 1 - \frac{1}{2} R^2$. The covariance falls one for one with δ_{π}, and the standard deviation only with $\sqrt{1 - R^2}$. Orthogonalising predictable noise, $\delta_{\pi} = 0$, would raise strength by $1/\sqrt{1 - R^2}$.

From the first-stage F to the loss of strength. The ratio of the orthogonalised to the raw first-stage F is approximately

$$\frac{F^{\perp}}{F} \approx \frac{(1 - \delta_{\pi})^2}{1 - R^2}.$$

Derivation. For a single instrument, $F = T\rho^2/(1 - \rho^2)$, exactly for the homoskedastic statistic. Hence $\rho = \sqrt{F/(F + T)}$, and, when both correlations are small, $F \approx T\rho^2$ and $F^{\perp}/F \approx (\rho^{\perp}/\rho)^2$, which equation (32) turns into the ratio above.

The rule of thumb. Three reported numbers give δ_{π}: the raw and orthogonalised first-stage F and the R^2 of the orthogonalisation regression. The correlations follow from the F, equation (32) gives $\delta_{\pi} = 1 - \sqrt{1 - R^2}\, \rho^{\perp}/\rho$, equation (14) gives $\hat{\rho}/\rho = \delta_{\pi}/\sqrt{R^2}$, and equation (11) gives the magnification bound $1/(1 - \delta_{\pi})$. Table 1 reports the results, with $T = 383$ and the R^2 of Appendix Table 3. Each raw wider-window instrument is re-estimated on the window of its orthogonalised version. For the high-frequency surprise, $\delta_{\pi} \approx 0.54$, six times the threshold $1 - \sqrt{1 - R^2} \approx 0.09$ in equation (33), and the bias that survives can be magnified up to 2.2 times. For the Cholesky-based instrument, the orthogonalised F also reflects that the orthogonalised series is zero in months without an FOMC announcement.

C The within estimator: derivations

C.1 The pooled instrument

In the data Ψ is unobserved, and the fitted direction supplies a stand-in. Each whitened level moment splits into its part along the fitted direction and the rest,

$$\hat{G}^{w} = \hat{q}\,\hat{\phi}' + \hat{R}, \qquad \hat{\phi}_j := \hat{q}'\hat{g}_j^{w}, \qquad \hat{R} := (I - \hat{q}\hat{q}')\hat{G}^{w}. \tag{34}$$

$\hat{\phi}$ collects the level relevances, and $\hat{R}$ what the moments contain beyond the target direction. At the population moments $R = \mathcal{Q}_{-p}\Psi'$ by equation (20). This is the contamination alone. Therefore, $\ker R = \ker \Psi'$. The kernel of $\hat{R}$ therefore estimates the clean combinations, and the feasible weights are

$$\hat{w} := P_{\ker \hat{R}}\, \hat{\phi}, \tag{35}$$

scaled such that $\hat{w}'\hat{\phi} = 1$. The pooled instrument is $z_t^{\star} := \sum_j \hat{w}_j z_{jt}$. Because $\hat{w}$ is orthogonal to the residual loadings, its proxy-SVAR estimand is exactly the within direction, $\hat{G}^{w}\hat{w} = (\hat{\phi}'\hat{w})\,\hat{q}$. The pooled instrument therefore gives the same impulse responses as the within estimator. It adds an instrument series and its first stage.[14] Because $\hat{R}$ is built from $\hat{q}$, it also inherits any error in the

[14] Proposition C.3 in Appendix C.8 states the result, including the equality in every bootstrap draw.

fitted direction.

C.2 Summary of the algorithm

The algorithm below is the sequence of computations run in Section 6, with sample quantities in place of population ones. Steps 1–3 estimate the direction, steps 4–5 build the pooled instrument, and steps 6–7 give the impulse responses and their uncertainty.

Algorithm 1 (Implementation of the within estimator)**.** Inputs: the VAR residuals $\hat{u}_t$ and $\hat{\Sigma}_u = \hat{L}\hat{L}'$ (estimated on the VAR window); instruments z_{jt}, each observed on its own support $S_j \subseteq \{1, \dots, T\}$ with $T_j := |S_j|$ observations, in the units in which block structure is asserted;[15] the partition $c(\cdot)$ of Section 5.5.

1. *Moments.* $\hat{g}_j = T_j^{-1} \sum_{t \in S_j} \hat{u}_t z_{jt}$, $j = 1, \dots, m$.
2. *Whiten and demean.* $\hat{g}_j^w = \hat{L}^{-1} \hat{g}_j$; with weights $\omega_j > 0$, $\tilde{\hat{g}}_j^w = \hat{g}_j^w - \sum_{k \in c} \omega_k \hat{g}_k^w / \sum_{k \in c} \omega_k$ for $j \in c$ (any weights remove d_c exactly); $\widetilde{G}^w = [\tilde{\hat{g}}_1^w, \dots, \tilde{\hat{g}}_m^w]$.
3. *Direction.* $\hat{q} = \arg\max_{\|q\|=1} q' \widetilde{G}^w \omega \widetilde{G}^{w\prime} q$, $\omega = \operatorname{diag}(\omega_j)$, sign fixed by $s_p' \hat{L} \hat{q} > 0$; $\hat{b}_p = \hat{L}\hat{q}$, and $\hat{\beta} = \hat{b}_p / (s_p' \hat{b}_p)$. Equal weights give the estimator of Section 5.3; the application uses precision weights $\omega_j = 1/\hat{v}_j$, with $\hat{v}_j$ the sampling variance of $\hat{g}_j^w$.[16]
4. *Loadings and residuals.* $\hat{\tilde{\phi}}_j = \hat{q}' \tilde{\hat{g}}_j^w$ (relevance deviations), $\hat{\phi}_j = \hat{q}' \hat{g}_j^w$ (level relevances), whitened within residuals $\hat{r}_j = \tilde{\hat{g}}_j^w - \hat{\tilde{\phi}}_j \hat{q}$, residual loadings $\hat{R} = \hat{G}^w - \hat{q}\hat{\phi}'$ as in equation (34), family-average contaminations $\hat{d}_c^w$ from the level residuals.
5. *Pooled instrument.* $\hat{w} = P_{\ker \hat{R}}\, \hat{\phi}$, equation (35) (or its precision-weighted form in equation (43)), scaled to $\hat{w}'\hat{\phi} = 1$; $z_t^\star = \sum_j \hat{w}_j \, (T/T_j) \, \tilde{z}_{jt} \mathbf{1}\{t \in S_j\}$, with $\tilde{z}_{jt}$ instrument j demeaned on its support and T the length of the moment window, such that $T^{-1} \sum_t \hat{u}_t z_t^\star = \hat{G}\hat{w}$ exactly and Proposition C.3 holds for the series itself;[17] its first-stage F is the reported strength.
6. *Impulse responses.* $\hat{\Theta}_h = \hat{\Phi}_h \hat{\beta}$; the shock series from the proxy-SVAR step with $z_t^\star$.
7. *Uncertainty.* The two-component variance and the level set for the direction; the moving-block bootstrap re-running steps 1–6, the weights included, on each draw, with the partition held fixed and the idiosyncratic contamination redrawn (Online Appendix OA.2).

The algorithm is not iterative. The pooled instrument of step 5 reproduces the direction of step 3 exactly (Proposition C.3). Hence, it does not alter it. The partition is the only input beyond the data, and the bootstrap holds it fixed. Online Appendix OA.2 gives the weak-identification-robust set for the direction, and Appendix C.6 the diagnostics reported with every fit.

[15] Exact block structure holds in the instruments' own units. Rescaling one instrument within a family changes its loading. Hence, the family's contamination is no longer common and the within transform no longer removes it. Standardising the instruments by their own standard deviations would therefore bias the estimator. The precision weights of step 3 make $\hat{b}_p$ invariant to rescaling a whole family, not one instrument within it.

[16] The variance $\hat{v}_j$ is computed under the residualised score (Online Appendix OA.2). With idiosyncratic contamination the efficient weights are $\omega_j = 1/(\hat{v}_j + \hat{\sigma}_v^2)$.

[17] With equal supports this is $\hat{w}' z_t$. Restricting to the common support would discard months and break this identity.

C.3 Clean strength

Section 5.6 states the decomposition of clean strength in words. Under exact block structure, the loading of a combination is $\Psi' w = \sum_c a_c \bar{\boldsymbol{\psi}}_c$, which depends on the weights only through their family totals a_c.

Proposition C.1 (Clean strength). *Under exact block structure the clean weight vectors* $\ker(\Psi')$ *are the within-family contrasts (family totals* $a_c = 0$*), of dimension* $m - C$*, plus the family-constant vectors whose totals cancel the family-average contaminations (*$\sum_c a_c \bar{\boldsymbol{\psi}}_c = 0$*), a space* $\mathcal{K}$ *of dimension* $C - r$*. With homoskedastic noise, clean strength, the concentration* $\mu^2(w^\star) = T\|P_\perp \phi\|^2/\sigma_\omega^2$ *of the best clean instrument, decomposes as*

$$\|P_\perp \phi\|^2 = \underbrace{\sigma_\phi^2}_{\text{contrasts: available for every } C} + \underbrace{\|P_{\mathcal{K}}\, \phi\|^2}_{\text{levels: generically nonzero iff } C>r}, \tag{36}$$

and $T\sigma_\phi^2/\sigma_\omega^2$ *is the concentration of the pooled contrast instrument* $\sum_c \sum_{j\in c} \tilde{\phi}_j z_{jt}$*, the strongest combination of within-family contrasts.*

Details are in Appendix C.4.

C.4 Proofs of the propositions of Section 5

Lemma C.1 (Observational equivalence). *Hold* B_{-p} *fixed, let* $B = [\, b_p \;\; B_{-p} \,]$ *be nonsingular and* $\phi \neq \mathbf{0}$*. For every* $\alpha \in \mathbb{R}^{n-1}$*, the pair* $(b_p + B_{-p}\alpha,\ \Psi - \phi\alpha')$ *generates the same* G *as* (b_p, Ψ) *in equation (16). Conversely, if* $(\tilde{b}_p, \tilde{\phi}, \tilde{\Psi})$ *generates the same* G*, then for some* $\kappa \neq 0$ *and* $\alpha \in \mathbb{R}^{n-1}$

$$\tilde{b}_p = \kappa\,(b_p + B_{-p}\alpha), \qquad \tilde{\phi} = \phi/\kappa, \qquad \tilde{\Psi} = \Psi - \phi\,\alpha'.$$

Hence b_p *is identified up to scale if and only if the maintained restriction on the contamination excludes* $\Psi - \phi\alpha'$ *for every* $\alpha \neq \mathbf{0}$*. With the contamination unrestricted it is never identified.*

Proof. The first part follows by substitution, $(b_p + B_{-p}\alpha)\phi' + B_{-p}(\Psi - \phi\alpha')' = b_p\phi' + B_{-p}\Psi' = G$. For the converse, because B is nonsingular, $\tilde{b}_p = \kappa\, b_p + B_{-p}a$ for unique $\kappa \in \mathbb{R}$ and $a \in \mathbb{R}^{n-1}$. Equating the two expressions for G gives $b_p\,(\kappa\tilde{\phi} - \phi)' + B_{-p}\big(a\tilde{\phi}' + \tilde{\Psi}' - \Psi'\big) = \mathbf{0}$. Both brackets therefore vanish. Hence $\phi = \kappa\tilde{\phi}$, which implies $\kappa \neq 0$, and with $\alpha := a/\kappa$, $\tilde{\Psi} = \Psi - \phi\alpha'$. If the restriction excludes every such $\Psi - \phi\alpha'$ with $\alpha \neq \mathbf{0}$, the only admissible alternatives are $\kappa\, b_p$. □

The relevance-proportional shift moves the target by $B_{-p}\alpha$ and compensates by removing $\phi_j\alpha$ from the loading of instrument j. The more relevant the instrument, the more of the shift its loading absorbs. A restriction on within-family averages of the deviations is left intact by this change, because the relevance deviations sum to zero within a family, which is why Assumption 1 conditions on the relevances.

Proof of Proposition 3 (identification). The relevance-proportional shift in the family model. By Lemma C.1, the relevance-proportional shift $b_p \mapsto b_p + B_{-p}\alpha$, $\boldsymbol{\psi}_j \mapsto \boldsymbol{\psi}_j - \phi_j\alpha$ leaves G unchanged. Writing $\phi_j = \bar{\phi}_c + \tilde{\phi}_j$, it splits into a change in the family mean and a change in the deviation:

$$\bar{\boldsymbol{\psi}}_c \mapsto \bar{\boldsymbol{\psi}}_c - \bar{\phi}_c\,\alpha, \qquad \boldsymbol{\nu}_j \mapsto \boldsymbol{\nu}_j - \tilde{\phi}_j\,\alpha.$$

The first is absorbed by d_c. Whether b_p is identified depends on whether the restriction on the deviations rules out the second.

(i) Since $\tilde{\phi}_j$ is a function of $\boldsymbol{\phi}_c$, under equation (17)

$$\mathbb{E}\left[\boldsymbol{\nu}_j - \tilde{\phi}_j \alpha \mid \boldsymbol{\phi}_c, c\right] = -\tilde{\phi}_j \, \alpha.$$

By Assumption C.1(i) some $\tilde{\phi}_j \neq 0$. Hence, the shifted deviations satisfy equation (17) only at $\alpha = \mathbf{0}$. By Lemma C.1, every alternative that generates G is such a relevance-proportional shift up to scale; rescaling ϕ leaves the loadings, and hence the restriction, unchanged. The only admissible alternatives are therefore $\kappa \, b_p$, and b_p is identified up to scale.

(ii) The family average of the shifted deviations is

$$\frac{1}{|c|} \sum_{j \in c} \left(\boldsymbol{\nu}_j - \tilde{\phi}_j \alpha\right) = \bar{\boldsymbol{\nu}}_c,$$

because the relevance deviations $\tilde{\phi}_j$ sum to zero within the family. The restriction $\mathbb{E}[\bar{\boldsymbol{\nu}}_c \mid c] = \mathbf{0}$ therefore holds for every α and does not pin down the target.

(iii) Let $\boldsymbol{\nu}_j = \alpha_0 \tilde{\phi}_j + \boldsymbol{\eta}_j$ and apply the relevance-proportional shift with $\alpha = \alpha_0$. The shifted loadings are

$$\boldsymbol{\psi}_j - \phi_j \alpha_0 = \left(\bar{\boldsymbol{\psi}}_c - \bar{\phi}_c \alpha_0\right) + \boldsymbol{\eta}_j,$$

a family mean plus a deviation $\boldsymbol{\eta}_j$ that satisfies equation (17). The pair $(b_p + B_{-p}\alpha_0, \{\boldsymbol{\psi}_j - \phi_j \alpha_0\})$ thus satisfies Assumption 1 and generates the same G; by part (i) applied to this pair, $b_p + B_{-p}\alpha_0$ is the only target, up to scale, consistent with Assumption 1, and no estimator can distinguish it from b_p. □

Proof of Proposition 4 (the two order conditions). (O1) Under exact block structure, G supplies nm moments. The unknowns are the direction ($n-1$ after normalisation), the m relevances, and the C family contaminations $d_c \in \text{range}(B_{-p})$, with $n-1$ entries each. The number of overidentifying restrictions is

$$nm - (n-1) - m - (n-1)C = (n-1)(m - C - 1), \tag{37}$$

which is nonnegative iff $m - C \geq 1$; it is also the degrees of freedom of the overidentification test (Online Appendix OA.2). The count is necessary but not sufficient. The within transform leaves $m - C$ contrasts, and they carry the target only if $\sigma^2_\phi > 0$ (Proposition C.2(i)). Some family must contain two instruments of unequal relevance.

(O2) A family-constant combination with family totals $a = (a_1, \ldots, a_C)'$ has contamination $\sum_c a_c \bar{\boldsymbol{\psi}}_c$ (Section 5.6). It is clean iff $a \in \ker[\bar{\boldsymbol{\psi}}_1, \ldots, \bar{\boldsymbol{\psi}}_C]$, a space of dimension $C - r$, which contains a nonzero vector iff $C > r$. Since $r \leq n-1$, $C > n-1$ is sufficient. The strength statement is Proposition C.1. □

Proof of Proposition C.1 (the decomposition). Split any weight vector into a family-constant part with the same family totals and a remainder,

$$w = w_L + w_0, \qquad (w_L)_j = a_c/|c| \;\text{ for } j \in c,$$

such that w_0 sums to zero within every family. The two parts are orthogonal. Under exact block structure, $\Psi' w = \sum_c a_c \bar{\boldsymbol{\psi}}_c$ (Section 5.6), which depends on w only through the totals. Every contrast w_0 is therefore clean, and w_L is clean iff its totals lie in $\ker[\bar{\boldsymbol{\psi}}_1, \ldots, \bar{\boldsymbol{\psi}}_C]$, which defines $\mathcal{K}$, of dimension $C - r$. Hence $\ker(\Psi')$ is the orthogonal sum of the contrasts and $\mathcal{K}$, and

$$\|P_\perp \boldsymbol{\phi}\|^2 = \|P_{\text{contrasts}}\, \boldsymbol{\phi}\|^2 + \|P_{\mathcal{K}}\, \boldsymbol{\phi}\|^2.$$

The projection of $\boldsymbol{\phi}$ on the contrasts has entries $\phi_j - \bar{\phi}_c$. Hence, its squared norm is σ^2_ϕ. The level term is nonzero for generic $\boldsymbol{\phi}$ iff $\dim \mathcal{K} = C - r > 0$. The concentration of the pooled contrast instrument is derived in Appendix C.8. □

C.5 Recovery by the leading eigenvector

This appendix states formally when the leading eigenvector recovers the target, as summarised in Section 5.4. Whether it does depends on how large the idiosyncratic contamination is relative to the within-family dispersion of relevance. The answer needs one further assumption.

Assumption C.1 (Relevance dispersion and regularity)**.** (i) $\sigma^2_\phi > 0$, *with* σ^2_ϕ *the within-family dispersion of relevance equation* (19). (ii) *Under Assumption 1, conditionally on the relevances the* $\boldsymbol{\nu}_j$ *are independent across instruments with a common conditional variance* $\Sigma_\nu := \mathrm{Var}(\boldsymbol{\nu}_j \mid \boldsymbol{\phi}_c, c)$*, the same in every family, and finite fourth moments.*

Part (i) requires relevance to vary within at least one family. Part (ii) matters only when the contamination has an idiosyncratic part. The tilt formula of Proposition C.2(iii) holds for any realised deviations. Its average over the deviations, the large-panel limit of Proposition C.2(ii), and inference need the deviations to be independent across instruments with a common variance. Finite fourth moments make the contamination spread, a sum of squared deviations, settle down as instruments are added, and make its variability estimable for inference.

The estimator computes the scatter of the whitened within-demeaned moments. At the population moments it is $S_0 := \widetilde{G}^w \widetilde{G}^{w\prime}$, with $\tilde{g}^w_j = \tilde{\phi}_j q_p + \mathcal{Q}_{-p} \tilde{\boldsymbol{\nu}}_j$ and $\widetilde{G}^w = [\tilde{g}^w_1, \ldots, \tilde{g}^w_m]$. S_0 is what the estimator would compute with an unlimited time sample on the same instruments. The realised deviations $\boldsymbol{\nu}_j$ of the m instruments remain. Two quantities summarise them: the cross-moment between relevance and idiosyncratic contamination, $\boldsymbol{\zeta} := \sum_j \tilde{\phi}_j \boldsymbol{\nu}_j$, and the spread of the idiosyncratic contamination, or contamination spread, $N := \sum_j \tilde{\boldsymbol{\nu}}_j \tilde{\boldsymbol{\nu}}'_j$.

The cross-moment $\boldsymbol{\zeta}$ measures how far, in this particular panel, the idiosyncratic contamination happens to line up with the relevance deviations. Condition (b) rules this out on average, but not in every realised panel. It tilts the target axis. The contamination spread N measures how widely the demeaned contamination is scattered in each direction. It does not tilt the target, but it creates rival axes that can replace it as the leading one. Without idiosyncratic contamination both are zero, and the scatter has a single axis, the target. Otherwise $\boldsymbol{\zeta}$ tilts that axis and N adds rival axes. Proposition C.2 asks whether the leading eigenvector of S_0 is still q_p, the whitened target.

Proposition C.2 (Recovery by the leading eigenvector)**.** *Let* $S_0 := \widetilde{G}^w \widetilde{G}^{w\prime}$ *at the population moments.*

(i) (Exact block structure.) *If* $\boldsymbol{\nu}_j \equiv \mathbf{0}$*, the population within scatter is* $\widetilde{G}\widetilde{G}' = \sigma^2_\phi\, b_p b'_p$*, of rank one. Therefore, whenever* $\sigma^2_\phi > 0$ *the leading eigenvector of* $\widetilde{G}\widetilde{G}'W$ *is* b_p *for any* $W \succ 0$*; in the* Σ_u^{-1} *metric* $S_0 = \sigma^2_\phi\, q_p q'_p$*,* q_p *is its unique leading eigenvector, and* $b_p = \Sigma_u^{1/2} q_p$*.*

(ii) (Idiosyncratic contamination, $m \to \infty$.) *Under Assumptions 1 and C.1(ii), as $m \to \infty$ with $\sigma_\phi^2/m \to \bar{\sigma}_\phi^2$ and $(m-C)/m \to \bar{\kappa} \in (0,1]$, $m^{-1}S_0 \to \bar{\sigma}_\phi^2 q_p q_p' + \bar{\kappa}\, \mathcal{Q}_{-p}\Sigma_\nu \mathcal{Q}_{-p}'$, and q_p is its unique leading eigenvector iff*

$$\bar{\sigma}_\phi^2 > \bar{\kappa}\, \lambda_{\max}(\Sigma_\nu) \qquad \textit{(the ordering condition)}, \tag{38}$$

in which case the leading eigenvector of S_0 converges to q_p.

(iii) (Idiosyncratic contamination, finite m.) *For any realised $\{\boldsymbol{\nu}_j\}$, identically,*

$$S_0 = \sigma_\phi^2\, q_p q_p' + q_p(\mathcal{Q}_{-p}\boldsymbol{\zeta})' + (\mathcal{Q}_{-p}\boldsymbol{\zeta})\, q_p' + \mathcal{Q}_{-p} N \mathcal{Q}_{-p}', \tag{39}$$

and, if $\sigma_\phi^2 > \lambda_{\max}(N)$ and $\|\boldsymbol{\zeta}\|$ is small relative to the gap ς below, its leading eigenvector is

$$q_p + \mathcal{Q}_{-p}\big(\sigma_\phi^2 I - N\big)^{-1}\boldsymbol{\zeta} + O\big(\|\boldsymbol{\zeta}\|^2/\varsigma^2\big), \qquad \varsigma := \sigma_\phi^2 - \lambda_{\max}(N) > 0, \tag{40}$$

the target plus a tilt that vanishes when $\boldsymbol{\zeta} = \mathbf{0}$. Under Assumptions 1 and C.1(ii), the tilt has mean zero over the draw of the $\{\boldsymbol{\nu}_j\}$ to first order (exactly if their distribution is symmetric), and is of order $\|\Sigma_\nu\|^{1/2}/\sigma_\phi$, hence $O(m^{-1/2})$ when σ_ϕ^2 grows in proportion to m.

The leading eigenvector recovers the target when, within families, relevance varies more than the idiosyncratic contamination. Contamination that happens to line up with relevance tilts the estimate. Contamination that is widely spread can replace the target as the leading axis. Details are in Appendix C.4. Part (i) is the benchmark. With no idiosyncratic contamination, every demeaned moment lies on the line through b_p. Identification then fails only if no family contains two instruments of unequal relevance.

Part (ii) adds instruments. Here $\bar{\kappa}$ is the limit of $(m-C)/m$, the fraction of instruments that still carry information after demeaning, since each family uses one of its instruments to estimate its mean. As m grows, $\boldsymbol{\zeta}/m \to \mathbf{0}$ while $N/m \to \bar{\kappa}\Sigma_\nu$. The tilt vanishes, and q_p is an exact eigenvector of the scatter in the limit. It is the leading eigenvector, and a unique one, if and only if the ordering condition in equation (38) holds. Within families, the dispersion of relevance must exceed the dispersion of idiosyncratic contamination for the eigenvector to recover the target.

When the condition fails, the estimator returns a contamination direction, however long the sample. The fit of the estimator cannot reveal it (Lemma C.2), but the signs of the estimated relevances can (Appendix C.6).

Part (iii) is the case that applies in practice, with a finite number of instruments. The tilt is zero only if, in this panel, the contamination does not line up with the relevance deviations. It is zero on average, shrinks as instruments are added, and is larger when relevance dispersion barely exceeds the contamination spread. Its direction is random and would average out over many panels. It therefore adds to the uncertainty of the estimate rather than biasing it. When the contamination spread is small relative to σ_ϕ^2, the tilt is approximately $\sigma_\phi^{-2}\mathcal{Q}_{-p}\boldsymbol{\zeta}$. It enters the uncertainty of the direction as a second variance component (Online Appendix OA.2). The contamination spread N does not tilt the target but adds rival directions, and the target leads only if $\sigma_\phi^2 > \lambda_{\max}(N)$. This is the ordering condition applied to the realised panel. In small panels it is harder to meet, because the largest eigenvalue of a realised spread tends to exceed its population counterpart.

Proof of Proposition C.2 (recovery by the leading eigenvector). Setup. Work in the orthonormal basis

$\mathcal{Q} = [\, q_p \;\; \mathcal{Q}_{-p} \,]$ of equation (20), in which the target is the first axis and the contamination space the remaining $n-1$. The whitened demeaned moments are $\tilde{g}_j^w = \tilde{\phi}_j\, q_p + \mathcal{Q}_{-p}\tilde{\nu}_j$. Two sums appear:

$$\sum_j \tilde{\phi}_j^2 = \sigma_\phi^2, \qquad \sum_j \tilde{\phi}_j \tilde{\nu}_j = \sum_j \tilde{\phi}_j \nu_j = \zeta,$$

where the second equality holds because the $\tilde{\phi}_j$ sum to zero within each family.

Expanding $S_0 = \sum_j \tilde{g}_j^w \tilde{g}_j^{w\prime}$ with the two sums above gives equation (39). In the basis $\mathcal{Q}$ it is a bordered matrix,

$$\mathcal{Q}' S_0\, \mathcal{Q} = \begin{pmatrix} \sigma_\phi^2 & \zeta' \\ \zeta & N \end{pmatrix},$$

in which the target axis carries σ_ϕ^2, the contamination space carries N, and ζ couples them. In particular q_p is an eigenvector iff $\zeta = \mathbf{0}$.

(i) Exact block structure. With $\nu_j \equiv \mathbf{0}$, $\tilde{g}_j = \tilde{\phi}_j b_p$. Hence, $\widetilde{G}\widetilde{G}' = \sigma_\phi^2\, b_p b_p'$. For any $W \succ 0$, $\widetilde{G}\widetilde{G}' W b = \sigma_\phi^2 (b_p' W b)\, b_p$, whose only eigenvector with a nonzero eigenvalue is b_p. In whitened coordinates $S_0 = \sigma_\phi^2\, q_p q_p'$.

(ii) Idiosyncratic contamination, $m \to \infty$. Divide the bordered matrix by m. The corner converges to $\bar{\sigma}_\phi^2$. The border vanishes. The vector ζ has mean zero and variance $\sigma_\phi^2 \Sigma_\nu$. Hence, $\zeta/m = O_p(m^{-1/2})$. For the lower block, $\mathbb{E}[\tilde{\nu}_j \tilde{\nu}_j' \mid c] = (1 - |c|^{-1})\Sigma_\nu$, and summing over j,

$$\frac{1}{m}\,\mathbb{E}[N] = \frac{1}{m}\sum_c (|c| - 1)\,\Sigma_\nu \;\longrightarrow\; \bar{\kappa}\,\Sigma_\nu,$$

with N/m converging to the same limit by the law of large numbers across instruments. Hence

$$\frac{1}{m}\,\mathcal{Q}' S_0\, \mathcal{Q} \;\longrightarrow\; \begin{pmatrix} \bar{\sigma}_\phi^2 & \mathbf{0}' \\ \mathbf{0} & \bar{\kappa}\,\Sigma_\nu \end{pmatrix}.$$

The limit is block diagonal, with eigenvalue $\bar{\sigma}_\phi^2$ on the target axis and $\bar{\kappa}\lambda_k(\Sigma_\nu)$ in the contamination space. The target is its unique leading eigenvector iff $\bar{\sigma}_\phi^2 > \bar{\kappa}\lambda_{\max}(\Sigma_\nu)$, which is equation (38). The leading eigenvector of $m^{-1}S_0$ then converges to q_p, because an eigenvector is continuous in the matrix at a simple eigenvalue.

(iii) Idiosyncratic contamination, finite m. Let λ_1 be the largest eigenvalue. Evaluating the Rayleigh quotient at the first axis gives $\lambda_1 \geq \sigma_\phi^2 > \lambda_{\max}(N)$. An eigenvector $(x_0, \boldsymbol{x})$ with eigenvalue λ_1 must have $x_0 \neq 0$. Otherwise $N\boldsymbol{x} = \lambda_1 \boldsymbol{x}$, which is impossible since $\lambda_1 > \lambda_{\max}(N)$. Normalising $x_0 = 1$, the second block row reads $\zeta + N\boldsymbol{x} = \lambda_1 \boldsymbol{x}$. Hence,

$$\boldsymbol{x} = (\lambda_1 I - N)^{-1}\zeta, \qquad \lambda_1 = \sigma_\phi^2 + \zeta'(\lambda_1 I - N)^{-1}\zeta.$$

The solution is unique. Hence, λ_1 is simple, and the leading eigenvector is, exactly,

$$q_p + \mathcal{Q}_{-p}(\lambda_1 I - N)^{-1}\zeta.$$

The second equation gives $0 \leq \lambda_1 - \sigma_\phi^2 \leq \|\zeta\|^2/\varsigma$. Replacing λ_1 by σ_ϕ^2, and normalising the

eigenvector to unit length, changes it by $O(\|\zeta\|^2/\varsigma^2)$, which gives equation (40).

Mean and order of the tilt. Expanding $(\sigma_\phi^2 I - N)^{-1} = \sigma_\phi^{-2} I + O(\|N\|/\sigma_\phi^4)$, the tilt is $\sigma_\phi^{-2}\mathcal{Q}_{-p}\zeta$ to first order in the deviations, and $\mathbb{E}[\zeta \mid \{\phi_j\}] = \mathbf{0}$ by equation (17). If the distribution of the ν_j is symmetric, the mean is exactly zero. The map $\nu \mapsto -\nu$ flips ζ, leaves N unchanged, and so flips the tilt. By independence and the common variance, $\mathrm{Var}(\zeta \mid \{\phi_j\}) = \sum_j \tilde{\phi}_j^2 \Sigma_\nu = \sigma_\phi^2 \Sigma_\nu$. Hence, $\|\zeta\| = O_p(\sigma_\phi \|\Sigma_\nu\|^{1/2})$. With $\|(\sigma_\phi^2 I - N)^{-1}\| \le \varsigma^{-1}$ and ς of the order of σ_ϕ^2, the tilt is $O_p(\|\Sigma_\nu\|^{1/2}/\sigma_\phi)$, which is $O(m^{-1/2})$ when σ_ϕ^2 grows in proportion to m. □

C.6 Diagnostics

The diagnostics below are computed from the fit of Algorithm 1 and reported with every estimate.

The ordering condition. The estimator labels the leading axis of the within scatter as the target. Hence, the estimate is the policy shock only if the ordering condition in equation (38) holds. The condition cannot be tested against the residual scatter of the fit.

Lemma C.2 (The residual test of the ordering condition cannot fail)**.** *Let $\hat{\lambda}_1 \ge \hat{\lambda}_2$ be the two largest eigenvalues of the within scatter $\hat{S}$, and $\hat{N}$ the scatter of the within residuals at $\hat{q}$. Then $\hat{\sigma}_\phi^2 = \hat{\lambda}_1 \ge \hat{\lambda}_2 = \lambda_{\max}(\hat{N})$ in every sample, and $\hat{\sigma}_\phi^2/m > \hat{\kappa}\,\lambda_{\max}(\hat{\Sigma}_\nu)$, with $\hat{\kappa} = (m-C)/m$, fails only if $\hat{\lambda}_1/\hat{\lambda}_2 \le 1 + 1/(m-C-1)$.*

Proof. $\hat{\sigma}_\phi^2 = \sum_j (\hat{q}'\tilde{g}_j^w)^2 = \hat{q}'\hat{S}\hat{q} = \hat{\lambda}_1$. The residuals are $P_{\hat{q}}^{\perp}\tilde{g}_j^w$. Hence, $\hat{N} = P_{\hat{q}}^{\perp}\hat{S}P_{\hat{q}}^{\perp}$, with eigenvalues $0, \hat{\lambda}_2, \ldots, \hat{\lambda}_n$. On the range of $P_{\hat{q}}^{\perp}$ the eigenvalues of the numerator of equation (OA.5) are at most those of $\hat{N}$. Hence, $\lambda_{\max}(\hat{\Sigma}_\nu) \le \hat{\lambda}_2/(m-C-1)$ with equal weights. □

The leading axis is always labelled the target. Hence, a failure of the condition does not show in the residuals. It shows in the level relevances $\hat{\phi}_j = \hat{q}'\hat{g}_j^w$. On target, $\Pr(\hat{\phi}_j < 0) \approx \Phi(-\phi_j/\sqrt{v_j})$, and the number of negative level relevances is approximately Poisson-binomial with these probabilities, evaluated at $\hat{\phi}_j/\sqrt{\hat{v}_j}$. If instead the leading axis is a contamination direction a, $\hat{\phi}_j \approx a'\bar{\psi}_c$, with signs set by the family-average contaminations. An excess of negative level relevances is therefore evidence against the ordering condition; their absence is not proof of it, since the family-average contaminations may happen to share a sign. The eigengap $\hat{\lambda}_1/\hat{\lambda}_2$ and the level set of Online Appendix OA.2 describe how sharply the leading axis is determined, not whether it is the target.

Do the families agree? J_T splits into a within-family and a between-family part. Fit each family with at least three instruments on its own, with direction $\hat{q}_c$, and let J_c be equation (OA.4) for that family alone, on $(n-1)(|c|-2)$ degrees of freedom. Then

$$J_{\text{between}} := J_T - \sum_c J_c, \qquad (n-1)(C-1) \text{ degrees of freedom.}$$

The between part asks whether the families agree on the direction. It detects contamination correlated with relevance that differs across families; a component common to all families cannot be detected, since it is observationally equivalent to a shifted target (Proposition 3(iii)). The within part tests exact block structure inside each family. When it rejects, residuals aligned with the family's

own $\hat{d}_c$ point to differences in scale across its members (footnote 15), and residuals orthogonal to it point to idiosyncratic contamination, which $\hat{\Sigma}_\nu$ measures. Contamination that scales along the family's direction, $\boldsymbol{\psi}_j = s_j \bar{\boldsymbol{\psi}}_c$, is detected by the within part when s_j is unrelated to relevance. It is not detected when s_j is proportional to relevance, since the family is then exactly block-structured around a tilted direction of its own. Because the families' residuals are correlated through $\hat{V}$, the χ^2 reference is approximate for each part, though not for J_T itself. Regressing $\hat{r}_j$ on $\hat{\bar{\phi}}_j$ is not a test. The coefficient is zero at the fitted direction by construction.

C.7 What averaging removes, and what it leaves

Simple pooling (Section 5.6) shrinks the idiosyncratic contamination at rate $m^{-1/2}$ and leaves the common component untouched. Write each loading as a panel-wide mean plus an independent deviation, $\boldsymbol{\psi}_j = \bar{\boldsymbol{\psi}} + \boldsymbol{\xi}_j$. A combination $w'z_t$ then carries the contamination

$$\Psi' w = \Big(\sum_j w_j \Big) \bar{\boldsymbol{\psi}} + \sum_j w_j \boldsymbol{\xi}_j. \tag{41}$$

The first term enters with the sum of the weights, which is one for any average. The averaged instrument therefore identifies the direction of $b_p + B_{-p}\bar{\boldsymbol{\psi}}/\bar{\phi}$, with $\bar{\phi}$ the mean relevance. No number of instruments reduces this tilt.

The idiosyncratic component shrinks. With $\boldsymbol{\psi}_j = \bar{\boldsymbol{\psi}} + \boldsymbol{\xi}_j$ as in equation (41), the $\boldsymbol{\xi}_j$ independent, mean zero and of variance Σ_ψ. The equal-weight average, $w_j = 1/m$, has contamination

$$\Psi' w = \bar{\boldsymbol{\psi}} + \bar{\boldsymbol{\xi}}_m, \qquad \bar{\boldsymbol{\xi}}_m := \frac{1}{m} \sum_j \boldsymbol{\xi}_j,$$

and by independence

$$\mathbb{E}[\bar{\boldsymbol{\xi}}_m] = \mathbf{0}, \qquad \mathrm{Var}(\bar{\boldsymbol{\xi}}_m) = \frac{1}{m^2} \sum_j \mathrm{Var}(\boldsymbol{\xi}_j) = \frac{\Sigma_\psi}{m}.$$

The average of m independent deviations has a $1/m$ share of the variance of one of them. Chebyshev's inequality turns this into a rate: for every $M > 0$,

$$\Pr\left(\sqrt{m}\, \|\bar{\boldsymbol{\xi}}_m\| > M \right) \le \frac{m\ \mathbb{E}\, \|\bar{\boldsymbol{\xi}}_m\|^2}{M^2} = \frac{m\ \mathrm{tr}\, \mathrm{Var}(\bar{\boldsymbol{\xi}}_m)}{M^2} = \frac{\mathrm{tr}\, \Sigma_\psi}{M^2}.$$

The bound does not depend on m. Hence, $\bar{\boldsymbol{\xi}}_m = O_p(m^{-1/2})$. Only a finite second moment is needed.

The common component does not. With $\bar{\phi} := m^{-1}\sum_j \phi_j$ the mean relevance, the moment of the averaged instrument is

$$\frac{1}{m}\sum_j g_j = \bar{\phi}\, b_p + B_{-p}\bar{\boldsymbol{\psi}} + B_{-p}\bar{\boldsymbol{\xi}}_m = \bar{\phi}\Big(b_p + \frac{B_{-p}\bar{\boldsymbol{\psi}}}{\bar{\phi}}\Big) + O_p(m^{-1/2}). \tag{42}$$

Provided $\bar{\phi}$ stays bounded away from zero, the proxy-SVAR estimand of the averaged instrument converges to the normalised $b_p + B_{-p}\bar{\boldsymbol{\psi}}/\bar{\phi}$. This is the contaminated impact vector equation (6) of a single instrument carrying the panel's mean loading. Adding instruments measures the wrong target more precisely.

C.8 The constrained-optimal instrument

The solution. Standardise the noise, with $v := \Sigma_\omega^{1/2} w$, $a := \Sigma_\omega^{-1/2}\phi$ and $M := \Sigma_\omega^{-1/2}\Psi$. The program in equation (26) becomes $\max_v (v'a)^2/v'v$ subject to $M'v = \mathbf{0}$. Let $P := I - M(M'M)^{+}M'$ be the projection onto the orthogonal complement of the columns of M. A vector is feasible if and only if $Pv = v$. Hence, $v'a = v'Pa$. By Cauchy–Schwarz the maximum, $\|Pa\|^2$, is attained at $v^\star \propto Pa$, which is feasible since $M'P = \mathbf{0}$. Returning to $w = \Sigma_\omega^{-1/2} v$ and normalising $w'\phi = 1$,

$$w^\star = \frac{\Sigma_\omega^{-1}\phi^\perp}{\phi^{\perp\prime}\Sigma_\omega^{-1}\phi^\perp}, \qquad \phi^\perp := \phi - \Psi(\Psi'\Sigma_\omega^{-1}\Psi)^{+}\Psi'\Sigma_\omega^{-1}\phi, \qquad \mu^2(w^\star) = T\,\phi^{\perp\prime}\Sigma_\omega^{-1}\phi^\perp.$$

Here $\phi^\perp$ is the residual of ϕ on the columns of Ψ in the Σ_ω^{-1} metric. The Moore–Penrose inverse means no rank condition on Ψ is needed. With $\Sigma_\omega = \sigma_\omega^2 I$, $\phi^\perp = P_\perp\phi$, which gives equation (27). Since $\|Pa\|^2 = \|a\|^2 - \|(I-P)a\|^2$, the constraint costs the part of relevance collinear with the contamination.

Proof that concentrations add, and of the first term of Proposition C.1. (i) Without the constraint, $\Psi = \mathbf{0}$ above, and hence $P = I$. The maximum is then $T\|a\|^2 = T\phi'\Sigma_\omega^{-1}\phi$, attained at $w \propto \Sigma_\omega^{-1}\phi = w^{\mathrm{GLS}}$. With independent noise Σ_ω is diagonal. Hence,

$$\mu^2(w^{\mathrm{GLS}}) = T\sum_j \frac{\phi_j^2}{\sigma_{\omega j}^2} = \sum_j \mu_j^2.$$

Since each single instrument is the combination $w = e_j$, the pool is at least as strong as the strongest.

(ii) Because the $\tilde{\phi}_j$ sum to zero within each family, the pooled contrast instrument is the combination $w = \tilde{\phi}$. Its family totals are zero. Hence, it is clean under exact block structure, and

$$w'\phi = \sigma_\phi^2, \qquad \mathrm{Var}(w'\omega_t) = \sigma_\omega^2\sigma_\phi^2, \qquad \mu^2(\tilde{\phi}) = T\sigma_\phi^2/\sigma_\omega^2.$$

□

The feasible instrument. In the data Ψ is replaced by the residual loadings $\hat{R}$ of equation (34), with $\ker R = \ker \Psi'$ at the population moments:

$$\hat{w} \;\propto\; \Sigma_\omega^{-1}\hat{\phi}^\perp, \qquad \hat{\phi}^\perp := \hat{\phi} - \hat{R}'(\hat{R}\Sigma_\omega^{-1}\hat{R}')^{+}\hat{R}\Sigma_\omega^{-1}\hat{\phi}, \qquad \hat{w}'\hat{\phi} = 1. \tag{43}$$

With $\Sigma_\omega \propto I$ this is the Euclidean projection in equation (35); with $\Sigma_\omega \propto \mathrm{diag}(\hat{v}_j)$ it is the precision-weighted form of the application. Both satisfy $\hat{R}\hat{w} = \mathbf{0}$. Hence, by Proposition C.3(ii) they deliver the same impulse responses. They differ only in the shock series and the strength they report, and only the precision-weighted form makes the reported strength invariant to rescaling an instrument. The rank of $\hat{R}$ is at most $n-1$ by construction. Hence, $\ker \hat{R}$ has dimension at least $m-(n-1)$ for any partition.

The pooled instrument and the within estimator. The estimand of the pooled instrument, $\hat{G}\hat{w}$ normalised on the policy coordinate, coincides with the within direction.

Proposition C.3 (The pooled instrument identifies the impulse responses of the within estimator)**.** (i) Population. *Under exact block structure, let w be any clean combination, $\Psi' w = 0$, with $w'\phi \neq 0$; in particular $w = w^\star$. Then*

$$Gw = (w'\phi)\, b_p, \qquad so \qquad \frac{Gw}{s_p' Gw} = \frac{b_p}{s_p' b_p} \quad and \quad \Phi_h \frac{Gw}{s_p' Gw} = \Phi_h \frac{b_p}{s_p' b_p} = \Phi_h \beta \ \ for\ every\ h,$$

the impact vector and impulse responses that the criterion in equation (21) identifies (Proposition C.2(i)). (ii) Sample. *For any $\hat{w} \in \ker \hat{R}$ with $\hat{\phi}'\hat{w} \neq 0$, and in particular for equation (35),*

$$\hat{G}^w \hat{w} = (\hat{\phi}'\hat{w})\, \hat{q} \qquad \Longleftrightarrow \qquad \hat{G}\hat{w} = (\hat{\phi}'\hat{w})\, \hat{L}\hat{q},$$

exactly. After normalisation the two routes deliver the same $\hat{\beta} = \hat{b}_p / (s_p' \hat{b}_p)$ and the same $\hat{\Theta}_h = \hat{\Phi}_h \hat{\beta}$ at every horizon, in the sample and in every bootstrap draw that uses the same $\hat{\Sigma}_u$ and moments.

Both parts follow from $Gw = b_p(\phi' w) + B_{-p}(\Psi' w)$ and $\hat{G}^w \hat{w} = \hat{q}(\hat{\phi}'\hat{w}) + \hat{R}\hat{w}$.

D Simulated impulse responses

Figure 4 shows in two dimensions how demeaning within families recovers the direction of the policy shock. This appendix shows what that does to the impulse responses, in a simulated three-variable VAR. Figure 4 uses its own simulated points, with the same structure of the moments.

Design. The VAR has lag order one, $y_t = A\,y_{t-1} + u_t$, with the policy variable third, and

$$A = \begin{pmatrix} 0.90 & 0 & -0.08 \\ 0.10 & 0.85 & -0.05 \\ 0.05 & 0.05 & 0.80 \end{pmatrix}, \qquad b_p = (-0.45,\ -0.30,\ 1)'.$$

Two families of six instruments have moments $g_j = \phi_j\, b_p + d_c + e_j$, with relevance ϕ_j spread evenly from 0.4 to 2.6 within each family, plus a small perturbation, and family contaminations $d_1 = (-0.5,\ 0.9,\ 0.35)'$ and $d_2 = (0.8,\ -0.6,\ -0.25)'$. The idiosyncratic contamination is $e_j \sim N(0, 0.05^2 I)$. Within each family, its part orthogonal to b_p is made uncorrelated with relevance and with its part along b_p. This imposes in the sample what Assumption 1(b) imposes in population. Hence, the figure shows identification rather than sampling error. Four directions are computed. Two are the mean moments of each family, used alone. The third is the first principal component of all twelve moments, without demeaning. The fourth is the first principal component after demeaning within families. Each is normalised to $+1$ on the policy variable on impact, with impulse responses $A^h b$. The whitening step is omitted. Hence, the example isolates the within transform.

Results. Each family alone identifies the true direction plus a bias from its contamination, as the bias identity in equation (7) predicts, and the two families err in opposite directions. Pooling without demeaning does not average the errors away. Its first principal component is pulled toward the contamination and lies further from the truth than either family alone. After demeaning, the first principal component is the true direction, and its impulse responses coincide with the true ones at every horizon (Figure 9).

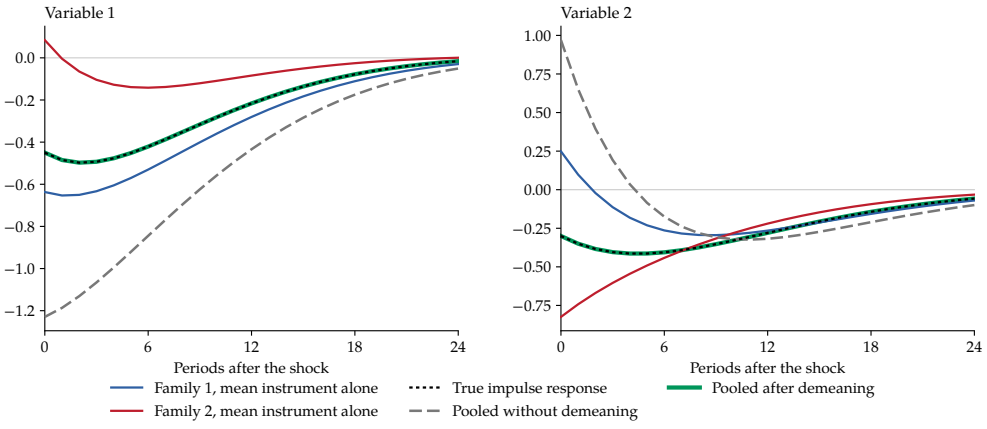


Figure 9: **Simulated impulse responses: demeaning within families recovers the true response.** Variables 1 and 2, after a policy shock that raises the policy variable by one unit on impact. Family 1 and 2: each family's mean instrument alone. Pooled: the first principal component of the moments, without and after demeaning within families. True: $A^h b_p$.

E Supplementary analyses for the empirical application

E.1 The role of the demeaning step

Figure 6 compares the baseline with pooling the twelve instruments without the demeaning step. Without it, the direction is the leading eigenvector of the precision-weighted scatter of the whitened moments, and the instrument is the unconstrained GLS pool, $w_j \propto \hat{\phi}_j / \hat{v}_j$. This scatter splits into a within-family part and a between-family part built from the family means. The split is exact because the family means use the same precision weights. The demeaning step discards the between-family part, which accounts for about four fifths of the scatter (84% along the direction estimated without demeaning, 78% along the baseline direction).

A pool that keeps the levels is therefore identified mostly by the four family means. Each mean carries its family's common contamination along with its average relevance (Section 6.4). The loading of that contamination on the policy rate is counted as relevance, and it raises the first stage. In the simulations of Online Appendix Section OA.4.2, where the truth is known, pooling without demeaning has a stronger first stage but a bias of about 0.26, against 0.017 for the baseline. The demeaning step gives up about five sixths of the first-stage F to remove the family-common contamination.

E.2 Weak instruments in the pool

Pooling keeps the weak instruments, because they can still provide identifying variation. Five of the twelve have a first stage below 10 on their own. In the pool, a weak member enters a single instrument with a small weight, not a first stage of its own. Therefore, it does not create the many-weak-instruments bias of two-stage least squares (Staiger and Stock, 1997). Dropping the weak members would remove the forward-guidance and asset-purchase factors, and with them most of the dispersion in relevance that sets the direction (Appendix Table 8).

E.3 Diversifying the families

Section 7.1 defines the three families used to reduce the dependence on the Swanson family: the information-robust family (IR), added as a fifth family; the Romer–Romer family diversified with the Aruoba and Drechsel (2024b) shocks (RR′); and the maturity ladder in the high-frequency family. This appendix gives the details and the results. All new members are orthogonalised before aggregation (PTA), like the baseline instruments. The Treasury surprises of the ladder come from the Acosta et al. (2025) database. Along the pooled direction, relevance ranges from 0.0065 to 0.0089 in the IR family and from 0.0043 to 0.0127 in RR′, against 0.0109 to 0.0123 in the baseline Romer–Romer family. In the ladder, it falls with maturity, from about 0.010 for the near-dated futures to 0.008 at two and five years and 0.004 at ten.

The information-robust family. The family holds three constructions, each orthogonalised announcement by announcement on the six Bauer and Swanson (2023b) controls. The first is the Miranda-Agrippino and Ricco (2021) surprise, which removes from the high-frequency surprise the part predicted by the Fed's Greenbook forecasts and their revisions. The second and the third apply the poor man's sign restriction of Jarociński and Karadi (2020) after the orthogonalisation, to the orthogonalised FF4 surprise and to the orthogonalised Bauer and Swanson (2023b) surprise. The

restriction keeps the surprise in the announcements where it and the S&P 500 surprise in the same window move in opposite directions, and sets it to zero otherwise. The orthogonalisation does not remove the co-movement with stock prices on average. The correlation of the orthogonalised surprise with the stock surprise is -0.54 for FF4 and -0.50 for the Bauer and Swanson (2023b) surprise, close to its value before the orthogonalisation. Yet 35% of the orthogonalised announcements still co-move positively and are set to zero in both series. Applying the restriction after the orthogonalisation ensures that it acts on the part of the surprise that enters the moments. Both sign-restricted series stay unpredictable from the controls, with $p = 0.94$ for FF4 and $p = 0.83$ for the Bauer and Swanson (2023b) surprise over 1988–2019. The three series cover different periods within the moment window, from 1988 for the third and 1994 for the first two, and each enters on its own support. All three stay unpredictable from the controls.

As a further robustness exercise, the Swanson (2021) rotation applied to the yield-curve surprises around the FOMC minutes (2001 onward) gives a sixth family of three factors.

The price of relevance heterogeneity. Diversifying a family has a cost. The features that make its members differ in relevance can also make them differ in contamination, which Assumption 1 rules out. The ladder is the clearest case. The ten-year yield is its least relevant member because it is the most exposed to term premia and to news about the longer-run outlook, which are likely to be contaminated differently from the near-dated futures. Each family's own overidentification test is the only test available, and with three to six members it has little power (for the ladder, $p = 0.84$). When the contamination moves with relevance in the same way in every family, nothing in the data warns of it (Online Appendix OA.4.4). The partitions below therefore rely less on the Swanson family but more on block structure where it is harder to defend, and are reported as robustness, not as the baseline.

Results. Each new family adds dispersion in relevance, and the maturity ladder contributes the most (Table 11). Alone, it lowers the Swanson family's share of the identifying variation to 56% and raises the high-frequency family's to 24%, and it leaves the estimate close to the baseline. Adding the IR family and replacing the Romer–Romer family by RR′ gives the two families 3% and 8% of the identifying variation. Both pass their own overidentification tests ($p = 0.26$ and 0.69), and exact block structure is not rejected ($p = 0.54$). The estimate barely moves, but the Swanson family still carries 62%. With the ladder, the IR family and RR′ together, the estimate is close to the baseline, with $F = 20.5$ and a peak output decline of 4.6% after fourteen months. The Swanson family's share falls to 51%. Without it, output still falls by up to 3.4% after sixteen months and prices are 0.3% lower after two years, against -2.2% and a 0.2% *rise* in the baseline partition. Confidence bands are about 15% wider. Adding the minutes factors lowers the Swanson share further, to 37%, but the first-stage F falls to 10.4 (6.7 without the Swanson family) and the impulse responses grow (peak output decline 6.4%), the magnification of Section 3.3. They are not a useful hedge. The dependence on the Swanson family can thus be reduced without altering the baseline estimates, at the price of a block-structure assumption that is harder to defend for the longest maturities.

E.4 The internal-instrument VAR and local projections

Internal-instrument VAR. The pooled instrument is stacked above the variables of the baseline, $w_t = (z_t^\star, y_t')'$ with y_t in (IP, CPI, EBP, Fed Funds), and the system

$$w_t = \tilde{c} + \sum_{\ell=1}^{6} \tilde{\Pi}_\ell \, w_{t-\ell} + \eta_t, \qquad \mathrm{Var}(\eta_t) = \Sigma_\eta = PP', \tag{44}$$

is estimated with six lags and a constant, with P the lower-triangular Cholesky factor of Σ_η. With $z_t^\star$ ordered first, the first recursive shock $e_1' P^{-1} \eta_t$ is the part of the instrument's innovation not predicted by the lags of the system, and its impact on the other variables is unrestricted. With $\tilde{\Phi}_h$ the Wold coefficients of equation (44) and s_p the coordinate of the funds rate in w_t, the response to a shock that raises the funds rate by one point on impact is the y-block of

$$\Theta_h^{\mathrm{int}} = \frac{\tilde{\Phi}_h \, Pe_1}{s_p' Pe_1}, \qquad h = 0, \ldots, 36. \tag{45}$$

This is the normalisation in equation (4) of the external-instrument estimator. On impact Pe_1 is proportional to $\mathrm{Cov}(\eta_t, \eta_{z,t})$, the external-instrument moment with $z_t^\star$ replaced by its innovation. The two estimators differ at later horizons because the lags of $z_t^\star$ enter the equations of y_t. They coincide when the instrument carries no information about future y_t beyond the lags of y_t, which holds when the shock is invertible and the instrument's noise is serially uncorrelated (Plagborg-Møller and Wolf, 2021).

Local projections. For each variable k of y_t and each horizon $h = 0, \ldots, 36$,

$$y_{k,t+h} = \alpha_{k,h} + \theta_{k,h} \, i_t + \sum_{\ell=1}^{6} \gamma_{k,h,\ell}' \, w_{t-\ell} + e_{k,t+h}, \tag{46}$$

with the funds rate i_t instrumented by $z_t^\star$ (Stock and Watson, 2018). The controls are those of equation (44). Hence, the two estimators condition on the same information. With $\tilde{x}_t$ a variable residualised on the controls, $\theta_{k,h} = \mathrm{Cov}(\tilde{y}_{k,t+h}, \tilde{z}_t^\star) / \mathrm{Cov}(\tilde{i}_t, \tilde{z}_t^\star)$ is the response to a one-point rise in the funds rate, the normalisation of the baseline. In population the two estimators identify the same impulse responses at every horizon (Plagborg-Møller and Wolf, 2021); in a finite sample, local projections trade the VAR's extrapolation of the dynamics for variance. The leads are restricted to December 2019. Hence, the sample shortens by one month per horizon. Confidence bands are 68% intervals from the moving-block bootstrap of the baseline, with the pooling weights held fixed.

Assumptions. Both estimators use the pooled instrument $z_t^\star$ of Section 5.6. They inherit its identifying assumptions, namely the family partition, Assumption 1 and the constraint that removes the contamination the families share. They add no restriction on the families. Each adds only the conditions of its own estimator. Both require $z_t^\star$ to be uncorrelated with past and future shocks once the controls are included (Stock and Watson, 2018). Ordering $z_t^\star$ first in the internal-instrument VAR relaxes the invertibility of the policy shock that the external-instrument VAR requires (Plagborg-Møller and Wolf, 2021). The pooling weights are estimated once and held fixed in both estimators.

F Appendix tables

Table 3: Predictability of the raw instruments and the pooled instrument from pre-announcement information. Each is regressed on a constant and the six Bauer and Swanson (2023b) controls of Section 2.1; HC1 standard errors in parentheses; *, **, ***: 10, 5 and 1%. Romer–Romer: the instrument before orthogonalisation. MAR: the Miranda-Agrippino and Ricco (2021) surprise. Cholesky: the recursive funds-rate innovation. Pooled $z^{\star}$: the baseline pooled instrument. The first three at meeting frequency, the last two monthly (months with an FOMC decision). Joint F: HC1 test that all six coefficients are zero.

	Monetary policy surprises	Romer–Romer	MAR	Cholesky	Pooled $z^{\star}$
NFP surprise	0.093**	0.113	0.101**	0.276**	0.237
	(0.039)	(0.101)	(0.044)	(0.128)	(3.915)
NFP 12M	0.005**	−0.002	−0.000	0.001	−0.069
	(0.002)	(0.009)	(0.002)	(0.012)	(0.249)
S&P 500 3M	0.079	−0.156	0.033	−0.056	0.401
	(0.059)	(0.184)	(0.056)	(0.209)	(6.194)
Slope 3M	−0.011	−0.097***	−0.006	−0.076***	−0.211
	(0.007)	(0.028)	(0.009)	(0.025)	(0.929)
BCOM 3M	0.122**	0.022	0.034	−0.041	−0.121
	(0.052)	(0.147)	(0.050)	(0.179)	(5.527)
Treasury skew	0.032***	0.117***	0.011	0.078*	0.330
	(0.011)	(0.038)	(0.010)	(0.041)	(1.125)
R^2	0.165	0.105	0.066	0.060	0.001
Joint F (p)	5.22 (0.000)	3.59 (0.002)	1.11 (0.356)	2.64 (0.017)	0.04 (1.000)
N	316 meetings	256 meetings	219 meetings	267 months	267 months
Sample	1988m2–2019m12	1988m2–2019m12	1994m2–2019m12	1988m2–2019m12	1988m2–2019m12

Table 4: First-stage F (HC0) by instrument, orthogonalisation and policy rate. Each instrument is used alone, in the VAR in (IP, CPI, EBP, policy rate) of Section 6.1, on the 383 months within 1988m1–2019m12 on which all its versions are observed. The raw Romer–Romer and Cholesky-based instruments are re-estimated on the window of their orthogonalised versions. **Bold**: $F > 10$ (Stock and Yogo, 2005). PTA: purge-then-aggregate. ATP: aggregate-then-purge. FWL: orthogonalised jointly with the instrument's own first stage.

Instrument (statement window)	Fed funds	2-year	1-year	Shadow
Monetary policy surprises (MPS)				
Raw	**14.5**	9.4	**18.6**	8.0
Orthogonalised (ATP)	7.1	2.8	6.4	5.7
Orthogonalised (PTA)	3.6	0.6	2.7	2.4
Romer–Romer				
Raw	**37.3**	5.7	**12.9**	**25.1**
Orthogonalised (ATP)	**17.5**	0.9	1.8	**10.4**
Orthogonalised (PTA, FWL)	**29.6**	2.9	5.8	**21.8**
Cholesky-based				
Raw	**965.7**	**39.6**	**92.3**	**573.3**
Orthogonalised (ATP, FWL)	**533.5**	**22.7**	**41.6**	**311.1**

Table 5: The instrument panel. Twelve orthogonalised instruments in four families. Support is the span of non-missing months in the moment window (January 1988 to December 2019); T_j its length. PTA: purge-then-aggregate; ATP: aggregate-then-purge; FWL: orthogonalised jointly with the instrument's own first stage.

Family	Instrument	Support	T_j
High-frequency surprises	FF4, orthogonalised (PTA)	1994m02–2019m12	311
	ED4, orthogonalised (PTA)	1994m02–2019m12	311
	Bauer–Swanson MPS, orthogonalised (PTA)	1988m02–2019m12	383
Swanson factors	Swanson (2024) federal funds factor, orthogonalised (PTA)	1988m02–2019m12	383
	Swanson (2024) forward guidance factor, orthogonalised (PTA)	1988m02–2019m12	383
	Swanson (2024) asset purchase factor, tightening sign, orthogonalised (PTA)	1988m02–2019m12	383
Romer–Romer	Romer–Romer, orthogonalised (PTA)	1988m02–2019m12	383
	Romer–Romer, orthogonalised (PTA, FWL)	1988m02–2019m12	383
	Romer–Romer, orthogonalised (ATP)	1988m02–2019m12	383
Recursive (Cholesky)	Cholesky, federal funds rate, orthogonalised (ATP, FWL)	1988m02–2019m12	383
	Cholesky, 1-year Treasury, orthogonalised (ATP, FWL)	1988m02–2019m12	383
	Cholesky, 2-year Treasury, orthogonalised (ATP, FWL)	1988m02–2019m12	383

Table 6: The baseline and its main variants. T: months in the moment window. F and μ^2: first stage and concentration of the pooled instrument. J_T: overidentification statistic with the full sampling covariance; J_{between}: its between-family part. $I^2 = (J_T - \text{df})_+ / J_T$: the share of J_T in excess of its degrees of freedom. $\hat{\lambda}_1/\hat{\lambda}_2$: eigengap of the weighted within scatter. Impulse responses to a $+1$ percentage-point shock to the policy indicator; the peak decline of output is its largest absolute response, with its horizon.

Specification	T	F	μ^2	J_T (df)	p	J_{between} (df)	I^2	λ_1/λ_2	IP_0	IP peak decline (h)	CPI_0	EBP_0
Baseline (funds rate)	384	25.1	21.2	14.5 (21)	0.848	4.2 (9)	0.00	3.7	-0.48	-3.97 (15)	-0.22	$+0.88$
Unweighted	384	8.6	18.7	14.6 (21)	0.845	4.2 (9)	0.00	2.5	-1.42	-6.23 (13)	-0.45	$+1.33$
Shadow rate	384	12.7	16.0	13.8 (21)	0.879	4.9 (9)	0.00	3.3	-0.50	-4.50 (14)	-0.29	$+1.06$
VAR 1985–2008	252	10.5	11.8	23.1 (21)	0.341	10.0 (9)	0.09	6.2	-0.83	-7.63 (29)	-0.13	$+1.01$

Table 7: The families in the baseline. Share of $\hat{\sigma}^2_\phi$: the family's share of the identifying variation. J_c: the family's own overidentification statistic against its own direction. Contamination share: the fraction of the family's mean moment, in the Σ_u^{-1} metric, orthogonal to the pooled direction (Section 6.4).

Family	$\lvert c\rvert$	share of $\hat{\sigma}^2_\phi$	J_c (df)	p	contamination share
High-frequency surprises	3	0.01	0.8 (3)	0.853	0.09
Swanson factors	3	0.71	4.4 (3)	0.224	0.14
Romer–Romer	3	0.00	1.8 (3)	0.608	0.35
Recursive (Cholesky)	3	0.28	3.3 (3)	0.350	0.80

Table 8: Single instruments and pools on the baseline VAR and moment window. F: first stage against the funds-rate innovation. Impulse responses to a $+1$ percentage-point funds-rate shock; the peak decline of output is its largest absolute response, with its horizon. Strong instruments only: the instruments with $F > 10$ on their own (Appendix E.2).

Identification	F	IP_0	IP peak decline (h)	CPI_0	CPI_{24}	EBP_0
Bauer–Swanson surprise, raw	14.5	-0.28	-1.93 (20)	-0.15	-0.11	$+0.25$
Bauer–Swanson surprise, orthogonalised	3.6	-0.52	-6.58 (13)	-0.83	-3.00	$+1.97$
Swanson funds-rate factor, PTA	10.6	-1.61	-5.88 (13)	-0.33	-1.56	$+1.09$
Romer–Romer, PTA	22.3	-0.13	-1.69 (23)	-0.11	$+0.05$	$+0.19$
MAR information-robust, PTA	6.3	-2.20	-8.35 (13)	-0.46	-2.53	$+1.69$
Cholesky funds rate, ATP-FWL	533.5	$+0.01$	-1.40 (30)	-0.03	$+0.34$	$+0.05$
HF and Swanson families only	4.7	-1.16	-6.56 (13)	-0.62	-2.45	$+1.61$
All 12 instruments pooled, no within transform	146.9	-0.19	-2.04 (19)	-0.14	-0.16	$+0.32$
Strong instruments only ($F > 10$), pooled estimator	49.9	$+0.11$	-2.07 (20)	$+0.07$	$+0.18$	$+0.34$
Strong instruments only ($F > 10$), no within transform	284.3	-0.08	-1.47 (28)	-0.07	$+0.25$	$+0.07$
Pooled estimate (baseline)	25.1	-0.48	-3.97 (15)	-0.22	-0.94	$+0.88$

Table 9: Variations of the panel. The baseline, the baseline with the information-robust family (IR) of Appendix E.3 added as a fifth family, the five-family panel with each family dropped in turn, and the repartitioned families (Section 7.1). m instruments, C families. p, J_T: test of exact block structure; p, between: its between-family part. Eigengap: $\hat{\lambda}_1/\hat{\lambda}_2$ of the weighted within scatter. Impulse responses to a $+1$ percentage-point funds-rate shock.

Panel	m	C	F	p, J_T	p, between	eigengap	IP_0	IP peak decline (h)	CPI_{24}	EBP_0
Baseline: four families of three	12	4	25.1	0.848	0.897	3.7	-0.48	-3.97 (15)	-0.94	$+0.88$
Adding a family										
Adding the information-robust family (IR)	15	5	21.2	0.725	0.788	3.6	-0.57	-4.29 (15)	-1.04	$+0.95$
Leaving one family out of the five-family panel										
Without the high-frequency family	12	4	21.2	0.565	0.758	4.3	-0.52	-4.19 (15)	-1.04	$+0.94$
Without the Swanson family	12	4	34.2	0.467	0.280	4.0	-0.04	-2.43 (18)	$+0.12$	$+0.40$
Without the Romer–Romer family	12	4	20.6	0.557	0.639	3.7	-0.58	-4.30 (15)	-1.04	$+0.95$
Without the Cholesky family	12	4	4.5	0.858	0.953	2.9	-1.40	-7.45 (13)	-2.71	$+1.80$
Repartitioned families										
Each Swanson factor in the family of its horizon	18	5	30.0	0.641	0.302	3.9	-0.61	-3.90 (15)	-0.79	$+0.78$
Path factor with the Treasury curve	18	5	29.7	0.646	0.673	3.9	-0.60	-3.90 (15)	-0.80	$+0.79$
Without the Swanson factors	15	5	64.0	0.338	0.121	4.3	-0.20	-2.56 (18)	-0.02	$+0.41$

Table 10: Further five-variable systems. Each row is a separate VAR in the baseline four variables plus the variable listed, with the four baseline families and the IR family as a fifth, and the baseline weights and windows. F: first stage of the pooled instrument. J_T: test of exact block structure. x_0 and x peak: impulse response of the added variable on impact and at its largest absolute value, with its horizon. IP_0, IP peak decline and CPI_{24}: responses of output and prices in the same system. Impulse responses to a $+1$ percentage-point funds-rate shock.

Added variable	F	J_T (df)	p	x_0	x peak (h)	IP_0	IP peak decline	CPI_{24}
In the figure: demand and the labour market								
Real consumption (%)	14.3	35.2 (36)	0.505	−0.01	−1.06 (16)	−0.45	−4.52	−0.94
Retail sales (%)	10.3	40.9 (36)	0.263	−2.72	−4.49 (12)	−0.79	−5.46	−1.47
Real personal income (%)	22.6	30.5 (36)	0.726	−0.18	−1.45 (21)	−0.04	−3.54	−0.93
Payroll employment (%)	13.0	31.6 (36)	0.678	+0.04	−1.56 (22)	−0.43	−4.18	−1.04
In the figure: credit and financial conditions								
Business loans (%)	20.1	38.3 (36)	0.364	+0.08	−8.70 (29)	−0.07	−3.12	−0.93
Consumer credit, non-revolving (%)	16.8	32.0 (36)	0.660	−0.37	−2.88 (25)	−0.19	−4.19	−1.24
Mortgage spread (pp)	5.3	35.5 (36)	0.494	+0.72	+0.92 (1)	−1.17	−6.30	−1.87
VIX (index points)	9.9	29.9 (36)	0.753	+13.87	+14.47 (1)	−0.49	−4.73	−1.85
Other candidates, not plotted								
Core CPI (%)	13.5	26.6 (36)	0.872	−0.03	−0.93 (36)	−0.18	−3.79	−1.38
PCE price index (%)	9.1	32.1 (36)	0.652	−0.30	−1.33 (36)	−0.60	−4.32	−1.55
Building permits (%)	15.4	38.6 (36)	0.351	−13.25	−20.89 (5)	−0.25	−4.60	−1.27
Capacity utilisation, manufacturing (pp)	13.9	39.0 (36)	0.336	+0.01	−2.88 (12)	−0.55	−4.92	−0.98
Crude oil price (%)	21.4	36.3 (36)	0.456	+5.74	−7.78 (4)	−0.38	−3.95	−1.23
Trade-weighted dollar (%)	15.9	43.2 (36)	0.192	+0.74	+2.26 (5)	−0.18	−3.75	−1.47
Yen per dollar (%)	20.4	36.2 (36)	0.460	−0.49	−2.29 (36)	−0.24	−3.63	−0.81
Canadian dollars per dollar (%)	11.6	44.6 (36)	0.155	+2.93	+5.48 (5)	−0.30	−4.32	−1.64
10-year Treasury yield (pp)	10.0	62.1 (36)	0.004	−2.56	−3.48 (1)	−0.99	−3.85	+1.28
2-year Treasury yield (pp)	7.2	64.8 (36)	0.002	−3.25	−4.62 (1)	−0.47	−2.82	+1.18
BAA–10-year spread (pp)	8.3	40.4 (36)	0.282	+0.94	+1.47 (2)	−1.39	−7.25	−2.50
Real M2 (%)	14.2	52.0 (36)	0.041	+0.73	+2.39 (22)	−0.51	−4.03	−0.90

Table 11: Diversifying the families. Same VAR, windows, precision weights and estimator as the baseline; families other than those listed as in the baseline. IR: information-robust family of three distinct series; RR′: Romer–Romer and two Aruoba–Drechsel shocks; ladder: two-, five- and ten-year Treasury surprises added to the high-frequency family; minutes: Swanson factors on minutes releases as a sixth family. m instruments, C families. Swanson share: the family's share of the precision-weighted within-family variation along the estimated direction. Angles between normalised impact vectors, in the Σ_u^{-1} metric: to the baseline in the left block, to the same partition with all families in the right block. J_T p: test of exact block structure.

				Swanson	angle		IP peak decline		without the Swanson family			
Partition	m	C	F	share	(°)	J_T p	(h)	CPI_{24}	angle (°)	F	IP peak decline (h)	CPI_{24}
Baseline	12	4	25.1	0.71	0.0	0.848	−3.97 (15)	−0.94	34.0	42.3	−2.17 (19)	+0.18
IR + RR′	15	5	26.9	0.60	4.6	0.450	−3.68 (15)	−0.89	27.1	70.7	−2.18 (19)	+0.01
Maturity ladder in the high-frequency family	15	4	17.0	0.56	10.6	0.849	−4.87 (14)	−1.25	16.3	34.2	−3.59 (15)	−0.39
Ladder + IR + RR′	18	5	20.2	0.49	6.2	0.457	−4.48 (14)	−1.15	15.3	45.1	−3.31 (16)	−0.42
Ladder + IR + RR′ + Swanson factors on minutes	21	6	10.2	0.37	14.0	0.607	−6.20 (14)	−1.69	7.5	7.7	−6.18 (14)	−1.32

G Appendix figures

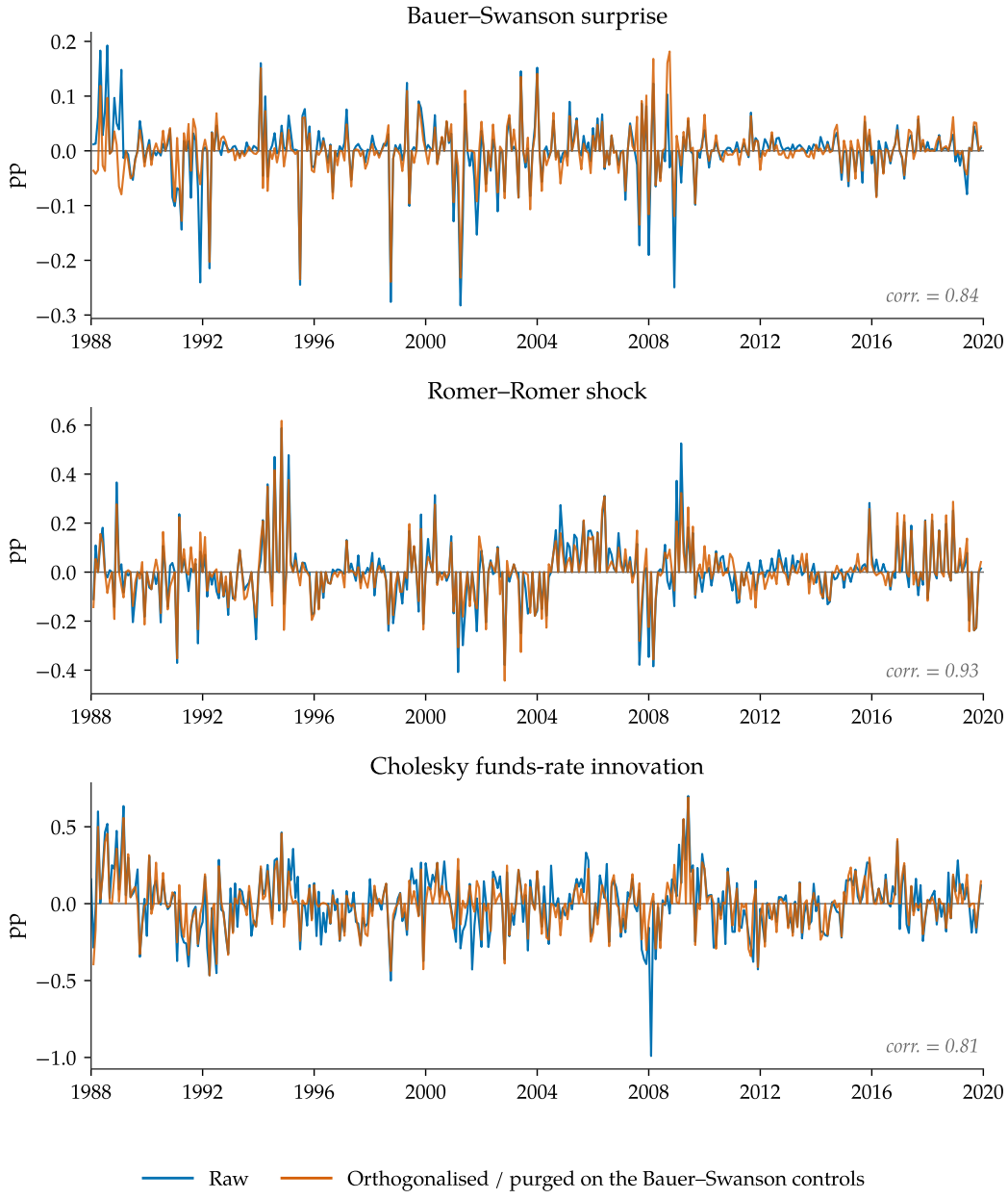


Figure 10: **The raw and orthogonalised instruments, January 1988 to December 2019.** Monthly series in percentage points. Top: the Bauer and Swanson (2023b) surprise (PTA). Middle: the Romer–Romer instrument (PTA-FWL). Bottom: the recursive (Cholesky) funds-rate innovation (ATP-FWL), zero in months without an FOMC announcement. The raw wide-window series are re-estimated on the window of the orthogonalised ones. Each panel reports the correlation of the two series.

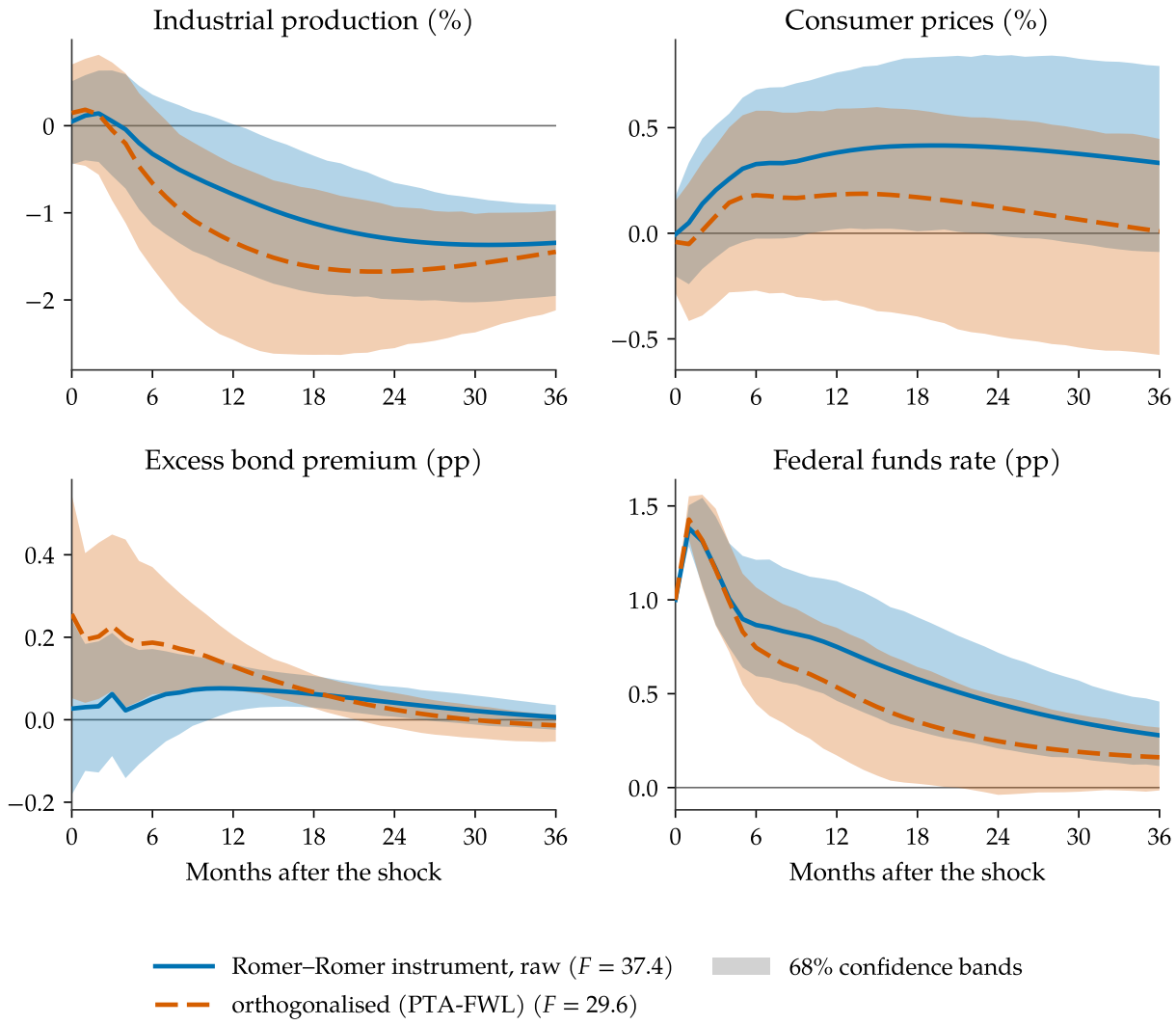


Figure 11: **Romer–Romer instrument: raw versus orthogonalised.** Single-instrument impulse responses to a +1 percentage-point funds-rate shock identified with the Romer and Romer (2004) instrument, raw (solid, $F = 37.4$) and orthogonalised on the six Bauer–Swanson controls at the announcement level, jointly with its first stage (PTA-FWL; dashed, $F = 29.6$). The raw instrument is the Romer and Romer (2004) residual, estimated on the same 256 meetings, February 1988 to December 2019, as the orthogonalised one, such that the two differ only by the controls, as in the raw row of Table 4. Baseline VAR; moments from January 1988 to December 2019. Shaded: 68% median-recentred percentile confidence bands from the moving-block bootstrap, 1,000 draws. The orthogonalisation costs about a fifth of the first-stage strength and enlarges the output and credit-spread impulse responses slightly.

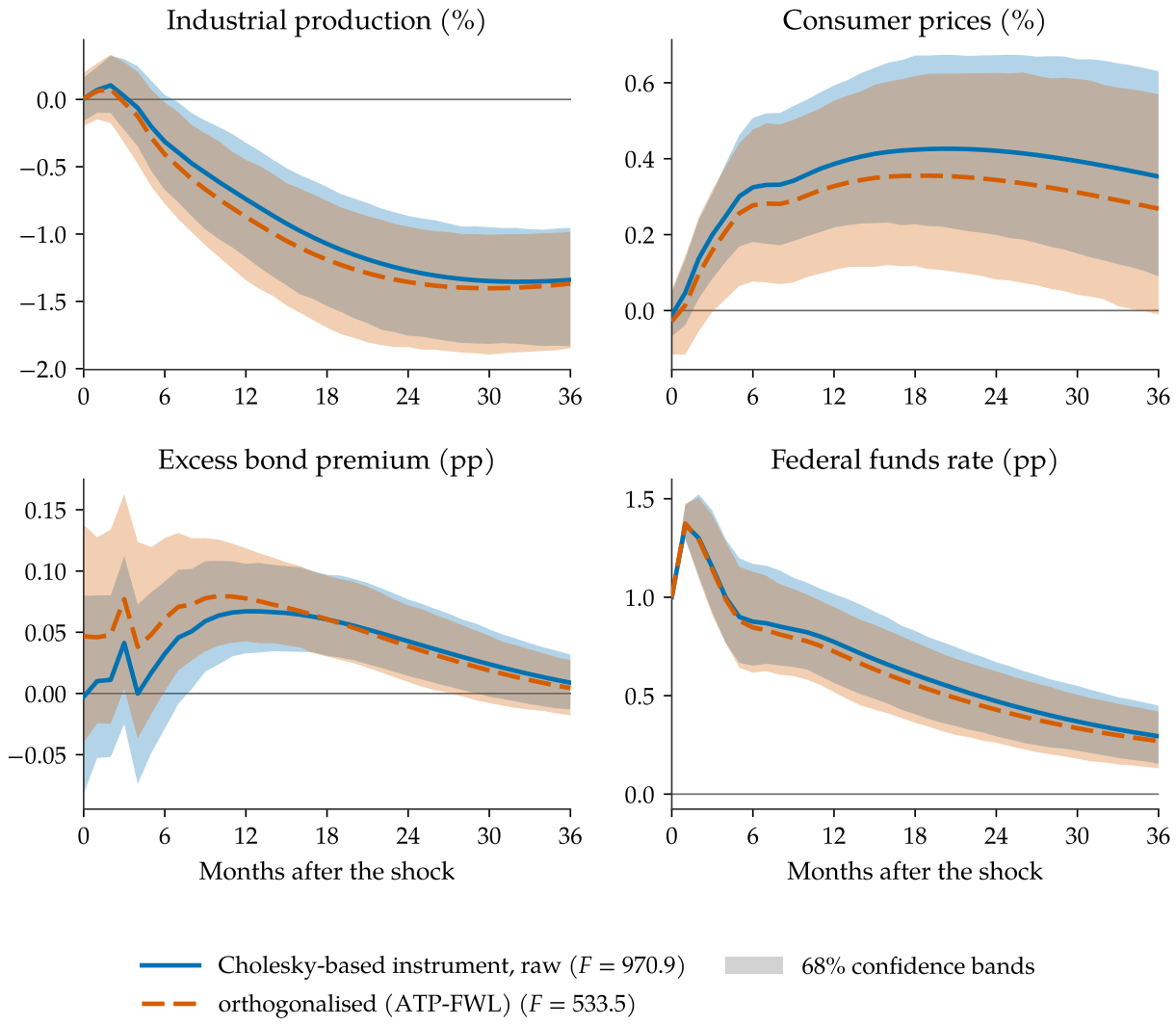


Figure 12: **Cholesky-based instrument: raw versus orthogonalised.** Single-instrument impulse responses to a +1 percentage-point funds-rate shock identified with the recursive funds-rate innovation of the instrument database, raw (solid, $F = 971$) and orthogonalised on the six Bauer–Swanson controls after monthly aggregation, jointly with its first stage (ATP-FWL; dashed, $F = 534$). The raw innovation is the database VAR's funds-rate innovation orthogonalised on the output and price innovations from January 1988 to December 2019, the window of the orthogonalised one, such that the two differ only by the controls, as in the raw row of Table 4. Baseline VAR; moments from January 1988 to December 2019. Shaded: 68% median-recentred percentile confidence bands from the moving-block bootstrap, 1,000 draws. The orthogonalisation removes close to half of the first-stage F and leaves the impulse responses essentially unchanged.

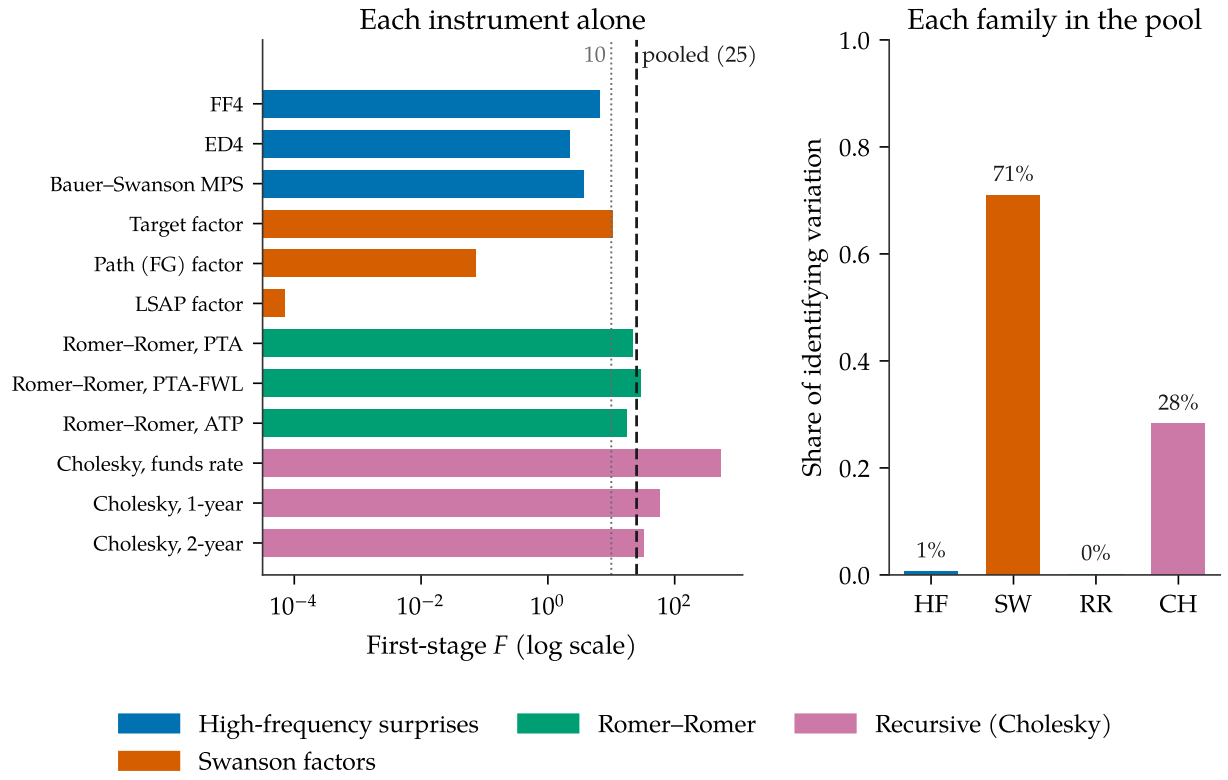


Figure 13: **What identifies the shock.** Left: first-stage F of each instrument entered alone against the funds-rate innovation on the moment window (log scale); the dotted line marks $F = 10$, the dashed line the pooled instrument. Right: each family's share of the identifying variation $\sum_{j \in c} \hat{\phi}_j^2 / \hat{v}_j$ in the precision-weighted within scatter, with the share in percent above each bar.

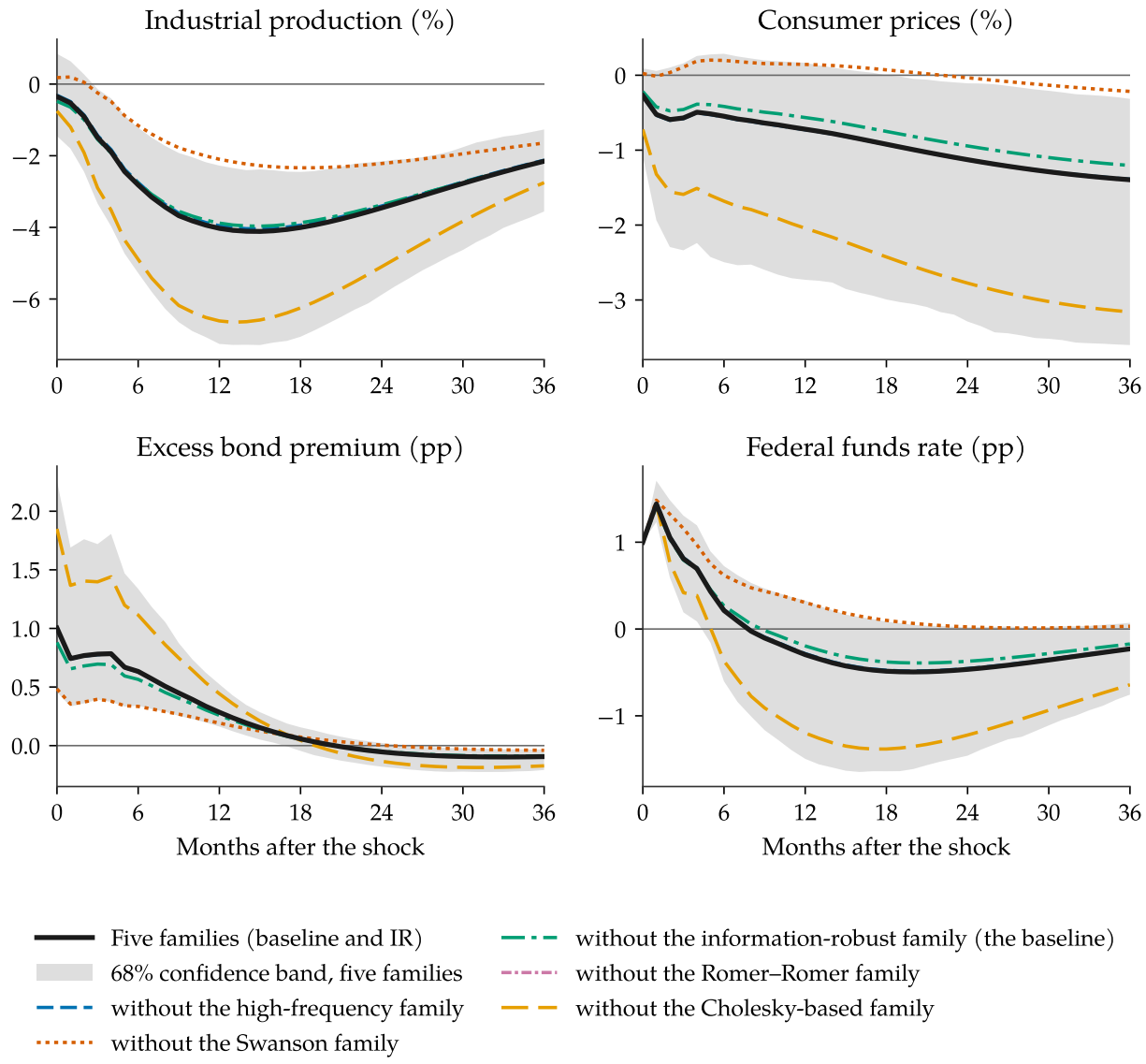


Figure 14: **Leaving one family out.** The five-family panel, the baseline families and the information-robust family (solid black, with its 68% confidence band), and the estimate with each family dropped in turn. With four families left, order condition 2 holds in every case. Dropping the information-robust family returns the baseline. Same VAR, windows and precision weights.

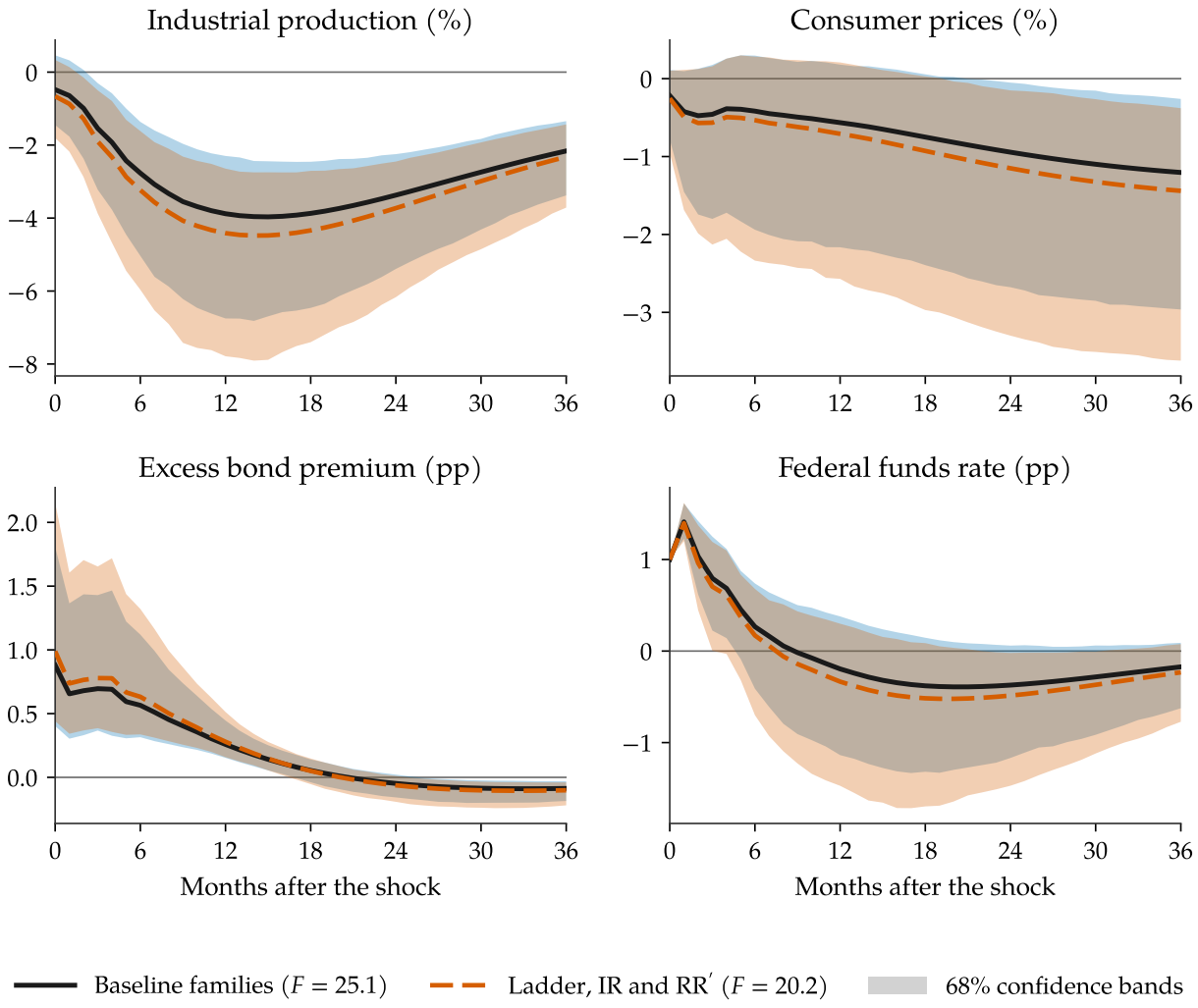


Figure 15: **Diversified families.** Baseline (solid, $F = 25.1$) against the partition with the maturity ladder in the high-frequency family, the information-robust family (IR) and the Romer–Romer family diversified with two Aruoba and Drechsel (2024b) shocks (RR′) (dashed, $F = 20.5$). Other families, VAR, windows, weights and bootstrap as in the baseline. Each with its 68% confidence band.

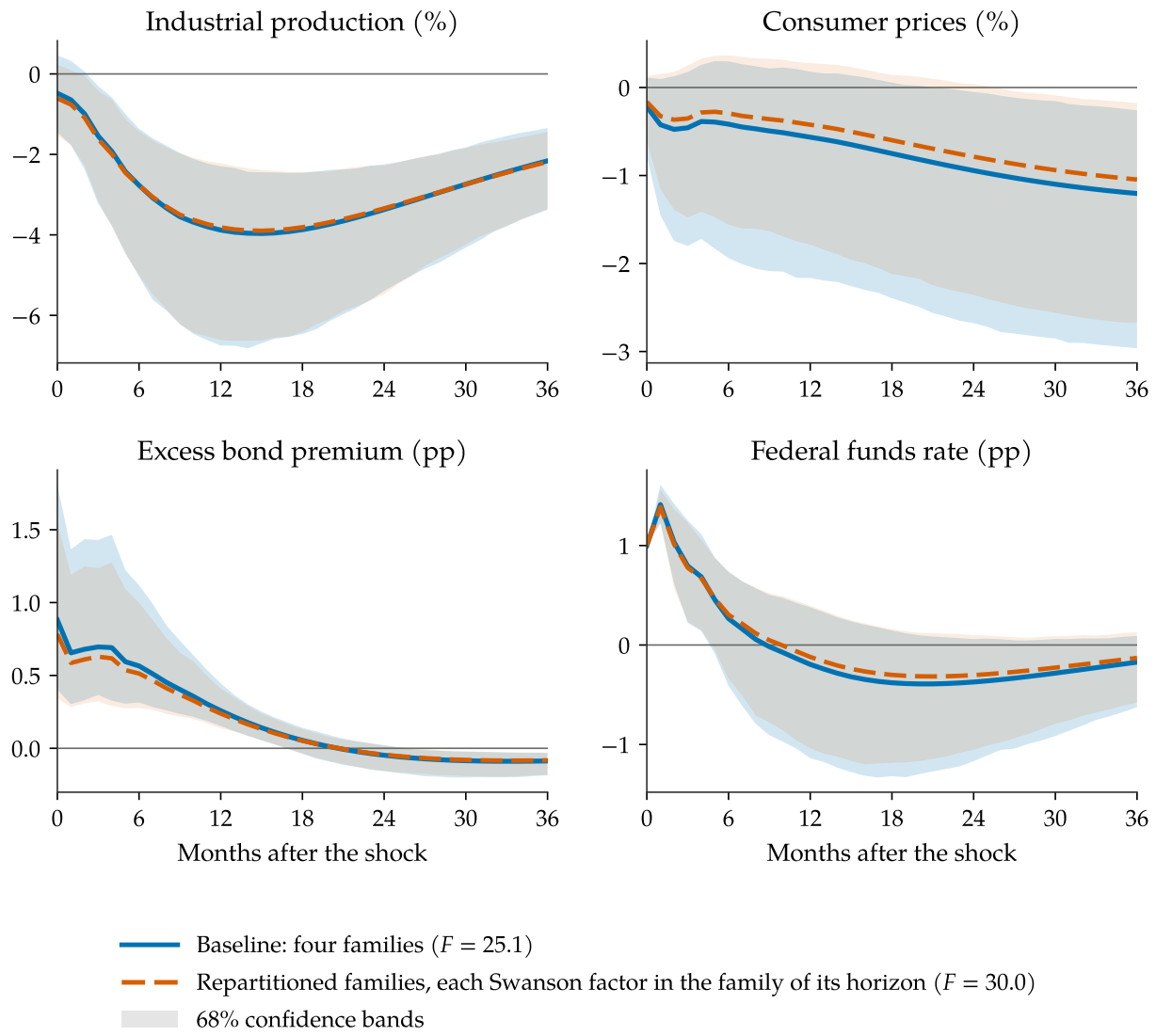


Figure 16: **Repartitioned families.** Impulse responses to a +1 percentage-point funds-rate shock. Solid: the baseline, four families. Dashed: five families formed by window and horizon (Section 7.1), the target and path factors with the near-dated futures and the asset-purchase factor with the Treasury curve. Same VAR, moment window, precision weights and bootstrap; 68% confidence bands.

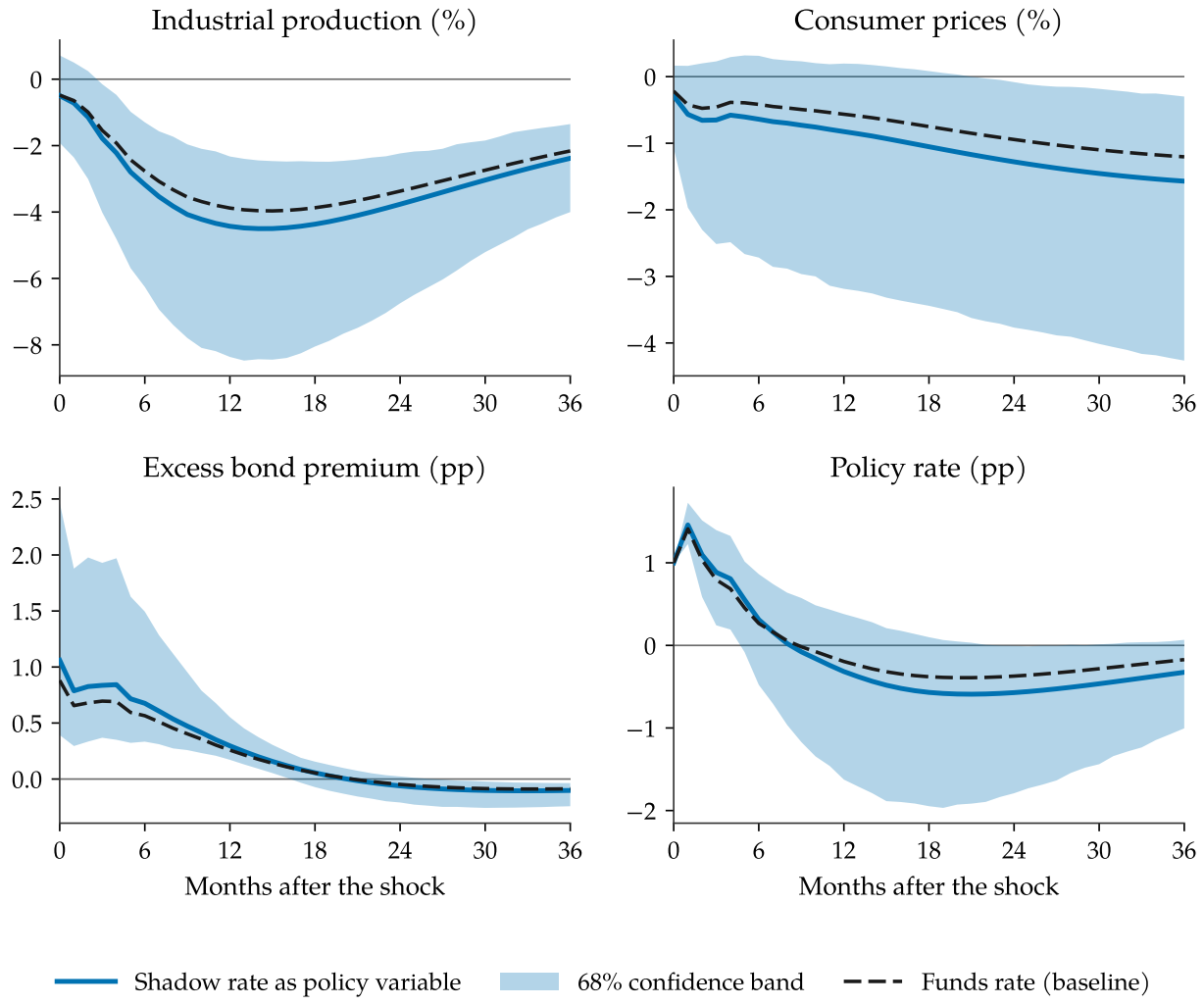


Figure 17: **The shadow rate as policy indicator.** Impulse responses to a +1 percentage-point shock to the shadow rate (the funds rate away from the zero lower bound, the Wu and Xia (2016) shadow rate at it), with 68% confidence bands; the funds-rate baseline is dashed. The bottom-right panel is the response of each specification's own policy rate.

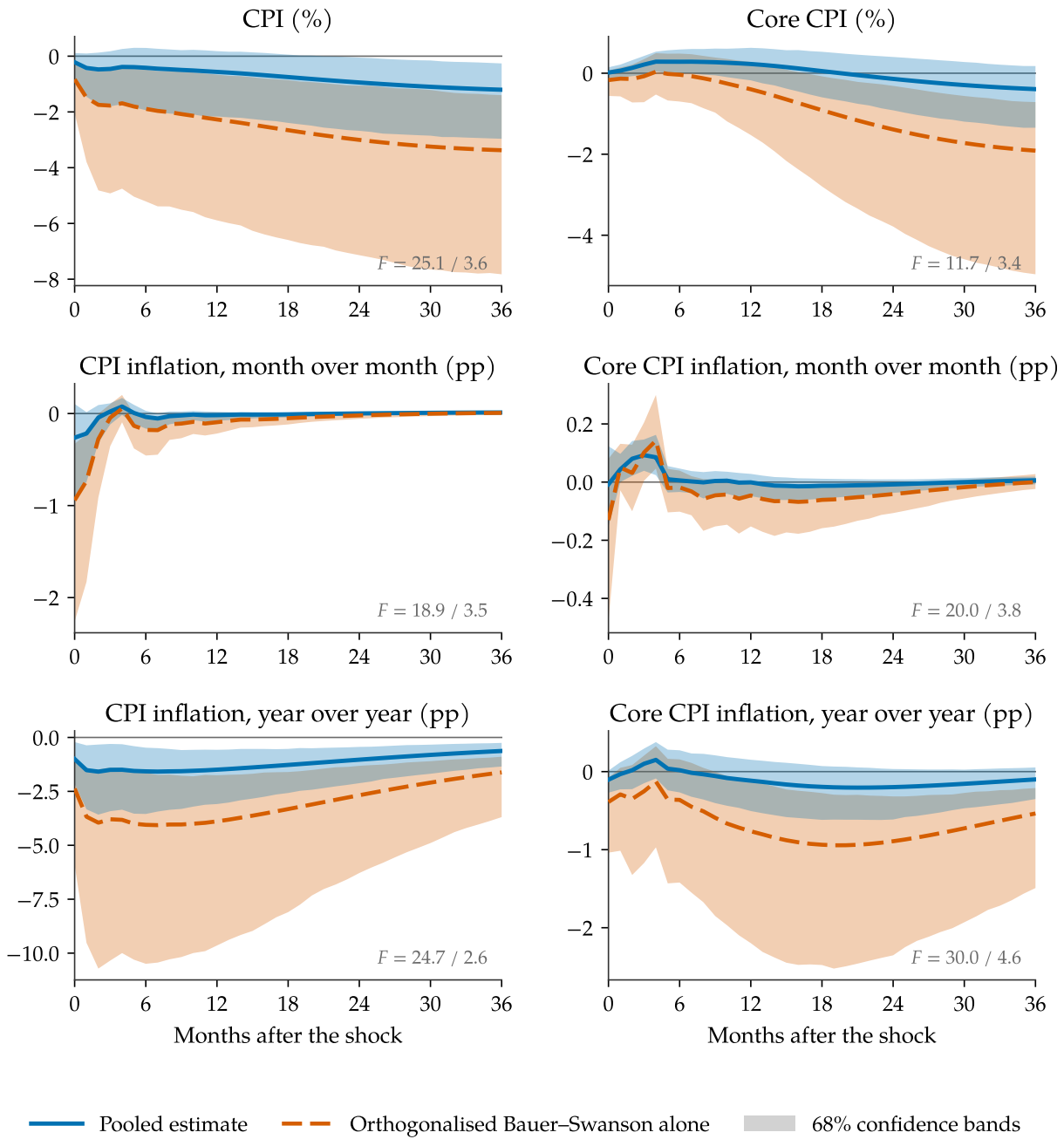


Figure 18: **The price measure: levels, month-over-month and year-over-year inflation.** Responses to a +1 percentage-point funds-rate shock of the CPI and core CPI: top, 100×log level; middle, month-over-month inflation, $100\Delta \log P_t$; bottom, year-over-year inflation, $100(\log P_t - \log P_{t-12})$, in percentage points. Each panel is from a separate four-variable VAR in which the price variable is the one shown. Pooled estimate (solid) and the orthogonalised Bauer and Swanson (2023b) surprise alone (dashed), each with its 68% confidence band. Same families, precision weights, windows and bootstrap as the baseline. F: first stage of the pooled instrument / of the orthogonalised surprise.

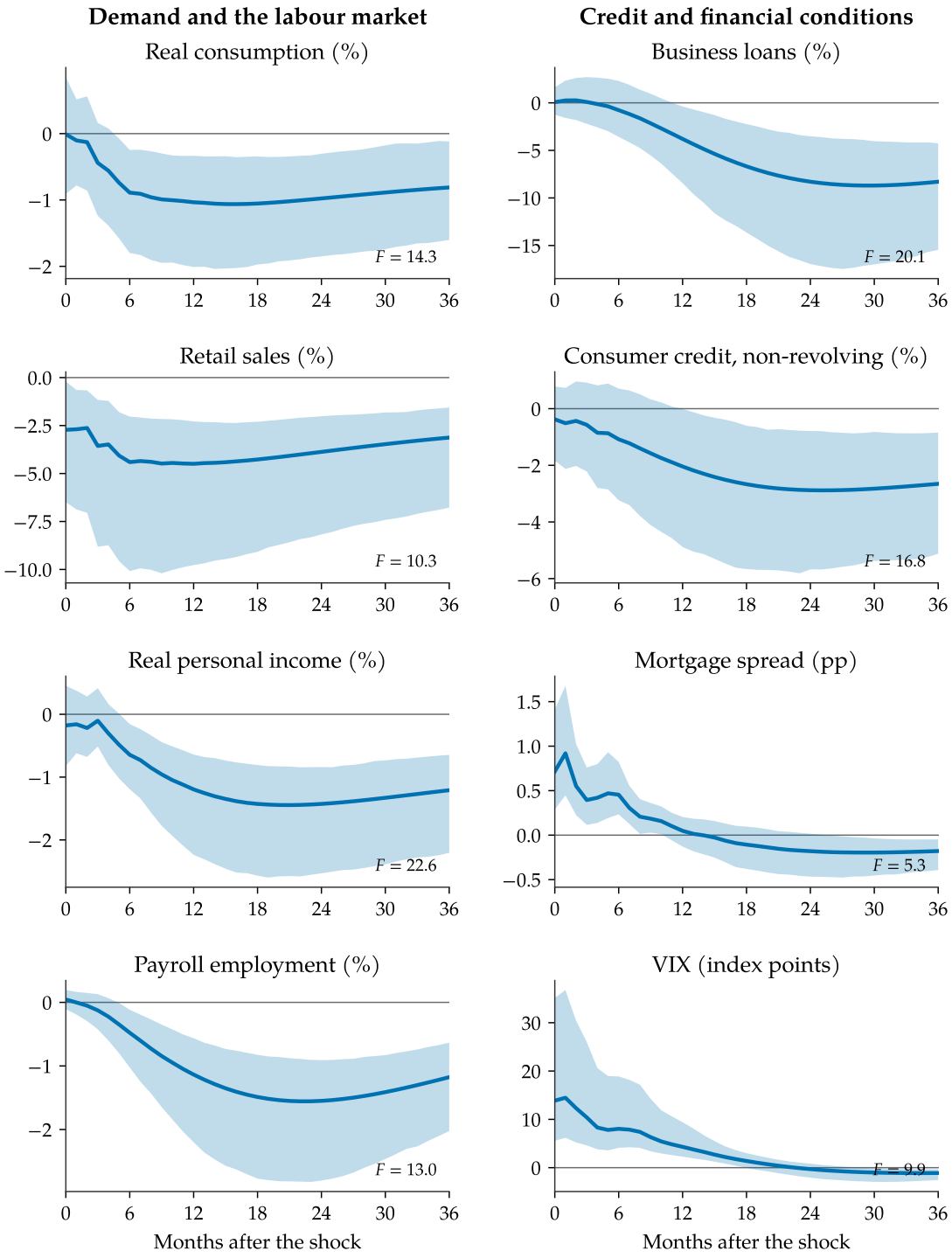


Figure 19: **Further variables, one at a time.** As in Figure 8: each panel is a separate five-variable VAR, and plots the impulse response of the added variable with its 68% confidence band. Left column: demand and the labour market. Right column: credit and financial conditions.

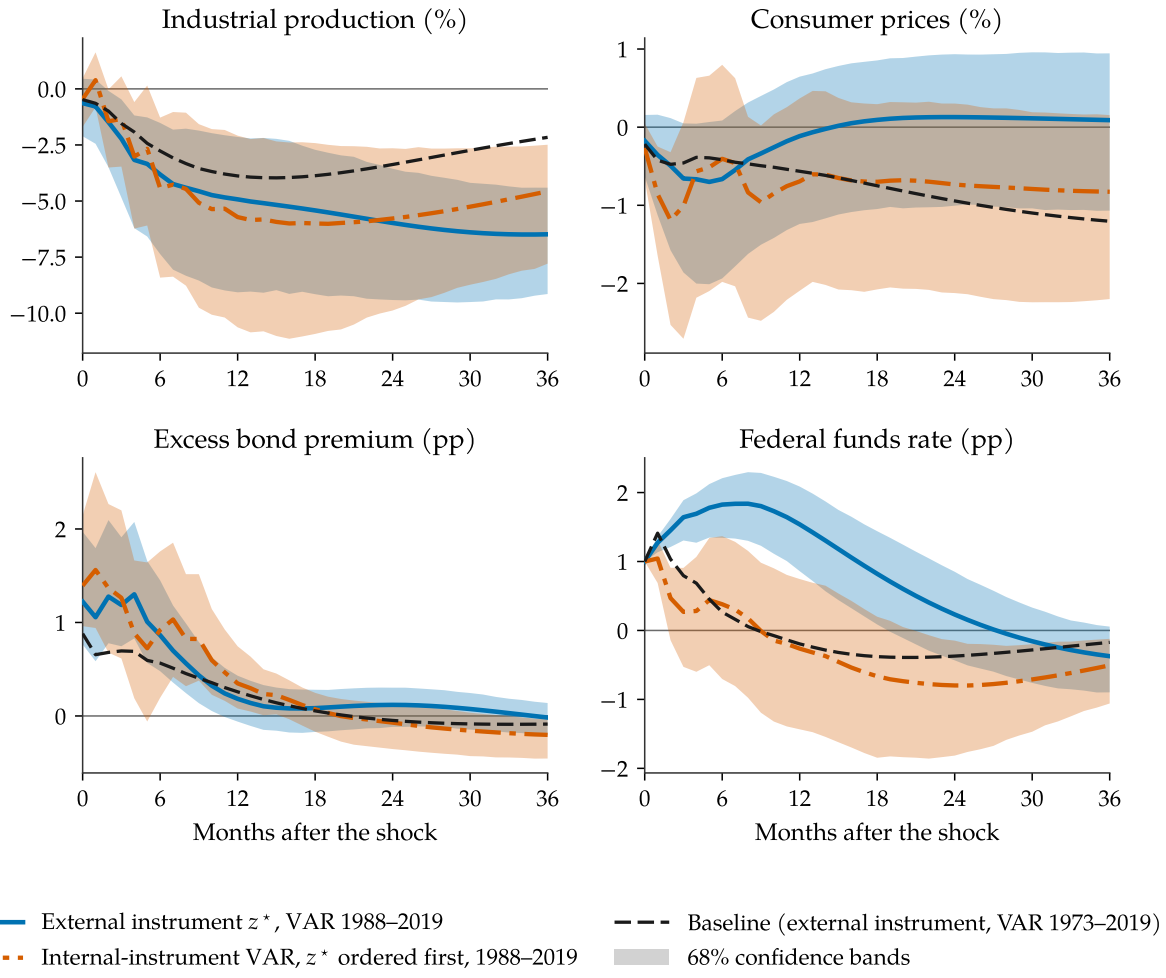


Figure 20: **The pooled instrument in an internal-instrument VAR.** Impulse responses to a +1 percentage-point funds-rate shock. Solid, with 68% confidence band: $z_t^\star$ as an external instrument on a VAR in (IP, CPI, EBP, Fed Funds) estimated from January 1988 to December 2019. Dash-dotted, with 68% confidence band: the internal-instrument VAR of Appendix E.4, with $z_t^\star$ ordered first, on the same sample. Dashed: the baseline (external instrument, VAR from January 1973 to December 2019; Figure 5). Six lags and a constant throughout. Bands: median-recentred percentile intervals from the moving-block bootstrap, 1,000 draws, with the pooling weights held at their estimates.

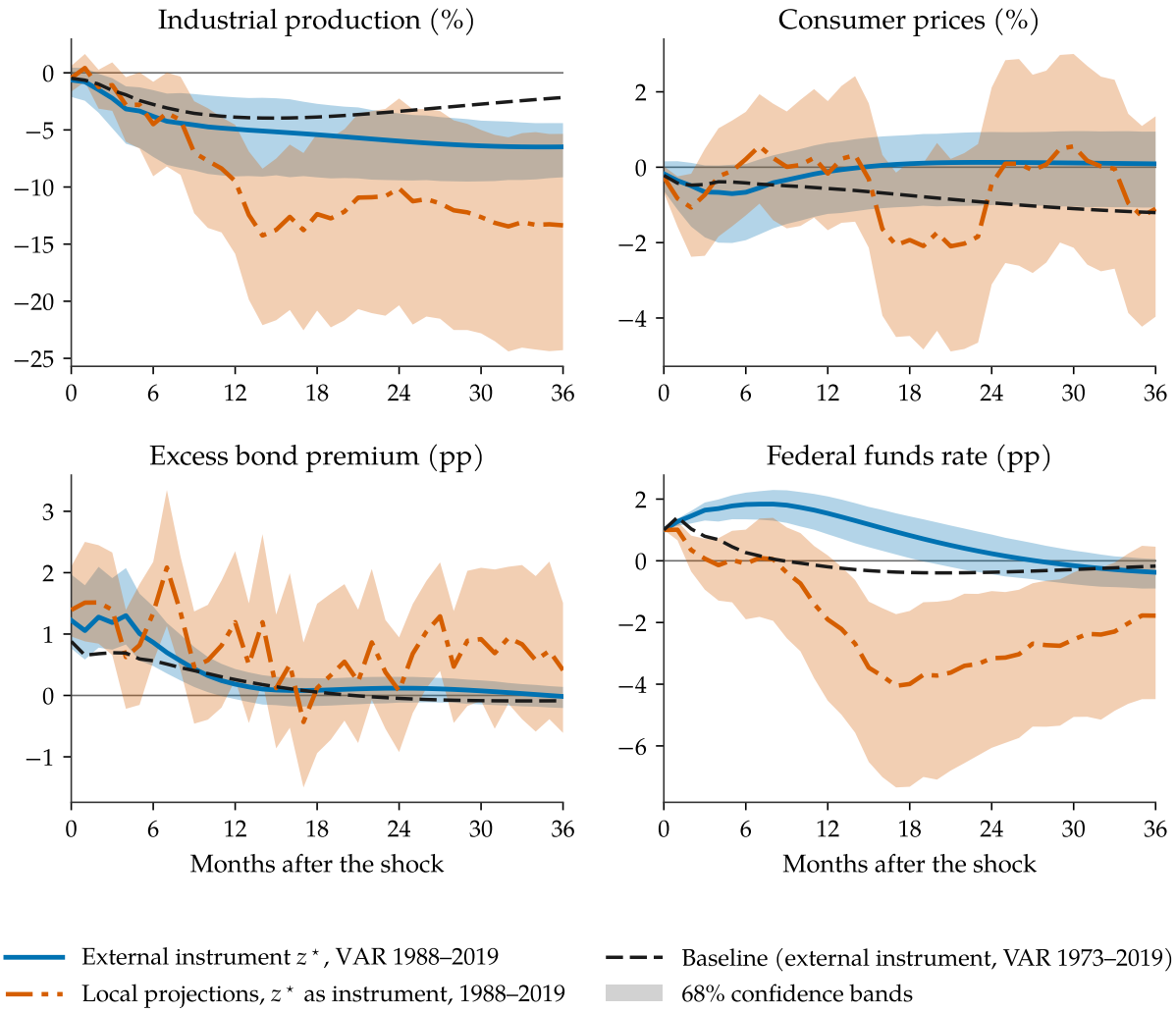


Figure 21: **The pooled instrument in local projections.** Impulse responses to a +1 percentage-point funds-rate shock. Solid, with 68% confidence band: $z_t^\star$ as an external instrument on a VAR estimated from January 1988 to December 2019, as in Figure 20. Dash-dotted, with 68% confidence band: the local projections of Appendix E.4, with the funds rate instrumented by $z_t^\star$, leads up to December 2019. Dashed: the baseline. Bands: median-recentred percentile intervals from a moving-block bootstrap of the time index, 1,000 draws, with the pooling weights held at their estimates.